

Talking Between Cells

CD Molecules · MHC · Cytokines · PRRs · The Inflammasome

PART 1 CD Molecules — the Vocabulary of the Cell Surface

PART 2 The Three Signals — How a T Cell Is Switched On

PART 3 MHC and HLA — the Display Case

PART 4 Cytokines — Properties, Receptors, Signalling

PART 5 What Cytokines Do — Differentiation, Survival, Death

PART 6 Innate Immunity — Triggers and Sensors

PART 7 The Inflammasome — Two Signals Again

PART 8 When It Goes Wrong — Storm, Disease, Therapy

PART 9 Master Tables, Self-Test and Glossary

SYNTHESIZED FROM

“Immune Molecules (CD molecules, cytokines and MHC)” — Lecture 3 · “Innate Immunity” — Lecture 4

Francis Kaming Chan, PhD · Liangzhu Laboratory, ZJU School of Medicine · 54 + 48 slides

Abbas & Lichtman · Janeway's *Immunobiology* · Kuby *Immunology*

For the mechanisms the slides state but do not explain

How to Use This Guide

These two lectures look like a vocabulary list — CD3, CD28, CTLA-4, HLA-DR, IL-2, γ c, JAK3, TLR4, MyD88, NLRP3, ASC, caspase-1, GSDMD — and a vocabulary list is exactly what you must not make of them. Every one of those names is **a word in a sentence that one cell says to another**, and the sentence always has the same shape.

The sentence every molecule belongs to

Nothing in these lectures acts alone. Every molecule occupies one of four slots, and if you can say which slot, you have understood it:

1. **Who is speaking, and from which cell?** — the ligand, and the cell that carries it.
2. **Who is listening, and from which cell?** — the receptor. A receptor with no named partner is a fact you have half-learned.
3. **What happens inside the listener?** — the signalling chain, which always ends at a **transcription factor**: NFAT, NF- κ B, AP-1, STAT, T-bet, GATA3.
4. **What breaks if it is gone?** — the deficiency, which is the proof that slot 3 is right. X-linked SCID, ALPS, MyD88 knockout, HLA-B27 disease.

The organising idea: nothing commits on one signal

This is the single observation that ties Lecture 3 to Lecture 4, and your lecturer builds both around it in the same words — “**signal 1 and signal 2**”.

SIGNAL 1

Is it there?

The **specific** signal. For a T cell, the TCR seeing its peptide-MHC. For a macrophage, a PRR seeing a PAMP. It says *something is here* — but on its own it is not enough.

SIGNAL 2

Is it dangerous?

The **permission** signal. For a T cell, CD28 meeting B7 on an activated APC. For the inflammasome, the activator — ATP, crystals, a pore-forming toxin. Without it the cell does **not** commit.

SIGNAL 3

What kind of response?

The **steering** signal — cytokines. Once the cell has committed, cytokines decide *which* effector it becomes: Th1, Th2, Th17, Treg. This is why cytokines are the bridge between innate and adaptive immunity.

Your lecturer draws this as a car. **Signal 1 is the ignition key** — without it nothing at all happens. **Signal 2 is the accelerator** — the key alone leaves the car idle, and in a T cell an idle car is not neutral but *anergic*, permanently switched off. **Signal 3 is the steering wheel** — it does not start the car, it decides where it goes. Keep that picture; the whole of Part 2 hangs on it, and Part 7 is the same picture drawn inside a macrophage.

Six devices built on that

Device	What it is for
Partner card	A CD molecule on its own is unlearnable; a pair is not. Every important surface interaction gets a card naming which molecule, on which cell, binds which molecule, on which cell — and what results . If you can produce the partner, you know the molecule.
Signal tags S1 S2 S3	Every molecule involved in switching a cell on is tagged with the signal it carries. The commonest exam question in this material is “which signal is this?”, so the answer travels with the molecule.

Device	What it is for
Cell tags T B APC MΦ	Which cell carries the molecule. Half of all mistakes in this topic are putting the right molecule on the wrong cell — CD28 is on the T cell, B7 is on the APC, and swapping them makes the sentence meaningless.
Signalling strips	Four chains are drawn as step-by-step strips — TCR, cytokine receptor, TLR and inflammasome. Each one ends at a transcription factor or an active enzyme, and each is drawn on the same template so the shared logic is visible.
Absence proves function	A red box giving the disease that appears when the molecule is missing. These two lectures are unusually rich in them: X-linked SCID, ALPS, MyD88 knockout, caspase-1 knockout, HLA-B27 disease, CAPS.
Alias decoder	The single biggest trap here is that one molecule has several names . CD21 is CR2 is the C3d receptor is the EBV receptor. CD152 is CTLA-4. CD25 is IL-2Rα. One table carries every alias in these two lectures, and you should read it before anything else.

Suggested study sequence

- 1. Before anything else (≈10 min).** §1.2 — the alias decoder. Half the confusion in this material is two names for one molecule.
- 2. First pass (≈50 min).** Parts 1–3: the partner cards, the three signals, MHC. Do not read the cytokine parts yet.
- 3. Second pass (≈60 min).** Parts 4–5, then 6–7. Notice at the top of Part 7 that you have seen this two-signal structure before.
- 4. Active recall.** Cover the right-hand side of every partner card and produce the partner. Then redraw the four signalling strips on blank paper.
- 5. Night before the exam (≈30 min).** Part 9 — master tables, then the self-test, then the glossary.

WHAT THE LECTURER SAYS TO TAKE AWAY

Lecture 3 ends with a slide titled *“Putting it all together”* naming three things, and Lecture 4 opens with four learning objectives. Together they are the examinable core of both:

- **CD molecules let us define different immune cell types.**
- **MHC molecules dictate the activation of T cells.**
- **Cytokines steer T cells to develop different immune functions.**
- **Innate immunity:** its triggers (DAMPs, PAMPs), its sensors (PRRs), its signalling pathways and its effectors (cytokines).
- **The inflammasome** as the hub of innate inflammation — its two-signal model, its adaptors, its caspases, and its diseases.

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PART ONE

01

CD Molecules — the Vocabulary of the Cell Surface

A leukocyte's surface is not a label; it is a **sentence about what that cell is doing right now**. Molecules appear when the cell is activated and disappear when it rests, which is exactly why we can read them. This Part gives you the reading system: what a CD number actually means, which molecule pairs with which, and — most importantly — the alias table, because the commonest reason this topic feels impossible is that one molecule has four names.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Define a leukocyte differentiation antigen and say why the definition contains the word “differentiation”.
- Explain what a CD number is assigned to — and why it is defined by antibodies rather than by structure.
- Translate between CD numbers and functional names in both directions.
- Name the partner of every surface molecule in this Part, and which cell each sits on.
- Explain why neither the BCR nor the TCR can signal on its own.

1.1 Leukocyte Differentiation Antigens

DEFINITION

Leukocyte differentiation antigens (LDAs) are **cell-surface molecules that may appear or disappear on the cell membrane during different stages of leukocyte differentiation and activation.**

READ THE DEFINITION AGAIN — THE IMPORTANT WORD IS “MAY”

A surface molecule is not a permanent name tag. It is **expressed conditionally**, and that conditionality is the entire diagnostic value of the system. If CD25 were always present on T cells it would tell you nothing; because it appears only after activation, its presence tells you the cell *has been activated*. The same logic runs through this whole topic: **CTLA-4 is induced only after the TCR has fired** (§2.2), **NLRP3 is induced only after a PRR has fired** (§7.2). In immunology, “when is it there?” is usually a more useful question than “what is it?”.

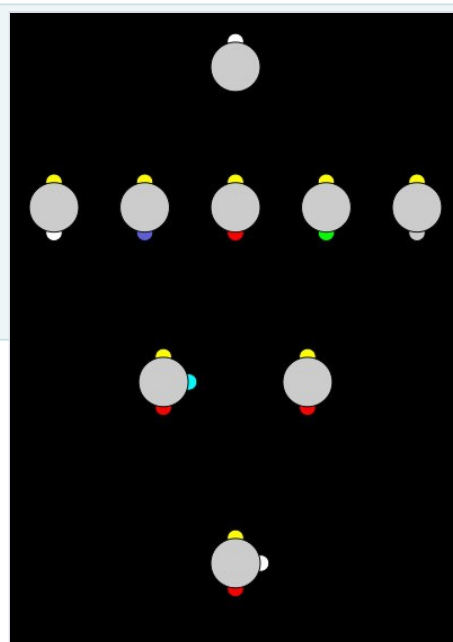
Two qualifications the slide makes

- **They are not confined to leukocytes.** The same molecules are found on **erythrocytes, endothelial cells and epithelial cells**. The name is historical — they were *found* on leukocytes — and the naming system has simply been kept as it expanded. Do not assume a CD number means “white cell”.
- **They are classified two different ways, and both are used.** By **function** — costimulation, chemotaxis, cell adhesion, complement fixation. By **structure** — the immunoglobulin superfamily, the TNF superfamily, the C-type lectins, the integrins. A question may come from either direction, so learn at least one family membership for each major molecule.

1.2 What “CD” Means — and the Alias Decoder

DEFINITION

A **cluster of differentiation (CD)** is the label given to a membrane molecule that is **recognised by a particular cluster of monoclonal antibodies**. When monoclonal antibodies raised in different laboratories all turn out to react with the same membrane molecule, they are **grouped together as one cluster**, and that cluster receives a number.



EXAM TRAP — THE CLUSTER IS A CLUSTER OF ANTIBODIES, NOT OF MOLECULES

Almost everyone first reads “cluster of differentiation” as though a group of related *molecules* shared a number. They do not. **One CD number = one membrane molecule**; the cluster is the set of antibodies that all hit it. The definition is therefore **operational** — it says how the molecule was *detected*, not what it *is* or what it *does*. That is why a CD number tells you nothing about function on its own, and why the alias table below exists at all.

The direct consequence: this entire nomenclature exists *because* hybridoma technology exists. Without Köhler and Milstein's monoclonal antibodies there would be no clusters to count. The CD system is the most-used practical product of the technique you met in Session 2.

Fig 1.1 The idea in one picture: cells that look alike are told apart by the **combination** of surface molecules they carry. No single CD identifies a cell; a pattern does.

- Numbering runs **CD1 to about CD350** in humans (≈ 350 as of the 2009 count quoted in the lecture; the list has continued to grow).
- A CD molecule may be a **receptor** or a **cytokine held on the cell surface** — CD256 (APRIL) and CD257 (BAFF) are cytokines that happen to have CD numbers.

ALIAS DECODER — READ THIS BEFORE ANYTHING ELSE

CD3	TCR signalling complex	$\gamma \delta \epsilon \zeta$ (η) chains; carries the ITAMs. The pan-T-cell marker
CD19	B-cell co-receptor subunit	The pan-B-cell marker ; the target of CAR-T therapy in B-cell leukaemia
CD21	CR2 = C3d receptor = EBV receptor	Four names, one molecule. You met it in Session 2 §6.3 as the link between complement and B-cell activation; Epstein-Barr virus uses the same door
CD25	IL-2Rα	The activation marker of T cells, and the third chain that converts an intermediate-affinity IL-2 receptor into a high-affinity one (§4.3)
CD35	CR1 — complement receptor 1	The erythrocyte immune-complex carrier of Session 2 §6.3, listed here as a myeloid CD molecule
CD40L	CD154	On the T cell. Its partner CD40 is on the B cell / APC. Mutation causes hyper-IgM syndrome
CD80 / CD86	B7-1 / B7-2	On the APC . Both bind CD28 and CTLA-4. “B7” and “CD80/86” are used interchangeably in the same sentence, constantly
CD152	CTLA-4	The brake . Almost always written CTLA-4, almost never CD152 — but the lecture slide uses the number
CD64	FcγRI — high-affinity IgG receptor	On macrophages. The receptor end of the opsonisation story from Session 2
CD79a / CD79b	Igα / Igβ	The signalling heterodimer of the BCR — the B-cell equivalent of CD3
CD11b / CD11c	CR3 / CR4 α chains	CD11b = macrophage marker (with CD18 = Mac-1 = CR3); CD11c = dendritic-cell marker
CD256 / CD257	APRIL / BAFF	TNF-family cytokines supporting B-cell survival. BAFF's receptor is BAFF-R = CD268

How to use this table. Do not try to memorise it as pairs of strings. Instead, for each row, say the **sentence** the molecule belongs to. “CD21 is on the B cell; it binds C3d that complement has deposited on antigen; the result is that the B cell needs a thousand times less antigen to respond.” Once you can say that sentence, the four names of CD21 are just four ways to start it — and you will never again wonder whether CR2 and CD21 are two different molecules.

1.3 Lineage Markers: Reading a Cell from Its Surface

TABLE 1.1 THE LECTURER'S TABLE OF MAJOR HUMAN CD MOLECULES, BY CELL AND BY JOB

Cell	Lineage markers	Signalling	Costimulation and other
T cells (T)	CD3 (all T) CD4 (helper) CD8 (cytotoxic)	CD3	CD28 (the accelerator) · CD152 = CTLA-4 (the brake) · CD154 = CD40L (the help signal to B cells)
B cells (B)	CD19, CD20	CD79a (Igα), CD79b (Igβ)	CD40 (receives help) · CD80 (B7-1), CD86 (B7-2) (give costimulation) · BAFF-R

Cell	Lineage markers	Signalling	Costimulation and other
Myeloid cells 	CD14 (the LPS co-receptor) CD11b — macrophages CD11c — dendritic cells	CD35 = CR1 · CD64 = FcγRI	CD256 = APRIL · CD257 = BAFF — surface cytokines that keep B cells alive

NOTE ON THE SLIDE

The lecture slide writes the BAFF receptor as “**CD168 (BAFF-R)**”. The BAFF receptor is **CD268** (TNFRSF13C); CD168 is RHAMM, an unrelated hyaluronan receptor. If you are reproducing the slide, reproduce it; if you are answering a question about the BAFF receptor, the number is **CD268**. The biology the slide is teaching — that myeloid cells present BAFF and APRIL to keep B cells alive — is unaffected.

CLINICAL CORRELATION — FOUR CD NUMBERS YOU WILL USE ON THE WARDS

- **CD4 count** — the single number that stages HIV infection and predicts which opportunistic infection to expect.
- **CD20** — the target of **rituximab**. A B-cell marker present on mature B cells but not on plasma cells, which is exactly why anti-CD20 depletes B cells without abolishing existing antibody.
- **CD19** — the target of **CAR-T** cells in refractory B-cell leukaemia and lymphoma.
- **CD3** — the immunohistochemical stain that says “this infiltrate is T cells”, used daily in histopathology.

1.4 The B-Cell Set: BCR Complex and Co-receptor

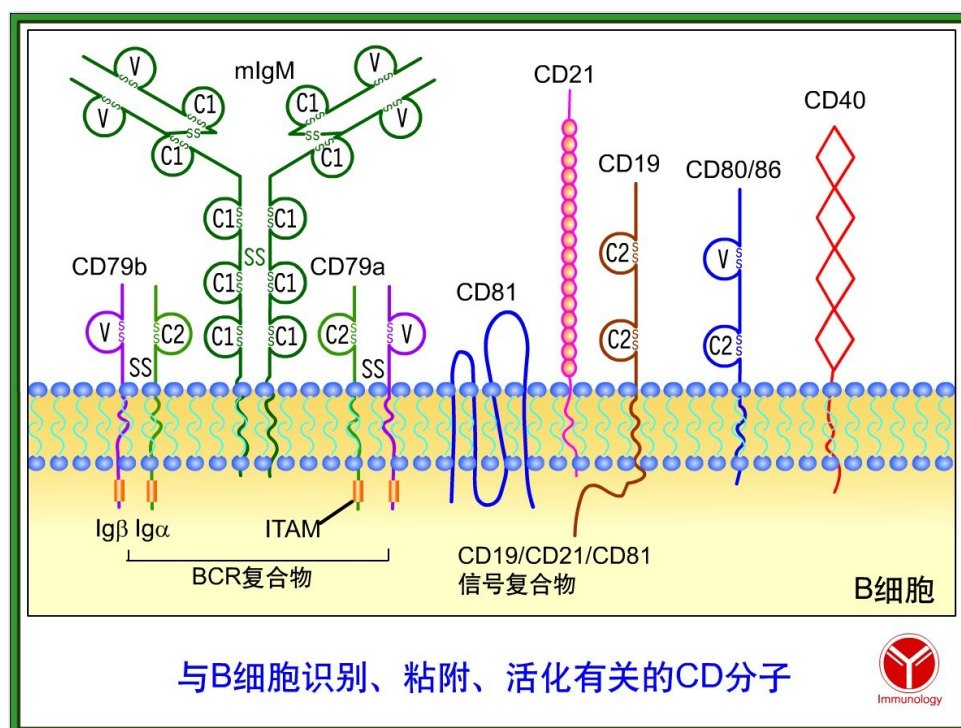


Fig 1.2 The B-cell surface. Left: the **BCR complex** — membrane IgM flanked by the Igα/Igβ (CD79a/CD79b) heterodimer carrying the ITAMs. Middle: the **co-receptor complex** CD19/CD21/CD81. Right: **CD40** and **CD80/86**, the molecules by which the B cell talks to T cells.

1.4.1 Why the BCR cannot signal by itself

Membrane IgM has a **cytoplasmic tail three amino acids long**. There is physically nothing inside the cell for a kinase to act on. The receptor can bind antigen beautifully and tell the cell nothing at all. The solution is to bolt a separate signalling unit onto it.

COMPLEX 1 OF 4 BCR + CD79a/CD79b

B CELL — THE ANTENNA

Membrane IgM (mIgM)

Binds antigen. Cytoplasmic tail 3 aa — cannot signal

NON-COVA-
LENTLY AS-
SOCIATED
WITH

B CELL — THE WIRE

CD79a / CD79b = Igα / Igβ

Heterodimer; long cytoplasmic tails containing IT-AMs

→ **Result:** antigen cross-links mIgM, the ITAMs of Igα/Igβ are phosphorylated by Lyn, Fyn and Blk, and the signal finally has somewhere to go. **The BCR recognises; CD79 transmits.**

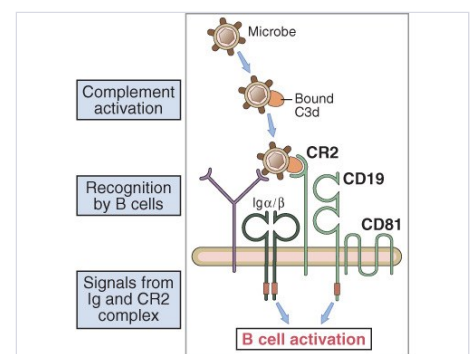
ITAM

Immunoreceptor tyrosine-based activation motif — a short cytoplasmic sequence with the consensus **YxxL/V**, repeated twice. Its two tyrosines are phosphorylated by Src-family kinases, and the doubly phosphorylated motif becomes a **docking site** for kinases carrying tandem SH2 domains: Syk in the B cell, ZAP-70 in the T cell.

REMEMBER IT AS

Antenna and wire. Every antigen receptor in the body is built the same way: a binding chain with no tail, bolted to a signalling chain with a long tail full of ITAMs. **B cell: mIg + CD79a/b. T cell: TCR + CD3.** If a question asks why anti-CD3 antibody activates T cells but anti-TCR-ligand does not, this is the answer — you are pulling on the wire, not the antenna.

1.4.2 The co-receptor: CD19 / CD21 / CD81



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Fig 1.3 Complement is activated on the microbe and leaves **C3d** bound to it. **CR2 (CD21)** grips that C3d while the BCR grips antigen on the same particle. Signals from the Ig complex and the CR2 complex arrive together, and the B cell is activated at far lower antigen concentrations.

PAIR 2 OF 4 CD21 (CR2) ↔ C3d on antigen

B CELL (B)

CD21 = CR2*with CD19 (the signalling subunit) and CD81*

BINDS

ON THE ANTIGEN ITSELF

C3d*the final breakdown product of C3b, left on the microbe by complement*

→ **Result:** the activation threshold of the B cell falls by up to **1000-fold**. This is the molecular bridge between innate and adaptive immunity — complement, an innate system, decides how easily an adaptive response starts. Epstein-Barr virus exploits the same receptor to enter B cells, which is why EBV infects B lymphocytes specifically.

1.4.3 The two molecules by which B cells and T cells talk

PAIR 3 OF 4 CD40 ↔ CD40L (CD154)

B CELL / APC (B) (APC)

CD40*TNF-receptor superfamily; constitutively present*

BINDS

T CELL (T)

CD40L = CD154*TNF superfamily; **induced** on activated helper T cells*

→ **Result:** the T cell delivers “help”. CD40 ligation drives the B cell into **proliferation, class switching and germinal-centre formation**. Loss of CD40L causes **X-linked hyper-IgM syndrome**: the patient makes IgM but cannot switch to IgG, IgA or IgE — the B cell is never told to.

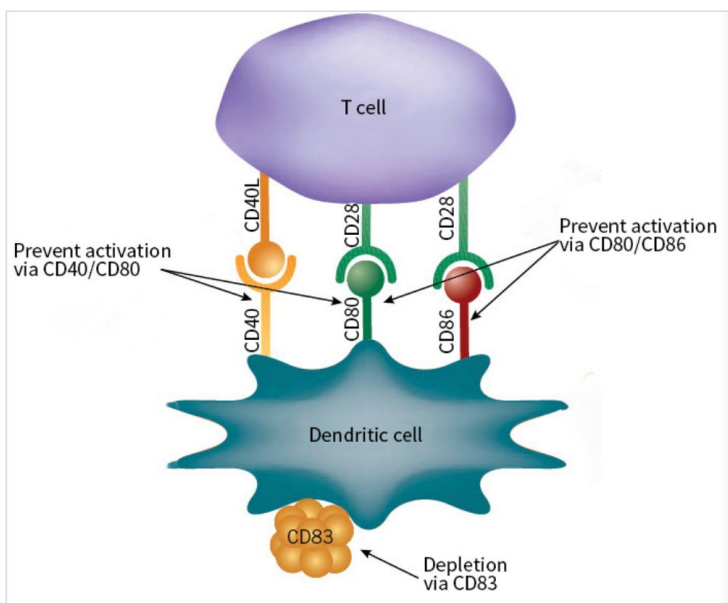


Fig 1.4 Seen from the other side: a dendritic cell presenting **CD40, CD80 and CD86** upwards to a T cell. B cells display exactly the same set — **a B cell is also an antigen-presenting cell**, and that is why the two figures in this section are the same figure drawn from opposite ends.

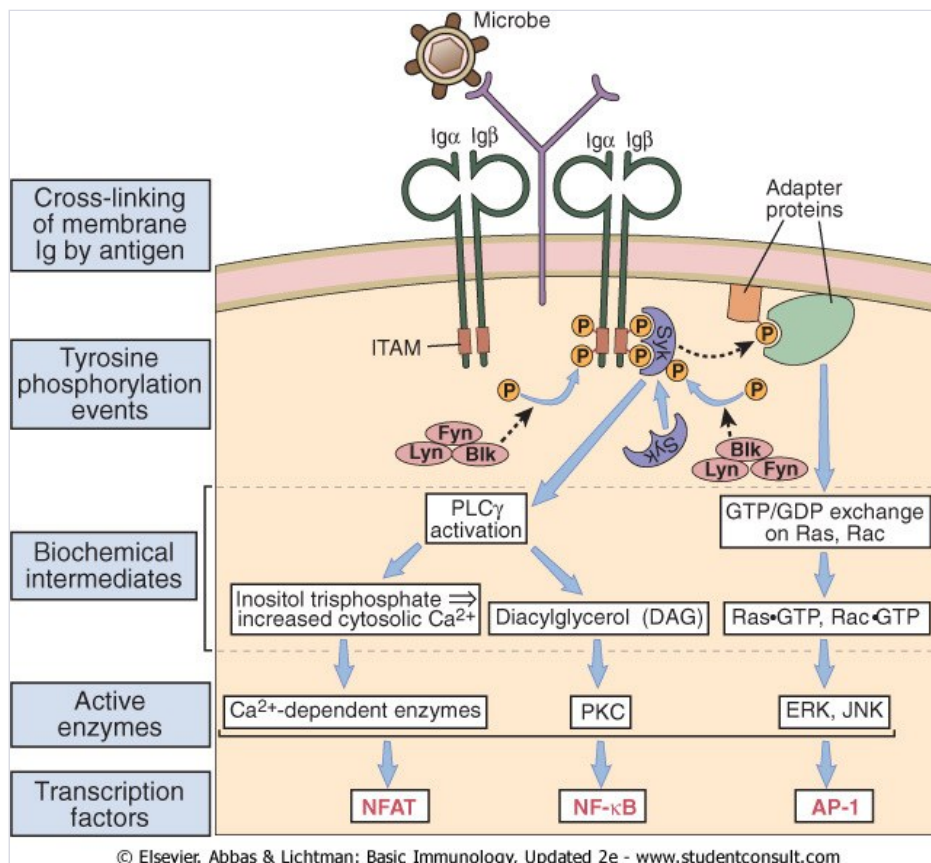


Fig 1.5 The B-cell chain end to end. **Cross-linking** of membrane Ig → **tyrosine phosphorylation** of the ITAMs by Lyn, Fyn and Blk → **PLCγ** and **Ras/Rac** → IP_3 with rising cytosolic Ca^{2+} , and DAG → **Ca²⁺-dependent enzymes**, **PKC**, **ERK** and **JNK** → the transcription factors **NFAT**, **NF-κB** and **AP-1**. Note the destination: three transcription factors. Every signalling figure in this guide ends the same way.

1.5 The T-Cell Set: CD3, CD4 and CD8

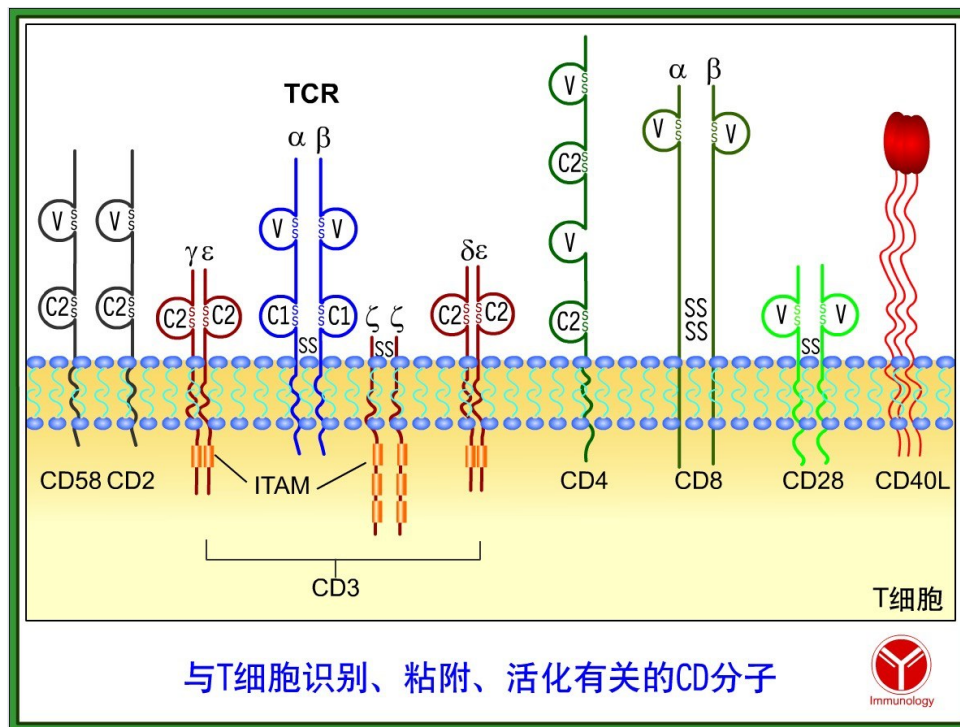


Fig 1.6 The T-cell surface, drawn to match Fig 1.2. The **TCR $\alpha\beta$** heterodimer sits in the middle surrounded by the **CD3** chains ($\gamma\epsilon$, $\delta\epsilon$ and the $\zeta\zeta$ pair) whose tails carry the ITAMs; **CD4** and **CD8** stand beside it as co-receptors; **CD28** and **CD40L** are the costimulatory molecules; **CD2** and **CD58** are adhesion molecules that hold the two cells together while all this happens.

1.5.1 CD3 — the wire of the T cell

MOLECULE CARD · S1 T

CD3

WHAT IT IS

A complex of **five proteins designated γ , δ , ϵ , ζ and η** , some of which form disulphide-bonded dimers. The usual arrangement is one **$\gamma\epsilon$** , one **$\delta\epsilon$** and one **$\zeta\zeta$** pair around the single TCR heterodimer.

WHAT IT CARRIES

Cytoplasmic **ITAMs** with the motif **$YxxL/V$** , which are the targets of tyrosine kinases.

Because: The $\zeta\zeta$ dimer alone carries three ITAMs per chain — six in that pair — while γ , δ and ϵ carry one each. A single TCR engagement therefore delivers **ten ITAMs** worth of signal, which is how a receptor with no intrinsic enzymatic activity can produce an all-or-none response.

WHAT IT DOES

Transduction of the signals that lead to T-cell activation — precisely the role $Ig\alpha/Ig\beta$ play in the B cell. The TCR has an almost non-existent cytoplasmic tail, exactly like membrane IgM.

PROOF, AND A DRUG

In vitro, T cells can be activated by an anti-CD3 antibody alone, with no antigen anywhere. Cross-linking CD3 is sufficient — which proves CD3 is the transmitter. The same fact became a drug: **mur-omonab-OKT3, an anti-CD3 monoclonal, was the first therapeutic monoclonal antibody ever licensed, used to reverse acute transplant rejection. Read its name with the decoder from Session 2 §4.1: **-momab** = fully murine.**

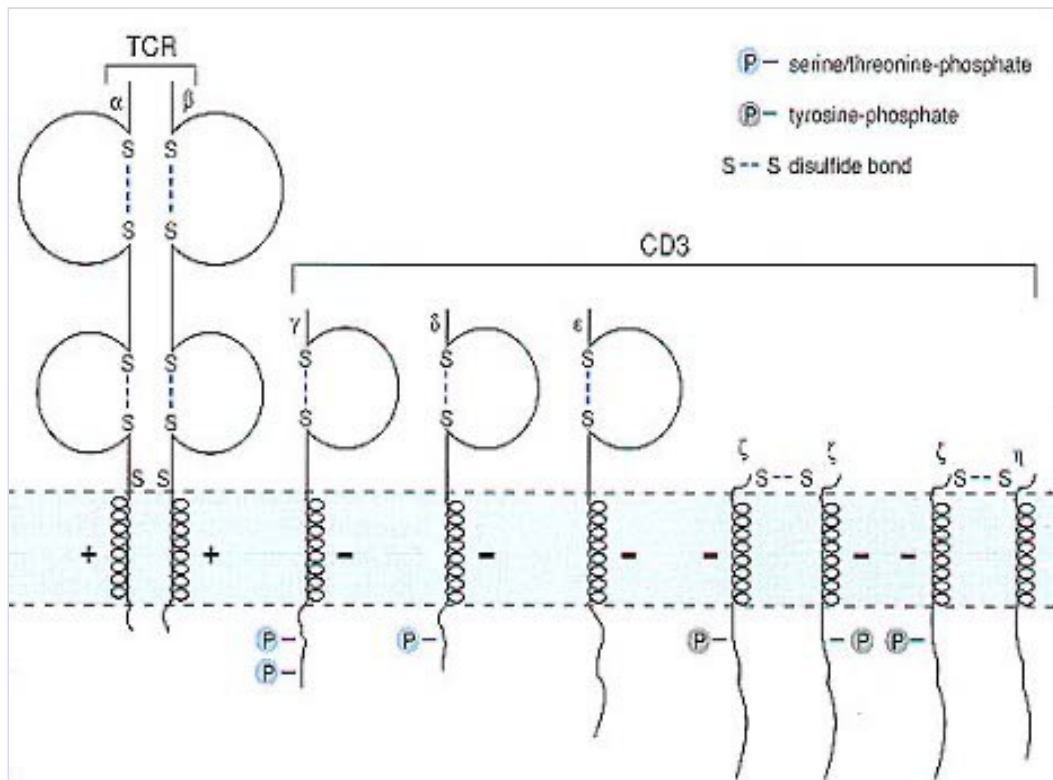


Fig 1.7 The CD3 complex drawn with its charges. The **TCR α and β chains carry positive charges** in their transmembrane segments; the **CD3 chains carry negative charges**. The complex is held together electrostatically inside the membrane — which is why no TCR reaches the surface unless CD3 is made with it, and why CD3 deficiency abolishes surface TCR expression entirely.

1.5.2 CD4 and CD8 — the co-receptors

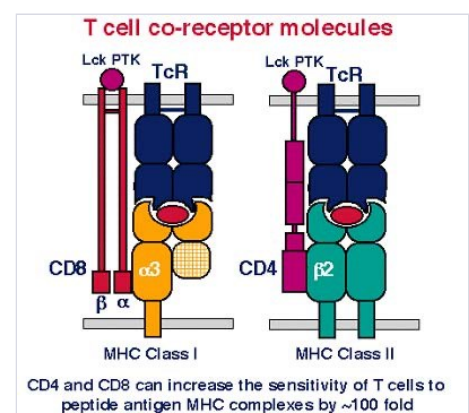


Fig 1.8 **CD8** (left) reaches down to the **$\alpha 3$** domain of MHC class I; **CD4** (right) reaches down to the **$\beta 2$** domain of MHC class II. Both carry **Lck** on their cytoplasmic tails. Together they raise the T cell's sensitivity to peptide-MHC roughly **100-fold**.

TABLE 1.2 CD4 VERSUS CD8

	CD4	CD8
Structure	Single transmembrane glycoprotein, four Ig-like domains	$\alpha\alpha$ homodimer or $\alpha\beta$ heterodimer , one Ig-like domain per chain
What it binds	The non-polymorphic region of MHC class II — its first and second domains contact the $\beta 2$ domain	The non-polymorphic $\alpha 3$ region of MHC class I , via the IgV-like domain of the α chain
Second job	Both bring the tyrosine kinase Lck to the TCR/CD3 complex. The co-receptor is not just a clamp; it delivers the enzyme that starts the signal	
Cell it marks	Helper T cells — and it is the HIV receptor	Cytotoxic T cells

REMEMBER IT AS

The rule of eight. **CD4 × MHC II = 8. CD8 × MHC I = 8.** Multiply the CD number by the MHC class and you always get eight. It is arbitrary, it is not biology, and it will save you in an exam — this pairing is asked about more often than almost anything else in the syllabus.

WHY “NON-POLYMORPHIC” IS THE LOAD-BEARING WORD

MHC molecules are the most polymorphic proteins in the human genome — that is the whole point of Part 3. But **CD4 and CD8 bind the parts that do not vary**. If the co-receptor bound a polymorphic region, it would have to be re-matched to every allele in every person, which is impossible. By binding $\alpha 3$ and $\beta 2$ — constant across all alleles — one CD8 works with every HLA-A, -B and -C molecule a person has. **Polymorphism is confined to the peptide-binding cleft, where it is needed; everything else is kept constant, where variation would break the machinery.**

PART TWO

02

The Three Signals — How a T Cell Is Switched On

A T cell that meets its antigen does not necessarily respond to it. It may respond, it may do nothing, or it may be **permanently disabled by the encounter** — and which of the three happens is decided not by the antigen but by a second molecule on the other cell. That is the most important idea in this Part, and it is the reason your immune system does not attack you.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Name the molecule that provides each of the three signals, and the cell it sits on.
- Explain what happens to a T cell that receives signal 1 without signal 2 — and why that is useful.
- Distinguish CD28 from CTLA-4 by structure, timing, affinity and effect.
- Trace the TCR signal from ITAM to transcription factor, and name the drug that blocks it.

2.1 Signal 1 — the TCR Sees Peptide–MHC

Signal 1 is the **specific** signal: the T-cell receptor recognising a peptide held in the groove of an MHC molecule on another cell. Everything about how that peptide got into that groove is Part 3. What matters here is that signal 1 is **necessary and not sufficient**.

Three molecules deliver it together, and you met all three in §1.5:

- the **TCR $\alpha\beta$ heterodimer** — recognises, cannot signal
- **CD3 S1** — signals, cannot recognise
- **CD4 or CD8 S1** — grips the non-polymorphic part of the MHC molecule and brings **Lck** to the party



Fig 2.1 Your lecturer's picture. **Key start = signal 1** (the TCR): without it nothing happens at all. **Gas pedal = signal 2** (costimulation): the key alone leaves the car standing. **Steering = signal 3** (cytokines): it does not start the car, it decides where it goes.

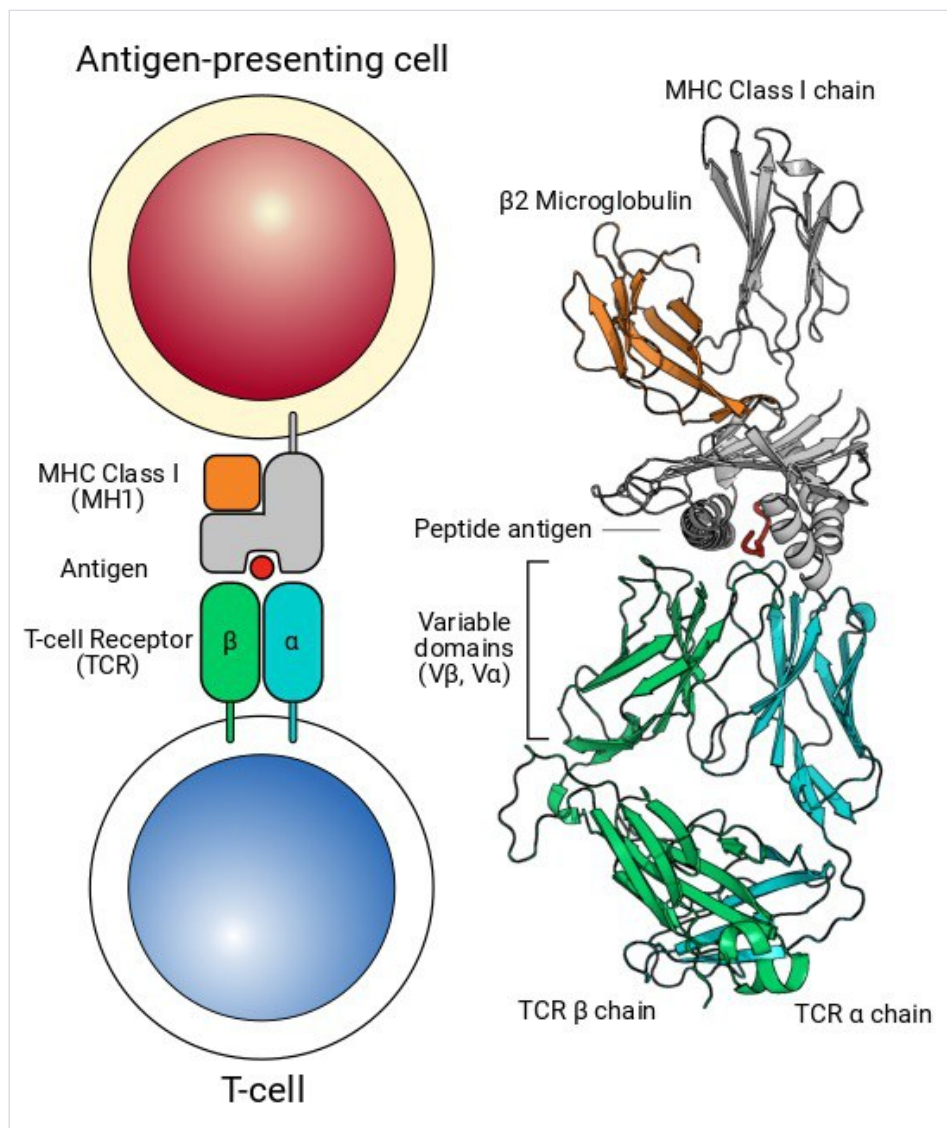


Fig 2.2 What signal 1 physically is. An antigen-presenting cell holds a **peptide** in the cleft of an **MHC class I** molecule (with its β_2 -microglobulin); the **TCR α and β chains** reach up and read both the peptide and the walls of the cleft. The ribbon model on the right shows the same thing at atomic scale: the variable domains $V\alpha$ and $V\beta$ sit over the peptide exactly as V_H and V_L sit over an epitope in an antibody.

2.2 Signal 2 — Costimulation, and the Brake

PAIR 4 OF 4 CD28 ↔ B7-1 / B7-2

T CELL **T**

CD28 **S2**

Ig superfamily; present on resting T cells

BINDS

APC **APC**

CD80 (B7-1) · CD86 (B7-2)

Up-regulated only when the APC has itself been activated — by a PRR, by a PAMP

→ **Result:** the T cell receives permission and commits — proliferation, differentiation, effector function, and production of **IL-2**. **The deep point is where B7 comes from:** the APC only displays it after *its own* innate sensors have detected something dangerous. Signal 2 is therefore innate immunity's veto over adaptive immunity — it is the molecular answer to “is this antigen actually a threat?”.

What happens with signal 1 alone

ANERGY

Anergy is the state of a lymphocyte that has received signal 1 without signal 2. It is **not** simply “no response”: the cell survives but becomes **functionally inactivated** — unresponsive to that antigen even if it is later presented properly with full costimulation. The lecture calls this **abortive T-cell activation**.

WHY THE SYSTEM IS BUILT THIS WAY

Your own tissues present your own peptides on MHC constantly, and a resting tissue cell carries **no B7**. So a self-reactive T cell that escaped thymic selection will meet its self-antigen on a cell that cannot costimulate — and it is **silenced by that meeting rather than activated by it**. The requirement for two signals is not an inefficiency; it is **peripheral tolerance**, and it is the answer to the question the lecturer poses on his slide: “*Why is it important to restrict T cell activation?*”

CTLA-4 — the brake that looks like the accelerator

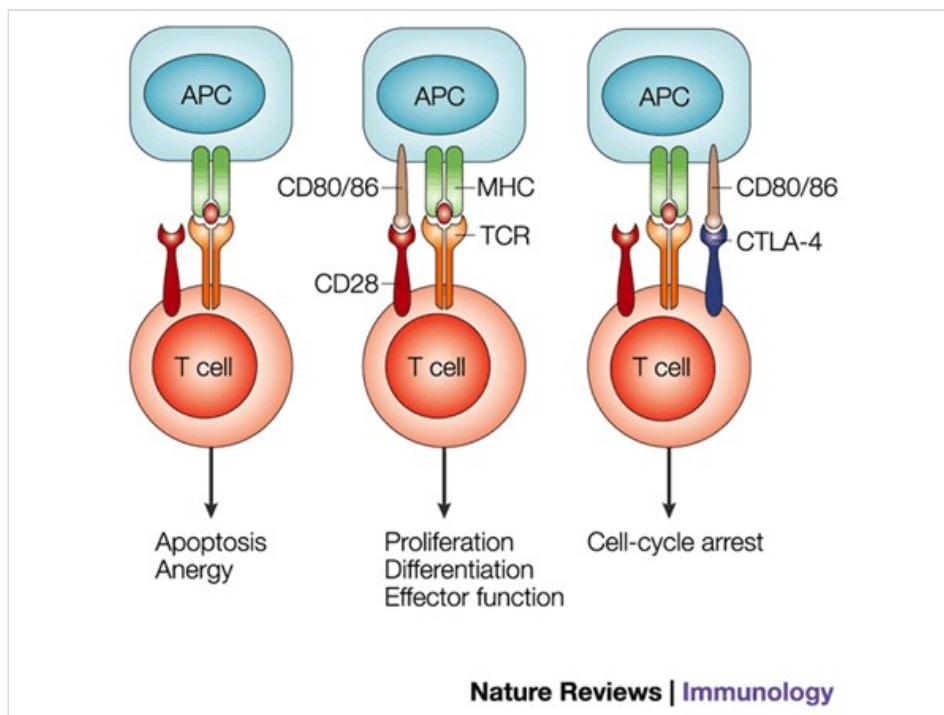


Fig 2.3 The three outcomes, side by side. **Left — MHC/TCR only:** apoptosis or anergy. **Middle — plus CD28 on CD80/86:** proliferation, differentiation, effector function. **Right — plus CTLA-4 on the same CD80/86:** cell-cycle arrest. Note that the middle and right pictures differ by one molecule on the T cell, and the APC is identical in both.

TABLE 2.1 CD28 VERSUS CTLA-4 — SAME LIGAND, OPPOSITE JOB

	CD28 — the accelerator	CTLA-4 (CD152) — the brake
Structure	Ig superfamily homodimer	Structurally related to CD28 — which is why both bind the same ligand
When present	Constitutive — already on the resting T cell, waiting	Induced — appears on the surface only <i>after</i> TCR stimulation
Ligand	The same: B7-1 (CD80) and B7-2 (CD86) on the APC. CTLA-4 binds them with much higher affinity , so once it appears it outcompetes CD28 for the same molecules	
Effect	Promotes activation — proliferation, differentiation, IL-2	Restricts activation — cell-cycle arrest, reduced IL-2
Logic	A self-limiting design: the act of switching a T cell on also switches on the molecule that will switch it off. The response is built to end	

— ABSENCE PROVES FUNCTION — CTLA-4 DEFICIENCY

Mice lacking CTLA-4 do not have a mild phenotype. They develop **massive lymphoproliferation with multi-organ lymphocytic infiltration and die within three to four weeks of birth**. In humans, heterozygous loss-of-function mutations cause **CTLA-4 haploinsufficiency**: lymphoproliferation, autoimmune cytopenias, enteropathy and hypogammaglobulinaemia. Removing one brake from a system with an intact accelerator is fatal — which tells you how continuously that brake is being applied in a normal person.

CLINICAL CORRELATION — RELEASING THE BRAKE ON PURPOSE

If CTLA-4 stops T cells attacking, then blocking CTLA-4 should let them attack — including attacking a tumour. **Ipilimumab**, an anti-CTLA-4 monoclonal, does exactly that in melanoma, and **James P. Allison** shared the **2018 Nobel Prize** for the idea. You met the naming rules and the Nobel in Session 2 §4.2; now you have the molecule. The same logic explains the toxicity: a patient on checkpoint blockade develops **autoimmune colitis, thyroiditis, hepatitis and dermatitis**, because you have removed a mechanism of peripheral tolerance from the entire body, not just from the tumour. **The side-effect profile is the proof that the mechanism is right.**

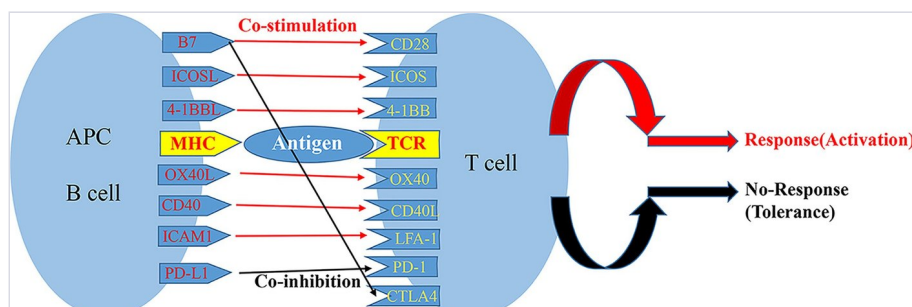


Fig 2.4 CD28/B7 is one pair among many. The upper half lists **co-stimulatory** pairs (B7–CD28, ICOSL–ICOS, OX40L–OX40, CD40–CD40L), the lower half **co-inhibitory** pairs (PD–L1–PD–1, B7–CTLA–4). The T cell adds up everything it is receiving, and the sum decides **response** or **tolerance**. This is why checkpoint blockade has more than one drug target.

2.3 Inside the Cell: from TCR to Transcription Factor

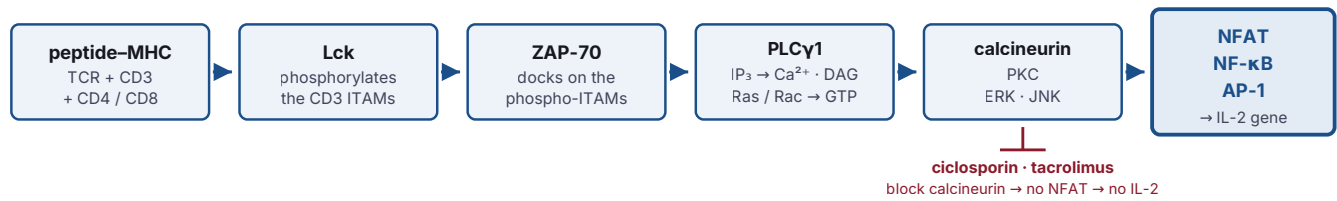


Fig 2.5 The TCR chain end to end, on the same template as every other signalling strip in this guide. Note where the transplant drugs act — **calcineurin is the target of ciclosporin and tacrolimus**, and blocking it is how we prevent graft rejection.

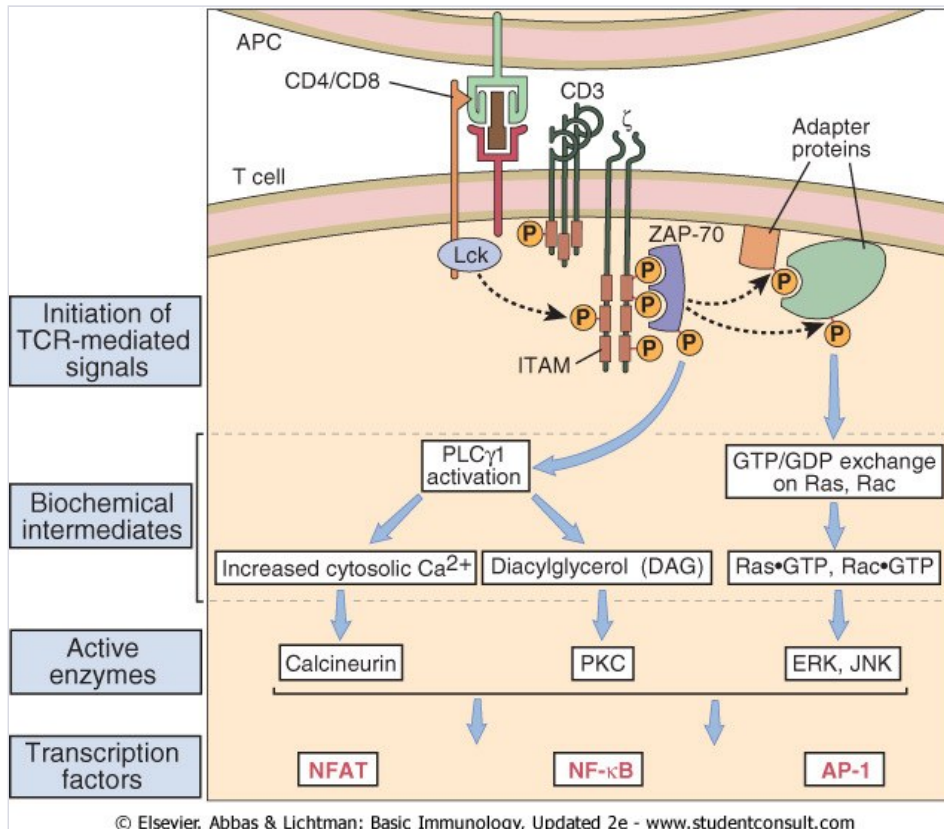


Fig 2.6 The same chain as the textbook draws it. Compare it with Fig 1.5 — the B-cell version — box by box. **They are the same diagram with different names:** Lck for Lyn, ZAP-70 for Syk, PLCγ1 for PLCγ2. The destination is identical: NFAT, NF-κB, AP-1. Learn one and you have learned both.

THE OUTPUT OF ALL THIS IS ONE CYTOKINE

The consequence of T-cell activation is the production of cytokines, and the first and most important is **IL-2**. It is a **trophic** cytokine — it sustains the survival and proliferation of the very cell that made it, an **autocrine** loop. Notice what that means for the three-signal model: **signal 3 is partly self-administered**. A T cell that has received signals 1 and 2 writes its own prescription for signal 3, which is why IL-2 appears again in §4.3 as the model cytokine receptor and in §5.4 as the survival factor of clonal expansion.

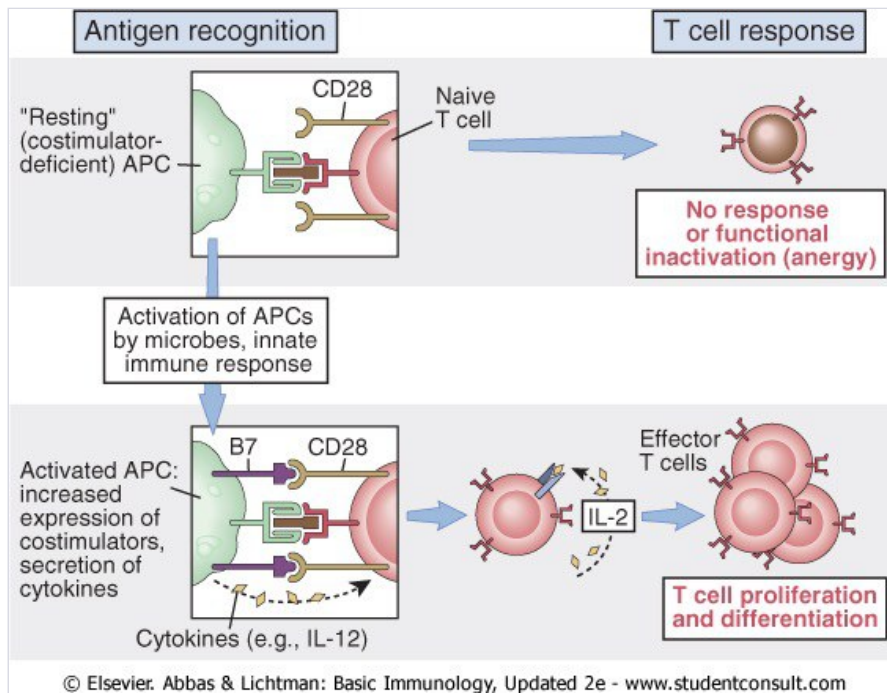


Fig 2.7 The whole of Part 2 in one figure. **Top:** a resting, costimulator-deficient APC presents antigen to a naïve T cell → **no response, or functional inactivation (anergy)**. **Bottom:** microbes and innate immunity activate the APC, which now expresses B7 and secretes cytokines → CD28 is engaged, IL-12 is released, and the T cell **proliferates and differentiates**. Same T cell, same antigen, opposite outcome.

2.4 Signal 3 — Cytokines Steer the Outcome

Signals 1 and 2 decide **whether** the T cell responds. They say nothing about **what kind** of response it mounts — and the kind matters enormously, because the response that clears a virus is useless against a worm and disastrous against a harmless pollen grain.

REMEMBER IT AS

Key, pedal, wheel. Signal 1 starts the engine; signal 2 makes it move; signal 3 decides the destination. A car with the key turned and no accelerator is not neutral — in a T cell, standing still means being scrapped. And a car moving with no steering ends up somewhere you did not want it.

Which cytokines do the steering, which transcription factors they switch on, and which effector cell results, is the whole of §5.1. The three cytokines to carry forward from here are the ones the APC releases at the same moment it displays B7: **IL-12** (drives Th1), **IL-4** (drives Th2) and **IL-6 with TGF- β** (drives Th17). Signal 2 and signal 3 usually arrive from the same cell, in the same encounter, and that is not a coincidence — the innate system that judged the antigen dangerous also judges *what kind* of danger it is.

PART THREE

03

MHC and HLA — the Display Case

A T cell cannot see a whole protein, and it cannot see anything in solution. It can only see a **fragment of a protein, held up by another cell** — and the molecule that does the holding is MHC. Two classes, two sources of peptide, two kinds of T cell. Almost every question in this Part is answered by asking one thing first: *where did the peptide come from?*

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Name the genes, the chains, the cells and the function of each MHC class without hesitating.
- Say which domains form the cleft and which bind the co-receptor — for both classes.
- Explain why class I peptides are 8–10 residues and class II peptides can exceed 20.
- Give four clinical uses of HLA typing and one HLA-linked disease association with its number.

3.1 Histocompatibility: the Words and the Loci

NAME DECODER — THE VOCABULARY OF TRANSPLANTATION

Histocompatibility antigen	A molecule that determines whether tissue is accepted	Split into major (the MHC itself, causing rapid rejection) and minor (ordinary polymorphic proteins whose peptides differ between individuals, causing slow rejection)
MHC	Major histocompatibility complex — a cluster of genes	On chromosome 6 in humans. The word names the gene cluster, not one protein
HLA	Human leukocyte antigens	The name of the human MHC locus. HLA = human MHC . They are the same thing said two ways
H-2	The mouse MHC locus	Every classic experiment in this field was done in mice, so the H-2 names appear constantly in the literature
Syngeneic	Genetically identical	Identical twins, or an inbred mouse strain. No rejection
Allogeneic	Genetically distinct, same species	The ordinary transplant situation. Rejection unless HLA is matched and the patient is immunosuppressed

The historical irony worth knowing. MHC molecules were discovered *because* they cause transplant rejection, and they are named after that. But rejecting transplants is not their function — nothing evolved to reject a kidney graft. Their real job is the one in this Part: **holding peptides up for T cells**. Rejection is a side-effect of being the most variable protein in the population. The name is a permanent reminder that we found the molecule before we understood it.



Fig 3.1 Nobel Prize 1980 — Baruj Benacerraf, Jean Dausset and George D. Snell, “for their discoveries concerning genetically determined structures on the cell surface that regulate immunological reactions.” They found the locus.



Fig 3.2 Nobel Prize 1996 — Peter C. Doherty and Rolf M. Zinkernagel, “for their discoveries concerning the specificity of the cell mediated immune defence.” They found **MHC restriction**: a cytotoxic T cell kills a virus-infected cell only if that cell carries the same MHC allele the T cell grew up with. A T cell recognises peptide **and** MHC, never peptide alone.

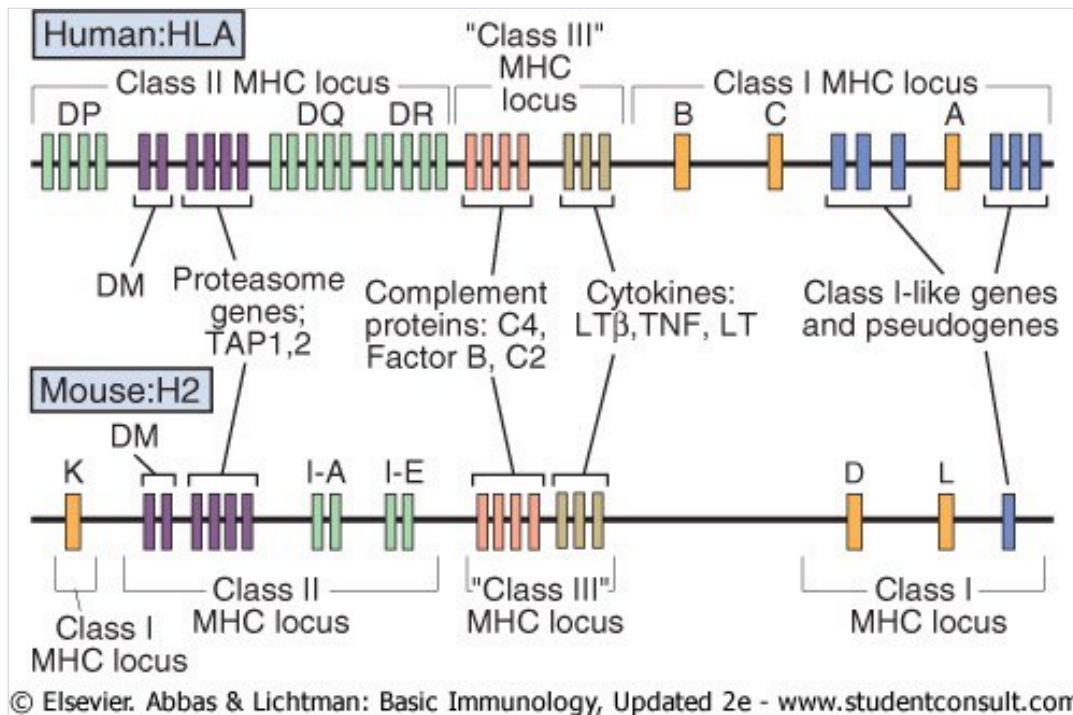


Fig 3.3 The locus, human above and mouse below. Read the human map from the left: **class II** (DP, DQ, DR) → **class III** → **class I** (B, C, A). Two things repay attention. First, **the order is different in the mouse** (class I K sits at the left-hand end), which is why you cannot transfer the map between species. Second, look at what sits in the **class III** region: the **complement proteins C4, factor B and C2**, and the **cytokines TNF, LTα and LTβ**. The gene cluster you learned as “transplantation genes” also carries half of the classical complement pathway from Session 2.

3.2 Class I — Genes, Structure, Function

MOLECULE CARD · MHC I

HLA class I

GENES

HLA-A, HLA-B, HLA-C — three loci, each with two alleles, both expressed. Every person therefore displays **six** different class I molecules.

DISTRIBUTION

Every nucleated cell in the body.

Because: This follows from the function. Any cell can be infected by a virus, so every cell must be able to report what it is making. The exceptions prove the rule: **mature red cells have no nucleus and no class I** — which is exactly why ABO and Rh, not HLA, govern ordinary blood transfusion.

STRUCTURE

α chain, 45 kDa — the only chain encoded in the MHC. Three domains: α1, α2, α3.

β₂-microglobulin, 12 kDa — a small invariant partner, and it is **encoded on chromosome 15, not in the MHC**.

Peptide-binding cleft: α1 + α2 — both walls come from the same chain.

CD8 binding site: α3 — non-polymorphic, membrane-proximal, Ig-fold.

FUNCTION

Recognise and present **ENDOGENOUS** antigens to CD8⁺ T cells. Endogenous means *made inside this cell* — viral proteins, tumour proteins, the cell's own proteins. The message is: “**here is what I am currently synthesising.**”

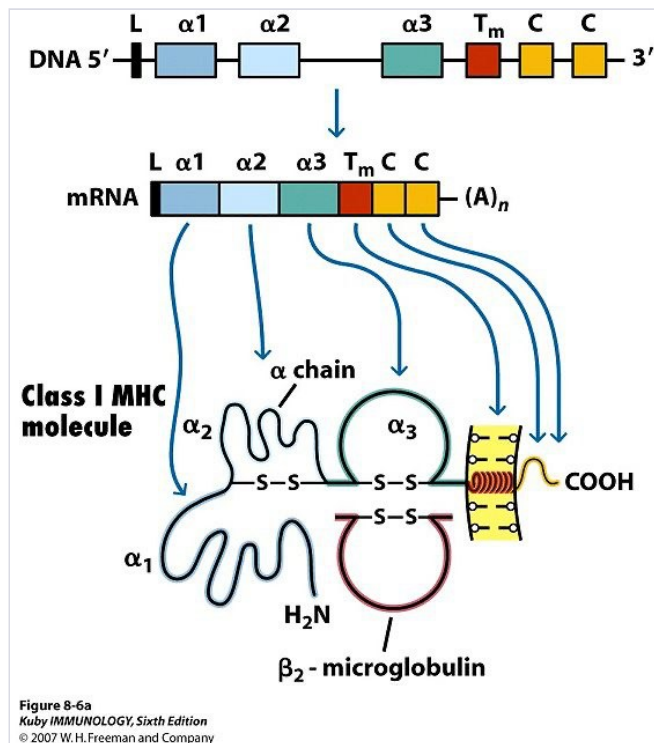
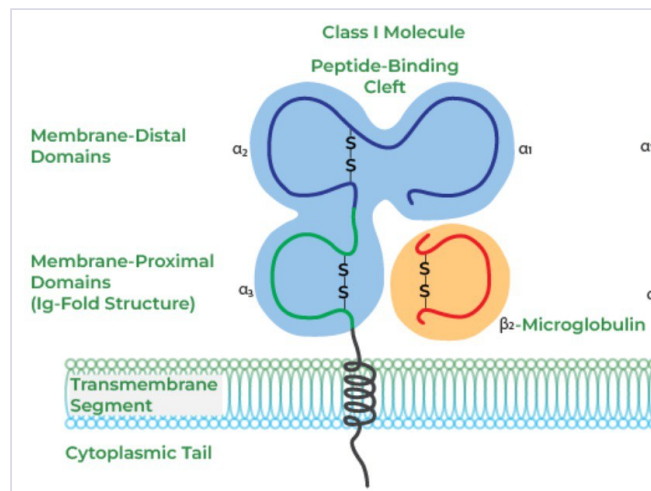


Fig 3.4 From gene to molecule. Each exon of the class I gene becomes one domain: **α1, α2, α3**, then the transmembrane segment and the cytoplasmic tail. β₂-microglobulin arrives from a gene on a different chromosome and folds underneath.

EXAM TRAP — B₂-MICROGLOBULIN IS NOT AN MHC GENE PRODUCT

The class I molecule has two chains, but **only one of them is encoded in the MHC**. β₂-microglobulin sits on **chromosome 15**, is **identical in everybody**, and is **not polymorphic**. Two consequences follow. First, β₂m cannot contribute to transplant mismatch. Second, a mutation in β₂m abolishes surface class I on every cell at once — it is a form of **bare lymphocyte syndrome type I**, in which CD8 T cells fail to develop. Class II, by contrast, has **both** chains encoded in the MHC.

3.3 Class II — Genes, Structure, Function

MOLECULE CARD · MHC II

HLA class II

GENES

HLA-DR, HLA-DQ, HLA-DP — and here **both** the α and the β chain are encoded in the MHC.

DISTRIBUTION

Professional antigen-presenting cells — B cells, dendritic cells, macrophages.

Thymic epithelial cells — where CD4 T cells are selected.

Activated T cells — in humans (not in mice), an activated T cell expresses class II. It is used as an activation marker.

STRUCTURE

α chain 35 kDa and β chain 28 kDa — two similar chains, not one big and one small.

Peptide-binding cleft: $\alpha 1 + \beta 1$ — one wall from each chain.

CD4 binding site: $\beta 2$ — non-polymorphic, membrane-proximal.

FUNCTION

Recognise and present EXOGENOUS antigens to CD4⁺ T cells. Exogenous means *taken in from outside* by phagocytosis or endocytosis and digested in the endosome. The message is: “**here is what I found out there.**”

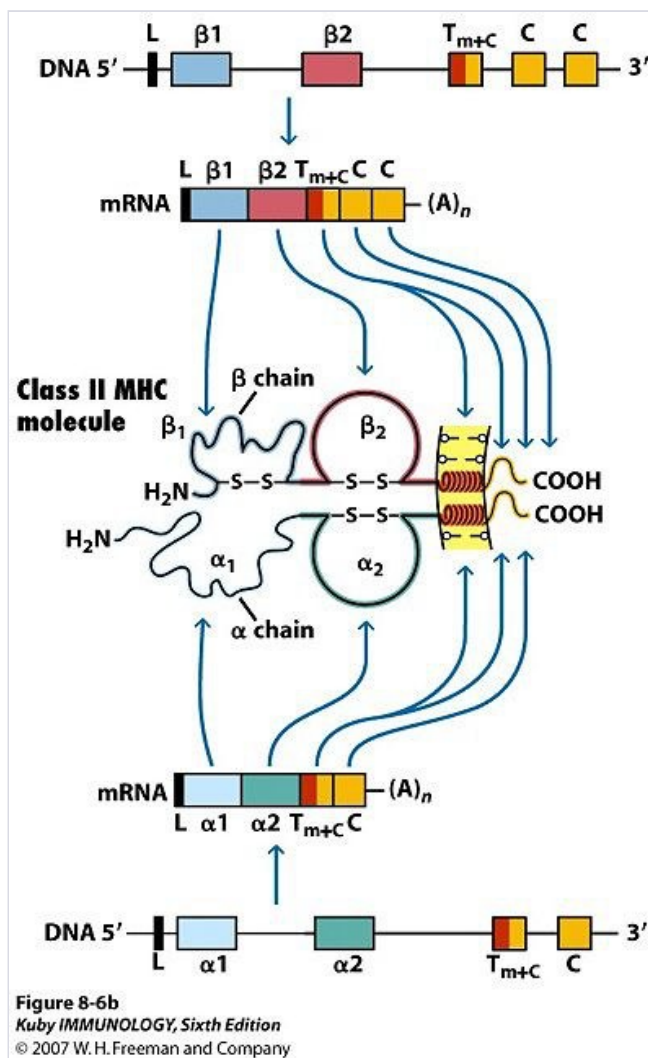


Fig 3.6 Two genes, two mRNAs, two chains — compare with Fig 3.4, where only one chain came from the MHC. The α and β chains are made separately and pair in the endoplasmic reticulum.

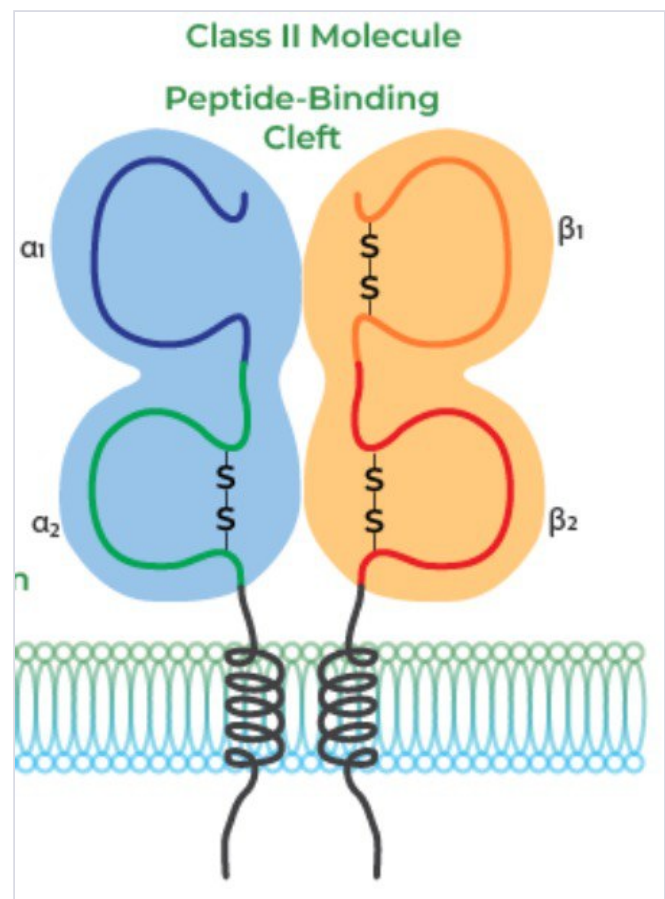


Fig 3.7 The class II molecule is **symmetrical**: $\alpha 1$ and $\beta 1$ above forming the cleft, $\alpha 2$ and $\beta 2$ below as the Ig-fold stalk, and **two** trans-membrane segments rather than one. Set this beside Fig 3.5 and the difference between the classes becomes visual rather than verbal.

3.4 The Peptide-Binding Cleft: Hot Dog and Pita



Fig 3.8 Class I = a hot dog. The bun is **closed at both ends**, so the sausage must be cut to fit: **8–10 amino acids**, no more. Conserved residues at each end of the class I cleft grip the peptide's N- and C-termini.



Fig 3.9 Class II = an open pita. The bread is **open at both ends**, so the filling can hang out: peptides of **13–25 residues**, and **over 20 is routine**. The cleft grips the middle of the peptide and lets the ends dangle.

TABLE 3.1 THE TWO CLEFTS

	Class I	Class II
Walls of the cleft	$\alpha 1 + \alpha 2$ — one chain	$\alpha 1 + \beta 1$ — two chains
Shape	Closed at both ends (a pocket)	Open at both ends (a groove)
Peptide length	8–10 amino acids	Over 20 can be accommodated
Where the peptide came from	Endogenous — cytosolic proteins, chopped by the proteasome, pumped into the ER by TAP	Exogenous — endocytosed proteins, digested in the acidified endosome
Read by	CD8⁺ cytotoxic T cells	CD4⁺ helper T cells

WHY THE SHAPES DIFFER — AND IT IS NOT ARBITRARY

Class I peptides are generated by the **proteasome**, a machine that cuts cytosolic proteins into fragments of a characteristic length and delivers them through TAP into the endoplasmic reticulum, where the class I molecule is waiting and folds *around* the peptide. A closed pocket is possible because the supply is size-matched.

Class II peptides are generated by **endosomal proteases** (cathepsins) chewing a protein at whatever bonds they can reach. The product is a ragged mixture of lengths. An open groove is the only design that can hold a peptide of unpredictable length — so **the shape of the cleft follows the machinery that fills it**.

3.5 HLA in Clinical Medicine

TABLE 3.2 THE LECTURER'S FOUR CLINICAL HEADINGS

Use	What it means in practice
1. Transplantation	HLA typing — determine the donor's and recipient's alleles and match as many as possible, especially HLA-A, -B and -DR. Cross-match — mix recipient serum with donor lymphocytes; if the recipient already has anti-donor HLA antibodies the cells are killed, and a positive cross-match is an absolute contraindication because it predicts hyperacute rejection .
2. Abnormal expression	Tumours down-regulate class I to hide from CD8 T cells — an effective escape, except that a cell with no class I is exactly what an NK cell kills (“missing-self”). Autoimmune disease increases class II on cells that should not have it, such as thyrocytes in Graves' disease, which start presenting their own antigens to CD4 T cells.
3. Disease association	The lecturer's example: ankylosing spondylitis and HLA-B27 , present in 58–97% of patients depending on the population studied, against roughly 8% of the general European population. Note what an “association” is and is not: B27 is neither necessary nor sufficient, it is a risk allele .
4. Medical jurisprudence	Because HLA is codominantly inherited as a haplotype block , a child's HLA type is one haplotype from each parent. Before DNA fingerprinting this was the standard tool for paternity testing and forensic identification , and HLA haplotypes are still used to trace family relationships in transplant donor searches.

CLINICAL CORRELATION — HLA ALLELES YOU MAY BE ASKED TO TEST FOR

- **HLA-B27** → ankylosing spondylitis, reactive arthritis, acute anterior uveitis.
- **HLA-DQ2 / DQ8** → coeliac disease. Their *absence* is the useful result: it essentially rules the diagnosis out.
- **HLA-B*57:01** → abacavir hypersensitivity. Testing before prescribing is **mandatory** in HIV care and has essentially abolished the reaction.
- **HLA-B*15:02** → carbamazepine-induced Stevens–Johnson syndrome, tested for in Han Chinese and several South-East Asian populations.
- **HLA-DR3 / DR4** → type 1 diabetes; **HLA-DR15** → multiple sclerosis.

REMEMBER IT AS

Inside-eight, outside-four. **Class I** shows what is made **inside** the cell, to **CD8**. **Class II** shows what came from **outside**, to **CD4**. Add the rule of eight from §1.5 ($CD4 \times II = 8$, $CD8 \times I = 8$) and you can reconstruct the entire table from two arithmetic facts.

04

Cytokines – Properties, Receptors, Signalling

There are more than a hundred named cytokines and you will never learn them as a list. You do not have to. They obey a small number of rules — about size, concentration, range, receptor architecture and signalling — and those rules are what gets examined. Learn the rules here; the individual cytokines and their jobs are Part 5.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- State the general properties of cytokines with numbers, not adjectives.
- Define and give an example of pleiotropy, redundancy, synergy and antagonism.
- Explain how adding a third chain converts the IL-2 receptor from low to high affinity.
- Explain why one mutation in one shared chain causes severe combined immunodeficiency.
- Draw the JAK-STAT pathway and name two ways the cell switches it off.

4.1 General Properties

TABLE 4.1 THE FIVE GENERAL PROPERTIES, WITH THE NUMBERS THAT MAKE THEM EXAMINABLE

Property	And what follows from it
Produced by immune effector cells and other cell types	Macrophages, lymphocytes, granulocytes and mast cells make them — but so do fibroblasts and endothelial cells . Cytokines are not the private language of the immune system; they are how any tissue calls for help.
Small proteins, mostly < 30 kDa	Small enough to diffuse quickly through tissue and to be filtered by the kidney, which is part of why their half-lives are minutes.
Act at picogram concentrations (pg/ml)	Roughly a million-fold lower than the plasma concentration of albumin. It follows that their receptors must have extremely high affinity — see §4.3 — and that a small change in production makes a large change in effect.
Mostly local, like hormones of the neighbourhood; a few systemic	The default is a short-range conversation between adjacent cells. When a cytokine <i>does</i> reach the circulation in quantity — TNF, IL-1, IL-6 — the result is fever, acute-phase response, and in the extreme case shock (Part 8).
Soluble or held on the cell surface	A membrane-bound cytokine can only act on a cell it is touching, which converts a diffusible signal into a contact-dependent one. Membrane TNF and CD40L work this way, and it is why the CD molecules of Part 1 and the cytokines of Part 4 are not two separate topics.

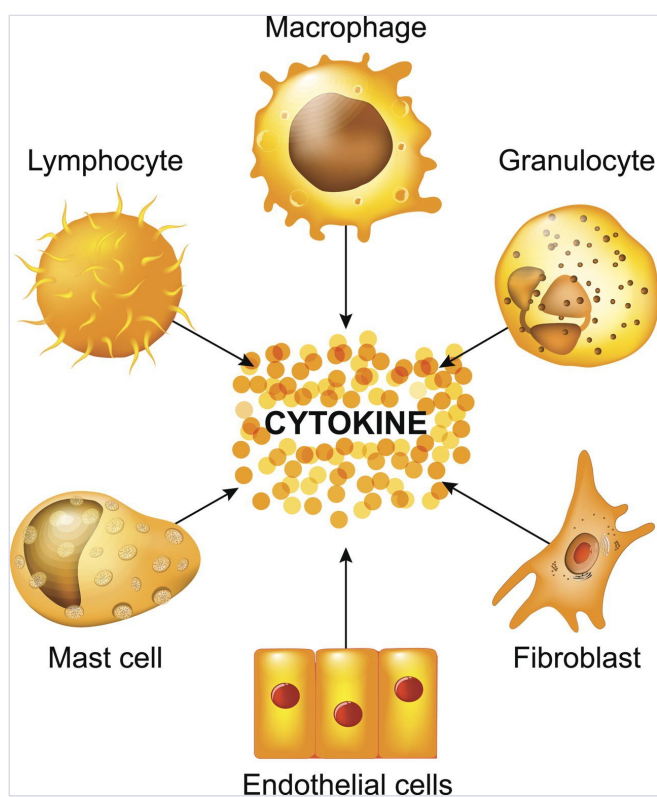


Fig 4.1 Who makes them: lymphocytes, macrophages, granulocytes, mast cells, fibroblasts and endothelial cells all feed the same pool.

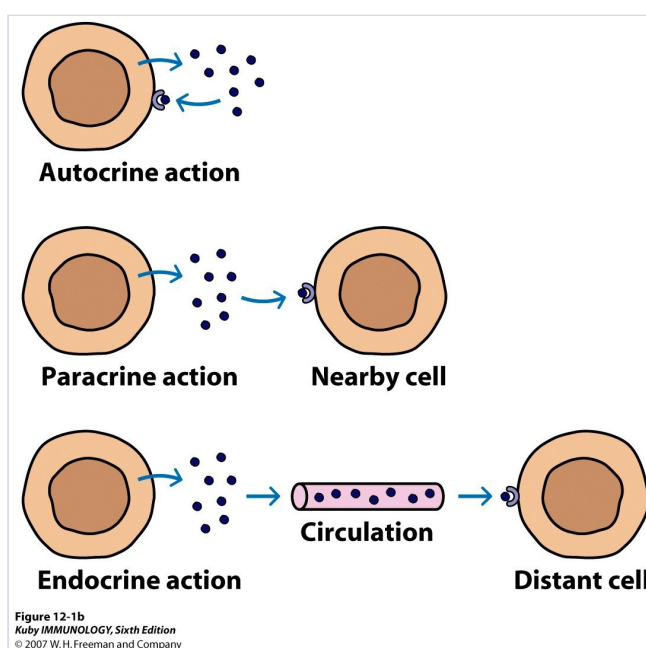


Fig 4.2 The three ranges. **Autocrine** — IL-2 made by a T cell acting back on that same T cell. **Paracrine** — IL-4 made by a T cell acting on a neighbouring B cell. **Endocrine** — carried in the circulation to a distant organ, the way a classical hormone works.

4.2 Functional Properties: Four Words You Must Be Able to Use

NAME DECODER — THE FOUR FUNCTIONAL WORDS

Pleiotropy	One cytokine, many different effects	<i>IL-2 controls proliferation, apoptosis and regulatory T-cell development — three outcomes that point in different directions, from one molecule</i>
Redundancy	Different cytokines, the same effect	<i>IL-15 and IL-2 can both sustain NK-cell proliferation and development. Removing one leaves the other doing the job</i>
Synergy	Two together do more than the sum of both alone	<i>IL-12 and IFN-γ together promote Th1 differentiation far more strongly than either alone</i>
Antagonism	One cytokine blocks another's effect	<i>IL-12/IFN-γ and IL-4 antagonise each other in controlling Th1 versus Th2 development. The two programmes actively suppress one another</i>

Worked example — why anti-cytokine drugs sometimes disappoint. Redundancy predicts it. If two cytokines do the same job, neutralising one leaves the other in place, and the patient does not improve. This is exactly why blocking a **shared receptor subunit** (§4.4) is often more effective than blocking a single cytokine — and why **ustekinumab**, which targets the p40 subunit common to IL-12 and IL-23, does more than an antibody against either cytokine alone. Redundancy also explains the opposite observation: single-cytokine knockout mice frequently look almost normal.

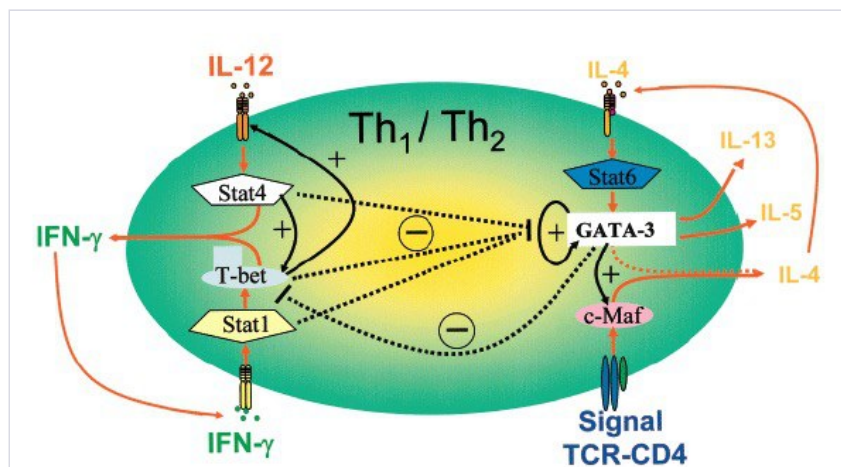


Fig 4.3 Antagonism drawn as a circuit. IL-12 $\rightarrow$ STAT4 $\rightarrow$ T-bet $\rightarrow$ IFN- γ (the Th1 arm) and IL-4 $\rightarrow$ STAT6 $\rightarrow$ GATA-3 $\rightarrow$ IL-4, IL-5, IL-13 (the Th2 arm). The minus signs in the middle are the point: each master transcription factor **suppresses the other**, and each arm's output cytokine **reinforces its own arm**. A circuit with positive feedback on each side and mutual inhibition between them is a **switch** — which is why a T cell commits to one lineage rather than drifting between them.

4.3 Receptor Oligomerisation and Binding Affinity

THE TWO SIGNALLING PRINCIPLES THE LECTURE STATES

- **Cytokines signal as oligomeric complexes.** A single receptor chain sitting alone in the membrane does nothing. Signalling begins when the ligand pulls **two or three chains together**.
- **Oligomerisation dictates affinity, specificity and diversity.** Which chains are recruited decides how tightly the ligand is held, which ligand fits at all, and how many different signals one small set of chains can generate.

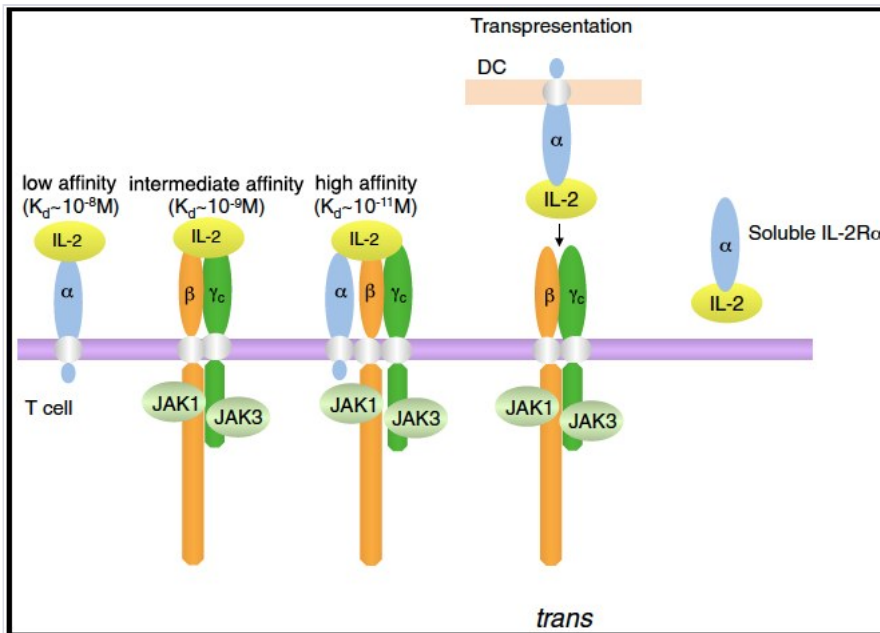


Fig 4.4 The IL-2 receptor in its three states, and the numbers are worth memorising. **α alone**: low affinity, $K_d \approx 10^{-8} \text{ M}$ — binds IL-2 but cannot signal, because α has no cytoplasmic signalling tail. **$\beta + \gamma$** : intermediate, $K_d \approx 10^{-9} \text{ M}$ — can signal; this is the resting form on NK cells and memory T cells. **$\alpha + \beta + \gamma$** : high affinity, $K_d \approx 10^{-11} \text{ M}$ — a hundred-fold tighter, and the form on a recently activated T cell. On the right, **soluble IL-2R α** shed into plasma, and **trans-presentation**, where one cell hands IL-2 to another.

WHY THIS DESIGN IS CLEVER

IL-2 circulates at picomolar concentrations. A resting T cell, carrying only the intermediate-affinity $\beta\gamma$ receptor, effectively cannot hear it. **The act of being activated induces CD25 (IL-2R α)**, which converts that same receptor to the high-affinity form — and now the cell can capture IL-2 at concentrations a hundred times lower.

The consequence is **selective amplification of exactly the right clone**: among a million T cells bathed in the same IL-2, only the ones that have just seen their antigen can use it. The cytokine is shared; the receptor is not. **This is how a clonal response stays clonal despite a diffusible growth factor.**

CLINICAL CORRELATION — CD25 AS A DRUG TARGET AND A MARKER

Basiliximab is an anti-CD25 monoclonal given at the time of renal transplantation. Read the name with the Session 2 decoder: **-ximab** = chimeric. It blocks the high-affinity receptor on *recently activated* T cells only — precisely the alloreactive clones — and leaves resting T cells untouched. **Soluble CD25** in serum is used as a marker of T-cell activation, and is one of the diagnostic criteria for **haemophagocytic lymphohistiocytosis**, a cytokine-storm syndrome you will meet in §8.2.

4.4 Shared Subunits and the Common γ Chain

Cytokines and their receptors are built like modular kit. **A single subunit can partner with several different subunits**, and each combination is a different ligand or a different receptor. This is how the immune system gets combinatorial diversity out of a limited parts list — and it is also how one mutation can be catastrophic.

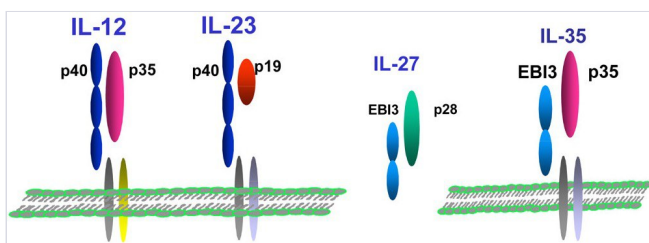


Fig 4.5 Shared subunits on the **ligand** side. p40 pairs with p35 to make **IL-12** and with p19 to make **IL-23**. EBI3 pairs with p28 to make **IL-27** and with p35 to make **IL-35**. Four cytokines with quite different jobs, built from five parts.

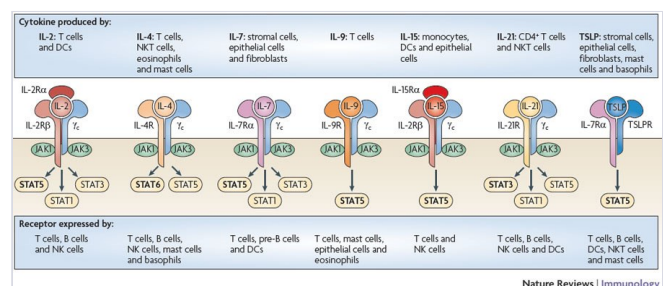


Fig 4.6 Shared subunits on the **receptor** side. The **common γ chain (γ_c , CD132)** is a component of the receptors for **IL-2, IL-4, IL-7, IL-9, IL-15 and IL-21**. Each receptor has its own private chain for specificity and shares γ_c for signalling — and γ_c is the chain that carries **JAK3**.

— ABSENCE PROVES FUNCTION — X-LINKED SCID, THE “BUBBLE BOY” DISEASE

The lecturer asks directly: *what happens if you carry a mutation that abrogates the function of the common γ chain?* Work it out rather than recall it. γ c is required by **six** cytokine receptors, so losing it removes six signals at once:

- **No IL-7 signal** → T cells never develop. **T⁻**
- **No IL-15 signal** → NK cells never develop. **NK⁻**
- **No IL-21 signal** → B cells are present in normal numbers but cannot make useful antibody. **B⁺ but non-functional**

The result is the classic **T⁻ B⁺ NK⁻ severe combined immunodeficiency**, X-linked because the γ c gene is on the X chromosome — which is why it presents in boys. Untreated, the infant dies of opportunistic infection in the first year, and the child photographed in the sterile isolator is the reason the disease is remembered as “bubble boy” syndrome. The cure is a **haematopoietic stem-cell transplant**, and X-SCID was one of the first diseases treated by gene therapy.

Milder versions exist and matter. A *hypomorphic* mutation — one that reduces rather than abolishes γ c function — gives a **less severe combined immunodeficiency** in which the patient survives longer but suffers **opportunistic infections such as severe, persistent Candida**. The same logic applies to **JAK3**, the kinase that γ c carries: JAK3 deficiency produces a clinically identical SCID, but **autosomal recessive** rather than X-linked, and therefore seen in girls as well.



Fig 4.7 A child with severe combined immunodeficiency in a sterile isolator — the image that named the syndrome.



Fig 4.8 The milder form: severe, persistent mucosal **Candida** infection. Deep fungal and viral infections in a child are a T-cell defect until proved otherwise.

4.5 JAK-STAT: the Shortest Path from Surface to Gene

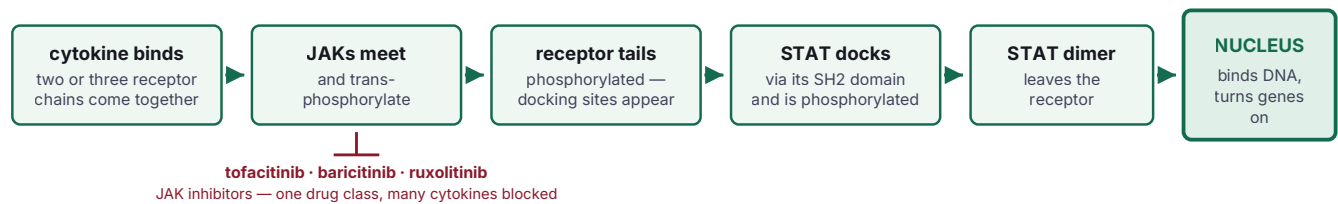


Fig 4.9 JAK-STAT is **the shortest signalling pathway in the cell**: from the outside of the membrane to DNA in five steps, with **no second messengers at all**. Compare it with the TCR strip of Fig 2.5, which needed calcium, DAG, PKC and three MAP kinases to reach the same destination.

TABLE 4.2 THE CAST OF JAK-STAT

Family	Members	What to know
JAK — Janus kinases	JAK1, JAK2, JAK3, Tyk2 — four	They sit pre-bound to the receptor tails, waiting. JAK3 is restricted to haematopoietic cells and associates specifically with γ_c — which is why a JAK3 inhibitor is comparatively lymphocyte-selective, and why JAK3 deficiency phenocopies X-SCID
STAT — signal transducers and activators of transcription	STAT1, 2, 3, 4, 5a, 5b, 6	The name is the mechanism: one protein that is both the signal transducer and the transcription factor. Which STAT is used determines which genes are switched on — STAT4 for IL-12 → Th1, STAT6 for IL-4 → Th2

NOTE ON THE SLIDE

The lecture slide says “**6 STATs — 1, 2, 3, 4, 5a, 5b**”. The canonical family in mammals has **seven** members, the seventh being **STAT6** — and STAT6 appears by name on the lecturer's own Th1/Th2 figure two slides earlier, as the transducer of IL-4. If you are asked to list them, list seven; if you are reproducing the slide, note that STAT5a and STAT5b are counted separately, which is why the list looks short.

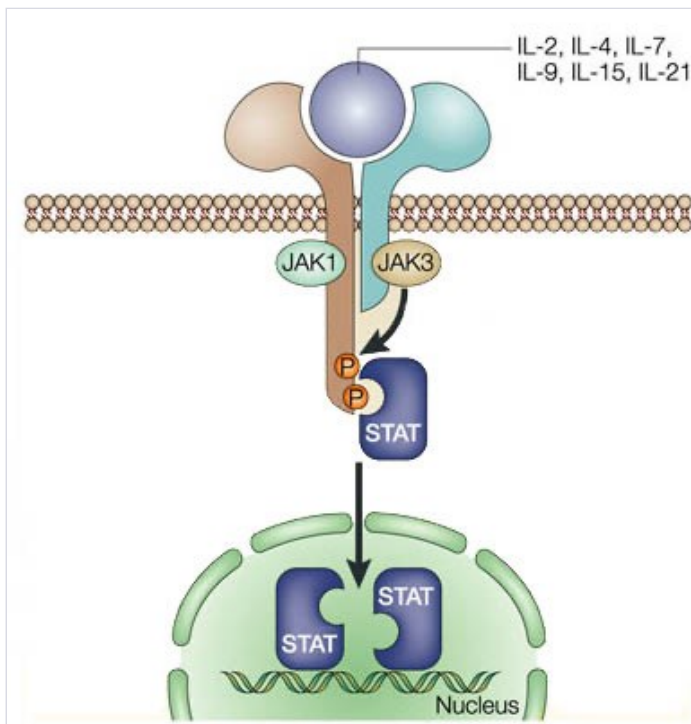


Fig 4.10 The γc family receptor as the textbook draws it: the private chain and γc , carrying **JAK1** and **JAK3** respectively, phosphorylate each other; a **STAT** docks, is phosphorylated, dimerises, and walks to the nucleus. The ligand list on the arrow — IL-2, IL-4, IL-7, IL-9, IL-15, IL-21 — is the γc family of §4.4.

4.6 Switching the Signal Off: SOCS, CIS and Phosphatases

A pathway with no off-switch is a disease. The cell has three, and the elegant part is that **the cytokine itself switches them on** — exactly the same self-limiting logic as CTLA-4 in §2.2.

TABLE 4.3 THREE WAYS TO STOP A CYTOKINE SIGNAL

Brake	Induced by	Mechanism
CIS cytokine-inducible SH2-containing protein	Many cytokines — classic feedback regulation	Occupies the docking site. Associates with βc and the erythropoietin receptor and prevents STAT from docking on the phosphorylated tail. The receptor is still signalling — there is simply nothing left to receive it
SOCS suppressor of cytokine signalling	Cytokines	Binds and inhibits the JAKs themselves , preventing STAT recruitment. Then it goes further: the SOCS box recruits a Cullin-family E3 ubiquitin ligase , and the JAK is destroyed in the proteasome . This is not damping — it is demolition
Phosphatases e.g. SHP-1	Constitutively present	Remove the phosphate. Every step of JAK-STAT is a phosphorylation, so a phosphatase simply reverses it. Fast, reversible, and requires no new protein synthesis

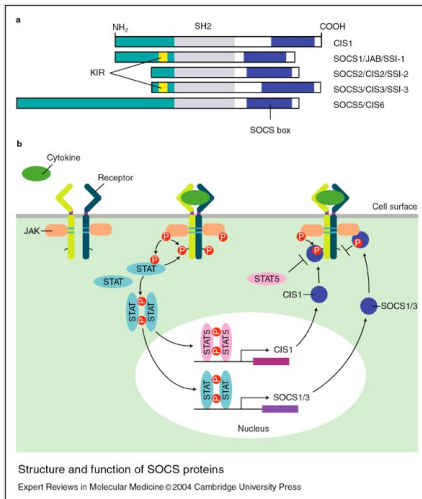


Fig 4.11 The SOCS family. Every member has the same three-part architecture: a variable N-terminus, an **SH2 domain** that finds the phosphotyrosine, and the **SOCS box** at the C-terminus. Below, the feedback loop: cytokine → STAT → nucleus → transcription of CIS and SOCS → inhibition of the receptor that started it.

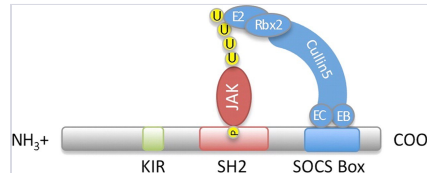


Fig 4.12 How the JAK is destroyed. The **SOCS box** binds **Cullin** and **Rbx2**, assembling an **E3 ubiquitin ligase** around the captured JAK. Ubiquitin chains are built on it, and the proteasome takes it apart. The cell does not merely silence the signal; it removes the machine.

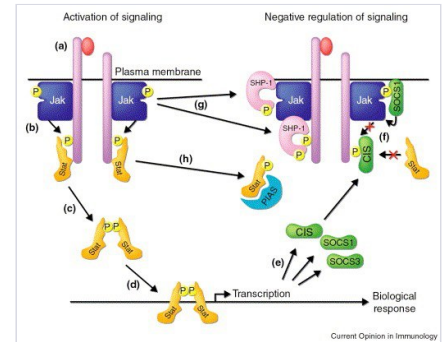


Fig 4.13 Activation (left) against negative regulation (right), on one diagram. Trace the letters: **(a–d)** is the normal pathway to transcription; **(e)** STAT induces CIS, SOCS1 and SOCS3; **(f)** CIS blocks the STAT docking site; **(g)** SHP-1 dephosphorylates the receptor and JAK; **(h)** SOCS targets the JAK for degradation. **Every off-switch is switched on by the signal itself.**

CLINICAL CORRELATION — DRUGGING THE PATHWAY IN BOTH DIRECTIONS

Blocking it: the **JAK inhibitors** — tofacitinib, baricitinib, upadacitinib (rheumatoid arthritis, ulcerative colitis, atopic dermatitis) and ruxolitinib (myelofibrosis, polycythaemia vera, graft-versus-host disease). Because so many cytokines converge on four kinases, one small molecule blocks a whole panel of cytokines at once — which is both the efficacy and the toxicity: infection, herpes zoster reactivation, cytopenias.

When it is stuck on: the **JAK2 V617F** mutation makes the kinase constitutively active without any cytokine, and causes **polycythaemia vera**, **essential thrombocythaemia** and **primary myelofibrosis**. A pathway whose brakes are described on this page is a pathway whose failure is a haematological malignancy.

05

What Cytokines Do — Differentiation, Survival, Death

Part 4 was the grammar; this is the vocabulary in use. Cytokines decide **what a lymphocyte becomes, which antibody class it makes, where lymphoid tissue is built, how long the cell lives and when it dies**. Five jobs, and the last two are the ones people forget: a cytokine that kills is still a cytokine.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- For each helper T subset, give the driving cytokine, the master transcription factor and the effector cytokines.
- Say which cytokine drives which antibody class, and why that matters in allergy.
- Distinguish IL-2 from IL-7 by when in a response each is needed.
- Explain why a Fas mutation causes lymphadenopathy and autoimmunity rather than immunodeficiency.

5.1 Steering T-Cell Differentiation

This is signal 3 from Part 2, in detail. A naïve CD4 T cell is a blank. What it becomes is decided by **which cytokine is in the room at the moment it is activated** — and that cytokine came from the innate immune cell that judged the threat. Each subset follows the same three-step recipe: **a driving cytokine → a master transcription factor → a set of effector cytokines**.

TABLE 5.1 THE HELPER T SUBSETS — THE TABLE TO KNOW COLD

Subset	Driven by S3	Master TF	Makes	Job — and the disease of too much
Th1	IL-12 (+ IFN- γ , IL-18)	T-bet <i>via STAT4</i>	IFN-γ , IL-2, TNF- β (LT- α)	Intracellular pathogens — activates macrophages to kill what they have swallowed. Excess → organ-specific autoimmunity, granuloma
Th2	IL-4	GATA-3 <i>via STAT6</i>	IL-4, IL-5, IL-13, IL-10	Helminths — eosinophils, mast cells, IgE. Excess → allergy and asthma
Th9	IL-4 + TGF-β	PU.1	IL-9	Mucus production , tissue inflammation; contributes to allergic airway disease
Th17	TGF-β, IL-6, IL-21, IL-23	RORγt	IL-17A, IL-17F , IL-6, IL-8, IL-21, IL-22, IL-26	Extracellular bacteria and fungi — recruits neutrophils. Excess → psoriasis, ankylosing spondylitis, IBD
Th22	TNF-α, IL-6	AHR	IL-22	Tissue inflammation and epithelial barrier repair; implicated in psoriasis and atopic dermatitis

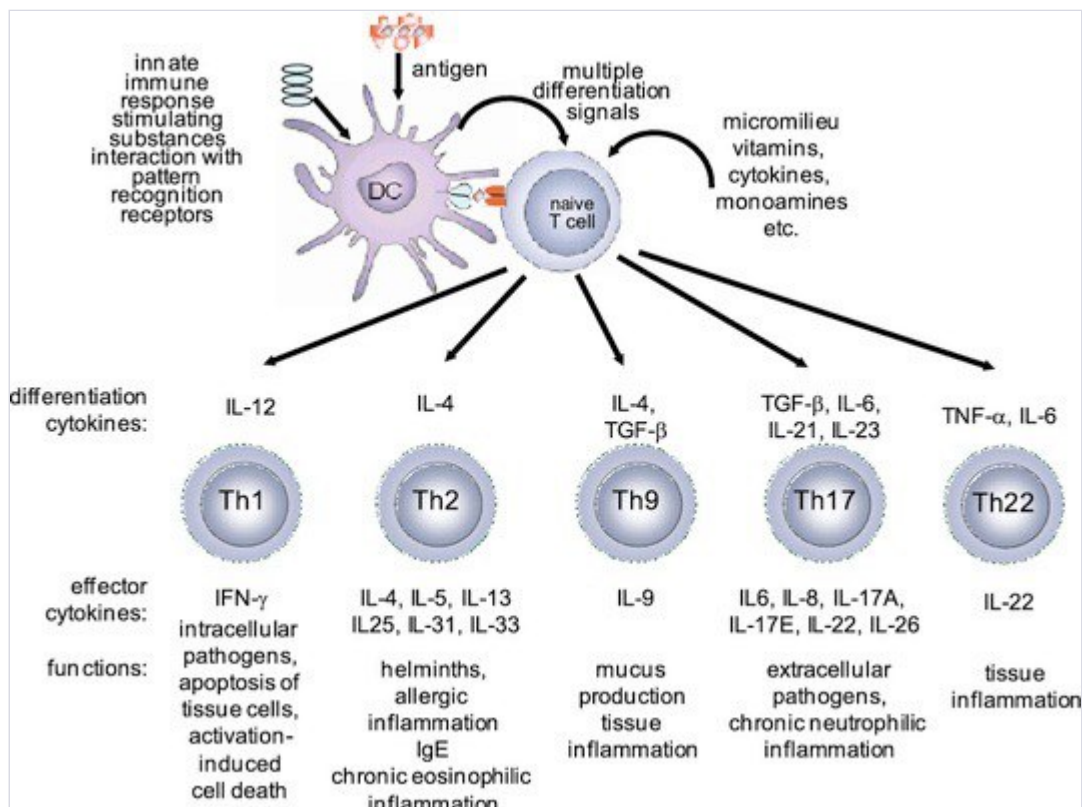


Fig 5.1 The lecturer's diagram, and it is laid out exactly as Table 5.1. A dendritic cell presents antigen to a naïve T cell; the **differentiation cytokines** across the top decide the lineage; the **effector cytokines** below each cell are what it then makes; the **functions** along the bottom are the jobs. Read down a column and you have one subset; read across a row and you have one kind of fact.

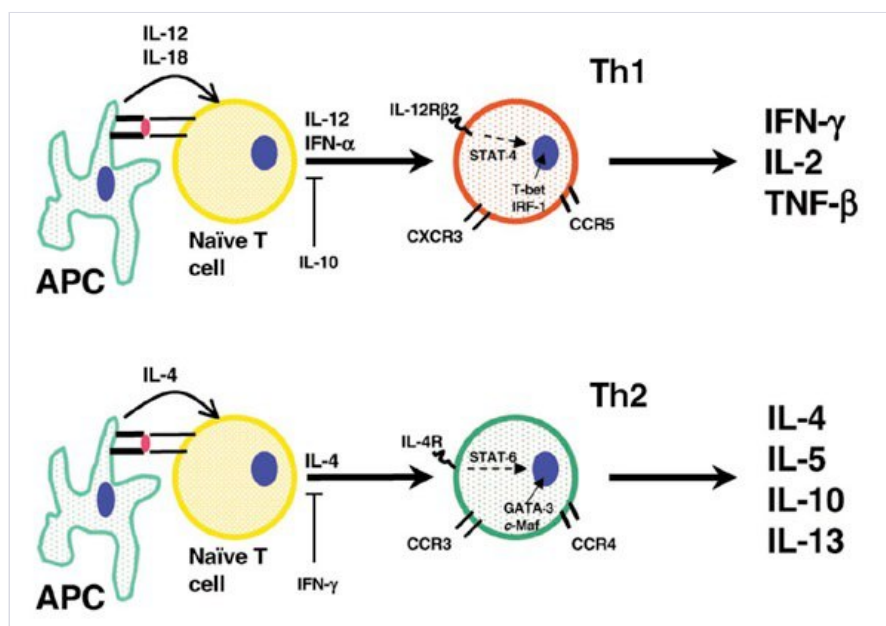


Fig 5.2 The two classic arms in full. **Top:** IL-12 and IL-18 from the APC $\rightarrow$ STAT4 $\rightarrow$ T-bet $\rightarrow$ a Th1 cell making IFN- γ , IL-2 and TNF- β , and expressing the chemokine receptors CXCR3 and CCR5. **Bottom:** IL-4 $\rightarrow$ STAT6 $\rightarrow$ GATA-3 and c-Maf $\rightarrow$ a Th2 cell making IL-4, IL-5, IL-10 and IL-13, expressing CCR3 and CCR4. Note the cross-arrows: IL-10 suppresses the Th1 arm and IFN- γ suppresses the Th2 arm.

REMEMBER IT AS

One-two, twelve-four. **Th1** is driven by **IL-12**; **Th2** is driven by **IL-4**. Then the transcription factors in alphabetical-ish order: **T-bet** for **Th1**, **GATA-3** for **Th2**, **ROR γ t** for **Th17**. And the outputs: **Th1** → **IFN- γ** (macrophages), **Th2** → **IL-4/5/13** (eosinophils and IgE), **Th17** → **IL-17** (neutrophils). Three subsets, three cell types recruited.

5.2 Controlling Isotype Switching

You learned in Session 2 that a B cell can change its heavy-chain constant region while keeping its specificity. What you were not told is **who decides which class it switches to**. The answer is the cytokine environment, and therefore the helper T subset of §5.1.

Role of cytokines in regulating expression of antibody classes							
Cytokines	IgM	IgG3	IgG1	IgG2b	IgG2a	IgE	IgA
IL-4	Inhibits	Inhibits	Induces		Inhibits	Induces	
IL-5							Augments production
IFN- γ	Inhibits	Induces	Inhibits		Induces	Inhibits	
TGF- β	Inhibits	Inhibits		Induces			Induces

Figure 9-13 Immunobiology, 7ed. (© Garland Science 2008)

Fig 5.3 The lecturer's table. Read the **rows**, not the cells: **IL-4** induces IgG1 and IgE while inhibiting IgM, IgG3 and IgG2a; **IFN- γ** does the mirror image, inducing IgG3 and IgG2a while inhibiting IgE; **IL-5** augments IgA production; **TGF- β** induces IgG2b and IgA. The Th2 cytokine makes the allergy antibody and the Th1 cytokine blocks it — the antagonism of §4.2 written out in antibody classes.

EXAM TRAP — IGG2A AND IGG2B ARE MOUSE SUBCLASSES

The table lists **IgG1**, **IgG2a**, **IgG2b** and **IgG3**. Those are the **mouse** IgG subclasses, because the switching experiments were done in mice. **Human IgG subclasses are IgG1, IgG2, IgG3 and IgG4** — there is no human IgG2a. The **biology transfers** and is what you should carry away: IL-4 → IgE, TGF- β → IgA, IFN- γ → opsonising and complement-fixing IgG subclasses. The **subclass numbers do not transfer**. In a human question, the human answer is: IL-4 and IL-13 drive IgE; TGF- β and IL-5 drive IgA; IFN- γ drives IgG1 and IgG3.

CLINICAL CORRELATION — BLOCKING THE IL-4 PATHWAY IN ATOPIC DISEASE

If IL-4 and IL-13 drive IgE, then blocking them should collapse the allergic response. **Dupilumab** is a monoclonal antibody against **IL-4R α** , the chain shared by the IL-4 and IL-13 receptors — a deliberate use of the shared-subunit principle from §4.4 to block two cytokines with one drug. It is now standard therapy for severe atopic dermatitis, asthma and chronic rhinosinusitis with nasal polyps. Read the name: **-umab**, fully human.

5.3 Building the Secondary Lymphoid Organs

Cytokines do not only run responses; they **build the buildings the responses happen in**. The TNF family members **lymphotoxin (LT- α , LT- β)** and **TNF** are required for the normal architecture of spleen, lymph nodes and Peyer's patches, and **LT in particular is a strong inducer of the chemokines that sort T cells and B cells into their separate zones**.

Receptor	Ligands	Effects seen in knockout (KO) mice				
		Spleen	Peripheral lymph node	Mesenteric lymph node	Peyer's patch	Follicular dendritic cells
TNFR-I	TNF- α LT- α_3	Distorted architecture	Present in TNF- α KO Absent in LT- α KO owing to lack of LT- β signals	Present	Reduced	Absent
LT- β receptor	TNF- α LT- α_2/β_1 LIGHT	Distorted No marginal zones	Absent	Present in LT- β KO Absent in LT- β receptor KO	Absent	Absent
HVEM	LT- α_3 LIGHT	Although both LT- α and LIGHT can bind HVEM, there is no known role for HVEM signaling in organogenesis				

Figure 7-38 Immunobiology, 6/e. (© Garland Science 2005)

Fig 5.4 Knockout evidence. Deleting the **TNF receptor** distorts splenic architecture and abolishes follicular dendritic cells; deleting **LT- β receptor** signalling additionally abolishes peripheral and mesenteric lymph nodes and Peyer's patches. **The animal has no lymph nodes at all**. This is the strongest possible statement that a cytokine can be a developmental morphogen, not merely a messenger.

5.4 Survival and Clonal Expansion

An immune response has a shape: a small number of cells expand enormously, most of them then die, and a few remain as memory. Cytokines control all three phases — and the experiment that proves it is one of the cleanest in immunology.

○ MECHANISM — THE SUPERANTIGEN EXPERIMENT AND THE “IL-2 PUMP”

- 1 Inject a **superantigen**, which activates every T cell carrying a particular **V β** gene segment regardless of its actual specificity.
WHY: This gives you a large, trackable population — you can count the V β -positive cells by flow cytometry day by day, which you could never do with a normal antigen-specific clone.
- 2 Those T cells undergo an **initial clonal expansion** over two to three days.
- 3 Expansion is followed by **contraction**: the population collapses back down.
WHY: This is the default. The cells were kept alive by a growth factor, and when the antigen is cleared the growth factor stops.
- 4 An **“IL-2 pump”** implanted in the animal rescues the contraction — the cells do not die.
*WHY: This is the decisive result. Adding back one cytokine prevents the death, which proves the death was caused by its **absence**, not by anything the antigen did. The deaths are therefore called **passive cell death** — death by neglect, or cytokine deprivation.*
- 5 Passive cell death is **regulated by the Bcl-2 family** — the intrinsic, mitochondrial apoptosis pathway.

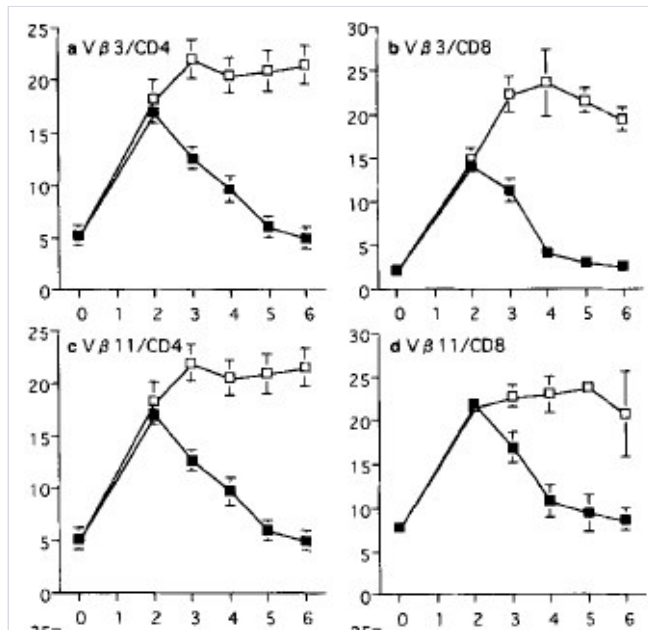


Fig 5.5 The data. Four panels — Vβ3 and Vβ11, CD4 and CD8. In each, the filled symbols fall away after day 2–3 (contraction) while the open symbols stay high: **the open symbols are the animals given IL-2.**

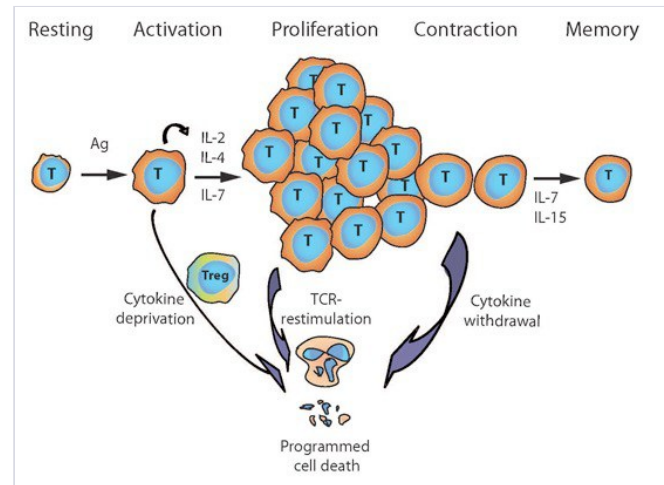


Fig 5.6 The five phases with their cytokines. **Activation and proliferation:** IL-2, IL-4, IL-7. **Contraction:** cytokine deprivation → programmed cell death, with a Treg helping to enforce it. **Memory:** IL-7 and IL-15, which keep the surviving cells alive for years without any antigen.

IL-2 VERSUS IL-7 — THE DISTINCTION THE LECTURE DRAWS IN ONE LINE

IL-7: survival during differentiation. IL-2: survival during proliferation. The experiment behind it: mice infected with LCMV generate a primary CD8 response in which **some effector cells express high levels of IL-7R and others do not**. Transfer only the **IL-7R^{hi}** cells into naïve mice and challenge, and you get robust expansion of antigen-specific CD8 cells; transfer the IL-7R^{lo} cells and you get almost nothing. **IL-7R expression marks the cells that will become memory.**

Two clinical consequences follow. **IL-7Rα (CD127) deficiency causes a T⁻ B⁺ NK⁺ SCID** — the T cells alone are missing, because only they depend absolutely on IL-7. And **IL-7 is being trialled as an adjuvant** to improve vaccine-induced memory, on exactly this reasoning.

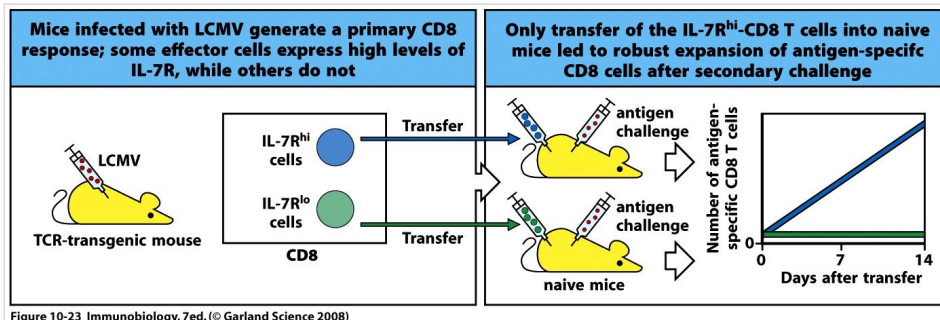


Figure 10-23 Immunobiology, 7ed. (© Garland Science 2008)

Fig 5.7 The transfer experiment. LCMV-infected TCR-transgenic mice make CD8 effectors, some IL-7R^{hi} (blue) and some IL-7R^{lo} (green). Transferred into naïve mice and challenged, **only the IL-7R^{hi} cells expand** — the blue line rises and the green line stays flat.

5.5 Death Cytokines: Fas, TNF and TRAIL

Section 5.4 was death by **neglect**. This is death by **instruction**. A subgroup of the TNF superfamily exists to tell a cell to kill itself, and the lecture names three ligands: **FasL**, **TNF** and **TRAIL** (TNF-related apoptosis-inducing ligand).

TABLE 5.2 TWO WAYS A LYMPHOCYTE DIES

	Passive — death by neglect	Active — death by instruction
Trigger	Cytokine withdrawal when antigen is cleared	FasL engaging Fas on a repeatedly restimulated T cell
Pathway	Intrinsic — mitochondrial, regulated by the Bcl-2 family	Extrinsic — death receptor, FADD, cas-pase-8
When it dominates	Low antigen load — the ordinary end of a resolving response	High antigen load — persistent or repeated stimulation
Failure causes	Lymphocyte accumulation	ALPS in humans; lpr and gld lupus-like disease in mice

— ABSENCE PROVES FUNCTION — ALPS, AND THE LPR AND GLD MICE

Mutations in **Fas** (the **lpr** mouse) or in **Fas ligand** (the **gld** mouse) cause the same **lupus-like autoimmune disease** — which tells you at once that the receptor and its ligand are in one pathway, because breaking either end gives the identical phenotype.

In humans the same defect is **ALPS** — **autoimmune lymphoproliferative syndrome**. The child in Fig 5.8 has the classic presentation: **massive, chronic, non-malignant lymphadenopathy with splenomegaly**, plus autoimmune cytopenias. The laboratory hallmark is an expanded population of **double-negative T cells** (CD3⁺ but CD4⁻CD8⁻) — cells that should have been deleted and were not.

Read what this proves. Losing a death pathway does not cause immunodeficiency — it causes **too much immune system**. Lymphocytes that cannot be deleted accumulate, and among them are the self-reactive ones that should have been removed. **Autoimmunity here is a failure of subtraction, not of selection.**

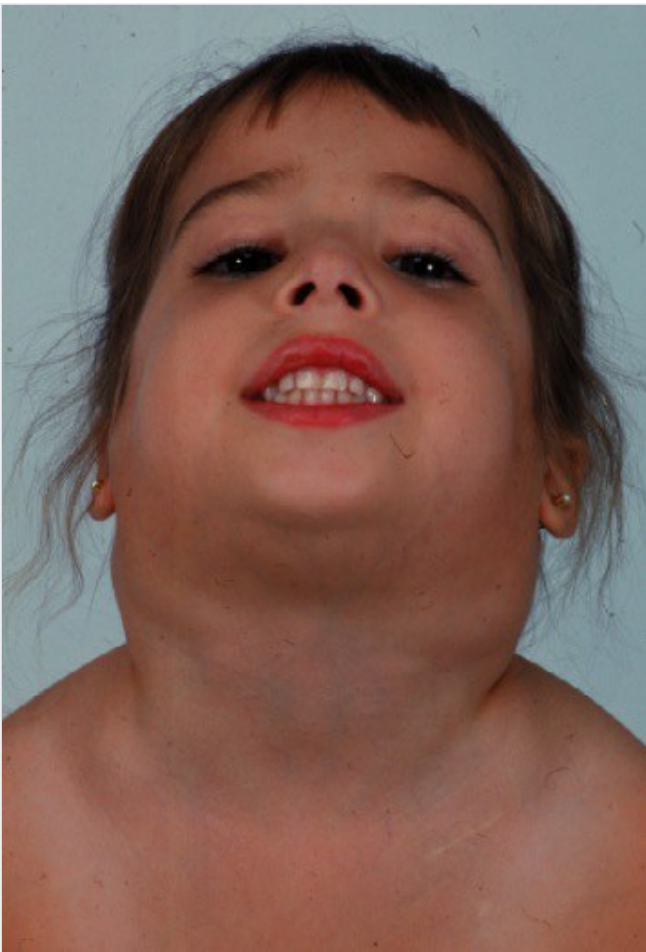


Fig 5.8 A patient with **ALPS** and a Fas mutation, showing gross cervical lymphadenopathy. Nothing here is malignant — these are normal lymphocytes that failed to die.

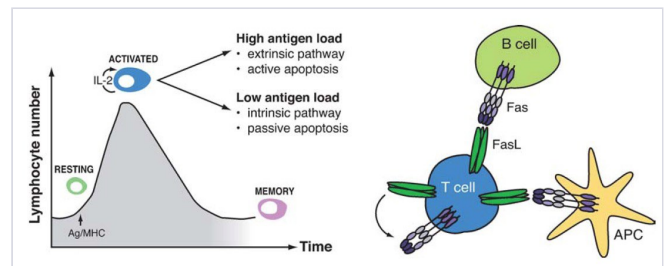


Fig 5.9 Left: the shape of a response — resting, activated, memory. Right: a T cell carrying **FasL** engaging **Fas** on a B cell, on another T cell or on an APC. **High antigen load → extrinsic, active pathway.** **Low antigen load → intrinsic, passive apoptosis.** Same endpoint, two routes chosen by how much antigen is left.

5.6 Cytokines as Drugs, and Viruses as Thieves

TABLE 5.3 CYTOKINES GIVEN AS MEDICINES

Cytokine	Trade names	Used for
Interferon-α	Infergen, Roferon, Intron-A, Peg-Intron	Antiviral — chronic hepatitis B and C (now largely superseded for hepatitis C by direct-acting antivirals), and some malignancies
G-CSF	Neupogen, filgrastim	Neutropenia — after chemotherapy, and to mobilise stem cells for harvest
Erythropoietin-α	Epoetin-alfa, Epogen	Anaemia — of chronic kidney disease and of chemotherapy
IL-2	Proleukin, aldesleukin	Antiviral or antitumour — high-dose IL-2 in metastatic renal cell carcinoma and melanoma; effective in a minority, and severely toxic (capillary leak syndrome)

Blocking a cytokine: anti-TNF

Marc Feldmann and Ravinder Maini proposed that **TNF sits at the top of a cascade of cytokines in rheumatoid arthritis** — that it is not one mediator among many but the one that drives the others. The prediction was testable: block TNF alone and the whole cascade should quieten. It did.

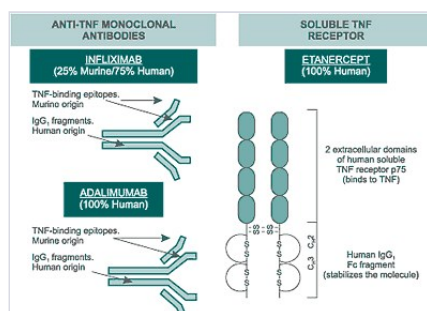


Fig 5.10 Two molecular designs. **Left — monoclonal antibodies:** infliximab (25% murine, 75% human — a chimera) and adalimumab (100% human). **Right — a soluble receptor decoy:** etanercept, two extracellular domains of human TNF receptor p75 fused to a human IgG₁ Fc, which stabilises the molecule and extends its half-life.



Fig 5.11 Plaque psoriasis before and during etanercept. The same drug is licensed for rheumatoid arthritis, psoriasis, psoriatic arthritis, ankylosing spondylitis and Crohn's disease — one cytokine, many diseases.

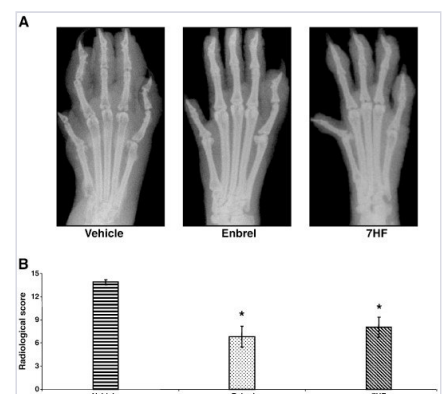


Fig 5.12 The harder endpoint: hand radiographs and radiological scores. Anti-TNF does not merely relieve symptoms — it **slows structural joint destruction**, which is why it changed the natural history of rheumatoid arthritis rather than just its comfort.

CLINICAL CORRELATION — WHAT YOU MUST CHECK BEFORE AN ANTI-TNF

TNF is required to **maintain granulomas**. Block it and a walled-off, dormant tuberculous focus can break open. **Screening for latent tuberculosis before starting anti-TNF therapy is mandatory**, and reactivation of TB is the signature adverse effect of the class. Hepatitis B reactivation and an increased risk of other intracellular infections follow the same logic. Once again the side-effect profile confirms the physiology: TNF's normal job is exactly what its blockade removes.

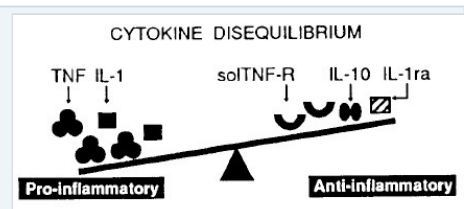


Fig 5.13 Cytokine disequilibrium. On one side TNF and IL-1; on the other the body's own inhibitors — soluble TNF receptor, IL-10 and IL-1ra. Inflammatory disease is the seesaw tipping. Every anti-cytokine drug is an attempt to add weight to the right-hand pan.

Viruses steal the whole system

TABLE 5.4 WHAT A POXVIRUS GENOME SPENDS ITS GENES ON

Class of stolen gene	Examples named on the slide	What it achieves
Soluble decoy cytokine receptors	vTNFR — CrmB, CrmC, CrmD, Crme · vIL-1βR · vIFN-γR · vIFN-α/βBP · vIL-18BP · vGM-CSF/IL-2BP	Mops the cytokine up before it reaches a real receptor. Exactly the design of etanercept — a soluble receptor used as a sponge
Viral cytokines	vIL-10 · vEGF · vVEGF · vSEMA · vCK (viral chemokine)	Actively suppresses the host response. vIL-10 is a working copy of our own anti-inflammatory cytokine, used against us
Chemokine-binding proteins	vCKBP1 · vCKBP2	Stops leukocytes being recruited — no chemokine gradient, no infiltrate
Membrane chemokine-receptor homologues	vCKR · vCD30 · membrane vTNFR	Diverts cell traffic , and in some cases lets the infected cell read host signals for its own benefit
Scale: a poxvirus genome is 130–300 kbp with about 200 open reading frames. A large fraction of those genes is spent on the list above. Herpesviruses do the same		

NOTE THE SYMMETRY WITH OUR OWN DRUGS

Etanercept is a soluble TNF receptor fused to an Fc. Crme is a soluble viral TNF receptor. They are the same invention. Poxviruses arrived at decoy-receptor therapy several million years before Feldmann and Maini did, and for the opposite purpose. When you meet a viral immune-evasion protein, ask what it is a copy of — the answer tells you which host molecule mattered enough to be worth stealing.

06

Innate Immunity — Triggers and Sensors

Adaptive immunity takes a week. You would be dead in a week. Everything that keeps you alive in the meantime is innate — and it is not, as it is often taught, a crude preliminary. It is the system that **decides whether the adaptive response happens at all**, because it supplies signal 2 and signal 3. Part 2 told you a T cell needs permission; this Part is about who grants it.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Give four differences between innate and adaptive immunity — without saying “fast” and “slow” only.
- List the five steps by which an inflammatory response is triggered, and match each cardinal sign to a mediator.
- Define PAMP, DAMP and PRR, and give three examples of each.
- Match at least six microbial ligands to their TLRs, and say which TLRs sit where.
- Draw the chain from TLR to NF- κ B and name the adaptor that all but one TLR uses.

6.1 Innate versus Adaptive

TABLE 6.1 THE TWO SYSTEMS COMPARED

	Innate	Adaptive
Timing	Early — minutes to hours	Late — days to a week
What it recognises	Not antigen-specific. It recognises general signals generated by infection — molecular patterns shared by whole classes of microbes	Antigen-specific — the T-cell receptor and the immunoglobulins of Session 2
Receptor repertoire	Germline-encoded , fixed, a few hundred receptors — you are born with the complete set	Somatically generated by V(D)J recombination, $\sim 10^{11}$ specificities
Memory	Classically none (though “trained immunity” modifies this)	Yes — the basis of vaccination
Role in the other system	Innate immunity provides the third signal — the inflammatory cytokine — that steers adaptive immunity , and the B7 that permits it. The relationship is not sequential; it is instructive	



Fig 6.1 Barrier. Skin and mucosa — a wall that keeps almost everything out and costs nothing to maintain.



Fig 6.2 Sentinels. Macrophages and dendritic cells patrolling the tissue behind the wall, ready within minutes.

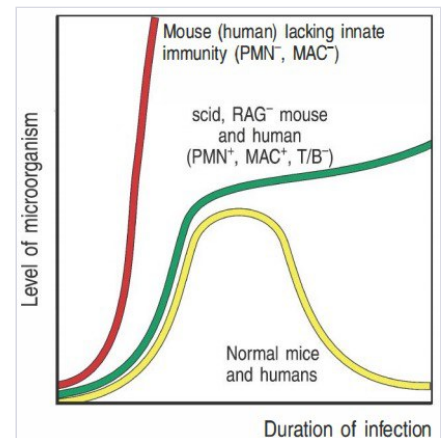


Fig 6.3 Why both are needed. Normal host (yellow) clears the organism. A host lacking **adaptive** immunity (SCID/RAG⁻, green) controls it for a while and then cannot finish. A host lacking **innate** immunity (no neutrophils or macrophages, red) is overwhelmed almost immediately.

6.2 What Inflammation Is, and How It Starts

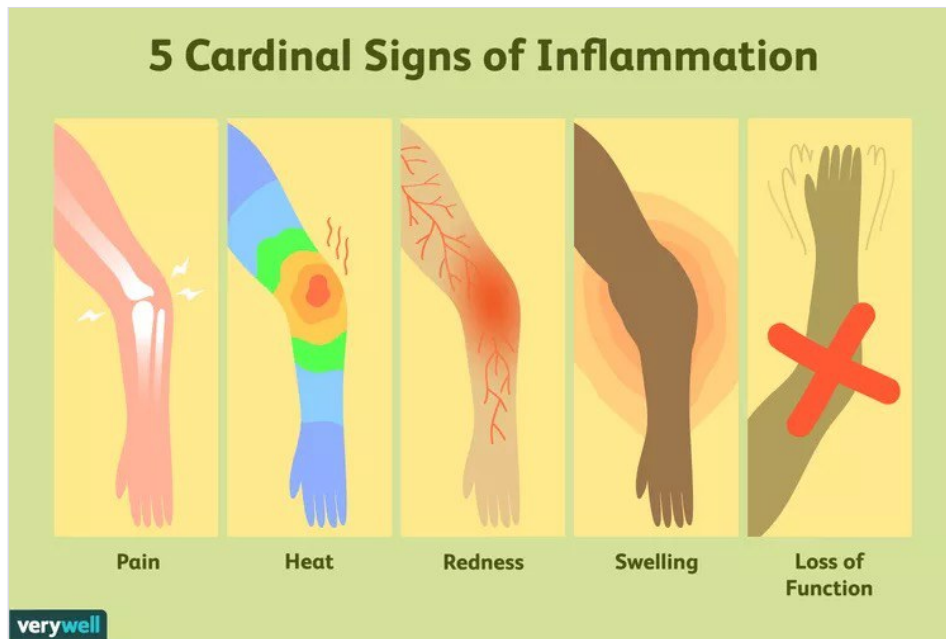


Fig 6.4 The five cardinal signs, named by Celsus in the first century and Galen in the second: **dolor** (pain), **calor** (heat), **rubor** (redness), **tumor** (swelling) and **functio laesa** (loss of function). Every one of them is now assignable to a molecule, and that is the point of this Part.

○ MECHANISM — THE LECTURER'S FIVE STEPS, WITH THE SIGN EACH ONE PRODUCES

- 1 **Breach of homeostasis** — epithelial barrier damage, pathogen infection, trauma, ischaemia.
WHY: Note the wording. It is not “infection”; it is a breach of homeostasis. That single choice of word is what allows the same pathway to explain gout, atherosclerosis and infarction as well as sepsis.
- 2 **Innate effectors sense the breach** — macrophages and dendritic cells detect **DAMPs** released by damaged host cells or **PAMPs** carried by the pathogen.
- 3 **Secretion of cytokines and chemokines** by those innate effector cells.
WHY: This is the amplification step. One macrophage that has seen one bacterium can now recruit thousands of cells. Everything after this point is done by cells that never saw the pathogen.
- 4 **Cytokines act on the vessels and the brain** — TNF causes **vasodilation** and increased permeability, giving **redness, heat and swelling**; IL-1 and IL-6 act on the hypothalamus to give **fever**. Prostaglandins and bradykinin sensitise nociceptors: **pain**.
- 5 **Chemokines recruit other immune cells** to the affected tissue — neutrophils first, then monocytes. Swelling plus pain plus infiltrate gives **loss of function**.

INFLAMMATION DOES NOT REQUIRE A MICROBE

The same five steps run when there is no pathogen at all. **Sterile inflammation** is triggered by non-infectious cell injury — infarction, crush injury, chemical burn, crystal deposition — because dying host cells release **DAMPs**, and a macrophage cannot tell a DAMP-driven emergency from a PAMP-driven one. It responds to both with gain of function: **phagocytosis and cytokine secretion**.

This is why the inflammasome, which you will meet in Part 7, is central to **gout** (uric acid crystals), **atherosclerosis** (cholesterol crystals) and **type 2 diabetes** (islet amyloid polypeptide) as well as to infection. **Sterile inflammation is where most of modern chronic disease lives.**

6.3 PAMPs and DAMPs — Non-Self and Danger

NAME DECODER — THREE ACRONYMS THAT CARRY THE WHOLE OF INNATE RECOGNITION

PAMP	Pathogen-associated molecular pattern	<i>A molecule made by microbes and not by us, essential to the microbe so it cannot be discarded: LPS, peptidoglycan, flagellin, lipoteichoic acid, unmethylated CpG DNA, double-stranded RNA. This is the “non-self” signal</i>
DAMP	Damage- (or danger-) associated molecular pattern	<i>A host molecule that is normally hidden inside a healthy cell and appears outside only when a cell dies badly: ATP, uric acid, HMGB1, mitochondrial DNA, heat-shock proteins, extracellular matrix fragments. This is the “danger” signal</i>
PRR	Pattern-recognition receptor	<i>The sensor that reads them. Germline-encoded — unlike a TCR, it is not made fresh in each cell. Found on the cell surface, in endosomes and free in the cytosol</i>

Why the two words exist. The old model was **self versus non-self** — the immune system attacks what is foreign. That model cannot explain why a sterile infarct causes violent inflammation, or why a transplanted kidney is attacked while a fetus is not. Polly Matzinger's **danger model** replaced it: the immune system responds to **damage**, whoever caused it. PAMPs and DAMPs are the two vocabularies of that single question — “**is something going badly wrong here?**” — and both are read by the same receptors and produce the same cytokines.

THREE PROPERTIES A GOOD PAMP MUST HAVE

- **Made by microbes, not by us** — otherwise the receptor would attack the host.
- **Essential to the microbe** — LPS is structural, flagellin is needed for motility. A pathogen cannot mutate them away without crippling itself. **This is how a fixed, germline receptor keeps up with a rapidly mutating pathogen.**
- **Shared by whole classes of microbes** — one receptor for all Gram-negatives, one for all flagellated bacteria. A few hundred receptors cover the microbial world.

6.4 Toll-like Receptors

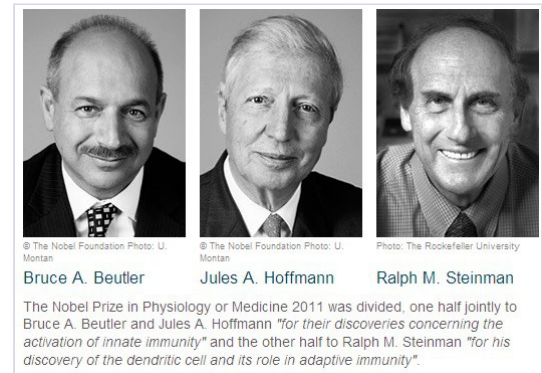


Fig 6.5 Nobel Prize 2011. One half jointly to **Bruce A. Beutler** and **Jules A. Hoffmann** "for their discoveries concerning the activation of innate immunity" — Hoffmann found Toll in *Drosophila*, Beutler found that the mouse LPS receptor was TLR4. The other half to **Ralph M. Steinman** "for his discovery of the dendritic cell and its role in adaptive immunity."

TABLE 6.2 THE LECTURER'S TLR TABLE — THE LIGAND, THE ORGANISM, THE RECEPTOR

Microbial component	Found in	TLR used
LPS (endotoxin)	Gram-negative bacteria	TLR4 (with MD-2 and CD14)
Diacyl lipopeptides	<i>Mycoplasma</i>	TLR6/TLR2 heterodimer
Triacyl lipopeptides	Bacteria and mycobacteria	TLR1/TLR2 heterodimer
Lipoteichoic acid	Group B <i>Streptococcus</i>	TLR6/TLR2
Peptidoglycan	Gram-positive bacteria	TLR2
Porins	<i>Neisseria</i>	TLR2
Lipoarabinomannan	Mycobacteria	TLR2
Flagellin	Flagellated bacteria	TLR5
CpG DNA (unmethylated)	Bacteria and mycobacteria	TLR9 — endosomal
Not yet defined	Uropathogenic bacteria	TLR11 — functional in mouse; a pseudogene in humans
Double-stranded RNA	Viruses	TLR3 — endosomal, and the only TLR that does not use MyD88
Single-stranded RNA	Viruses	TLR7 (and TLR8) — endosomal

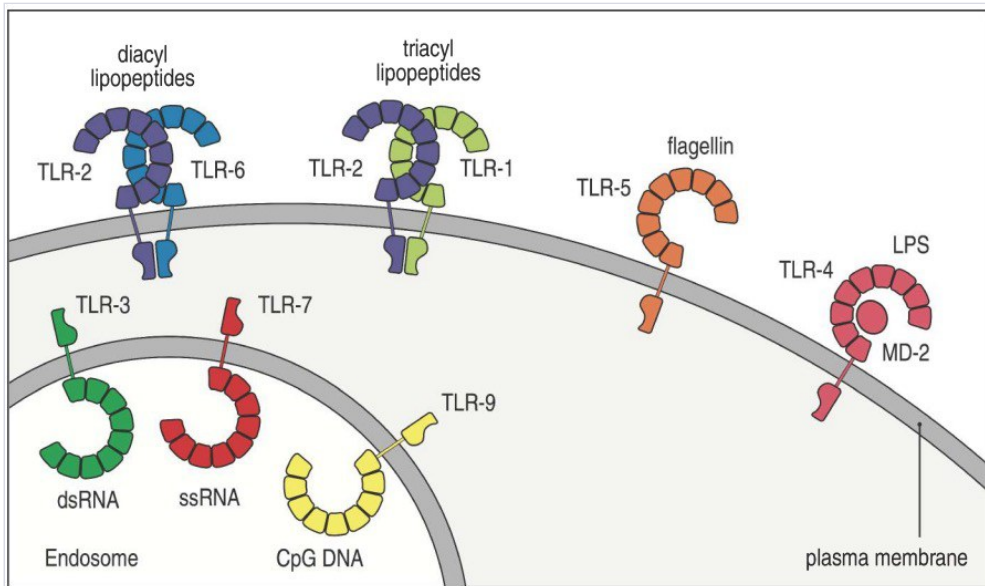


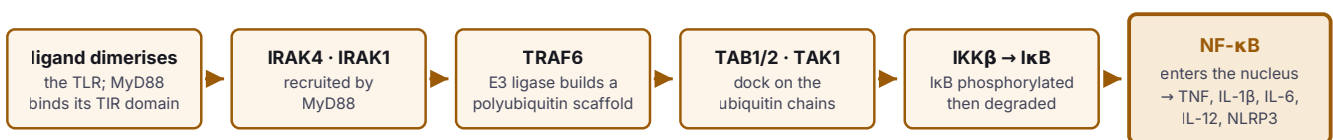
Fig 6.6 Where each TLR sits, and why. At the **plasma membrane**: TLR1/2, TLR2/6, TLR4 (with MD-2) and TLR5 — all reading **surface molecules of intact microbes**, which are outside the cell. In the **endosome**: TLR3 (dsRNA), TLR7 (ssRNA) and TLR9 (CpG DNA) — all reading **nucleic acids**, which only appear once a microbe has been swallowed and broken open.

THE LOCATION IS THE SAFETY MECHANISM

Nucleic acids are not foreign. Your own DNA and RNA are chemically almost identical to a microbe's, so a receptor for DNA cannot rely on chemistry alone to avoid attacking you. **It relies on geography.** Nucleic-acid sensors are confined to the **endosome**, a compartment your own DNA never enters — so anything found there arrived inside something that was swallowed.

When that geography fails, you get autoimmunity. In **systemic lupus erythematosus**, immune complexes containing self-DNA are taken into the endosome of a plasmacytoid dendritic cell, where TLR7 and TLR9 read them as viral — and the resulting **type I interferon signature** is one of the defining laboratory features of the disease. This is also why **hydroxychloroquine**, which raises endosomal pH and disables these TLRs, is a mainstay of lupus treatment.

6.5 From TLR to NF- κ B



The last box is **Signal 1 of the inflammasome (§7.2)** — hold on to it
 NF- κ B does not release IL-1 β ; it only writes the pro-form. Something else must cut it

Fig 6.7 The TLR chain, drawn on the same template as Fig 2.5 and Fig 4.9. Note the unusual mechanism in the middle: **TRAF6 signals by building a polyubiquitin chain that acts as a scaffold**, not by phosphorylating anything. Ubiquitin here is a signal, not a death sentence.

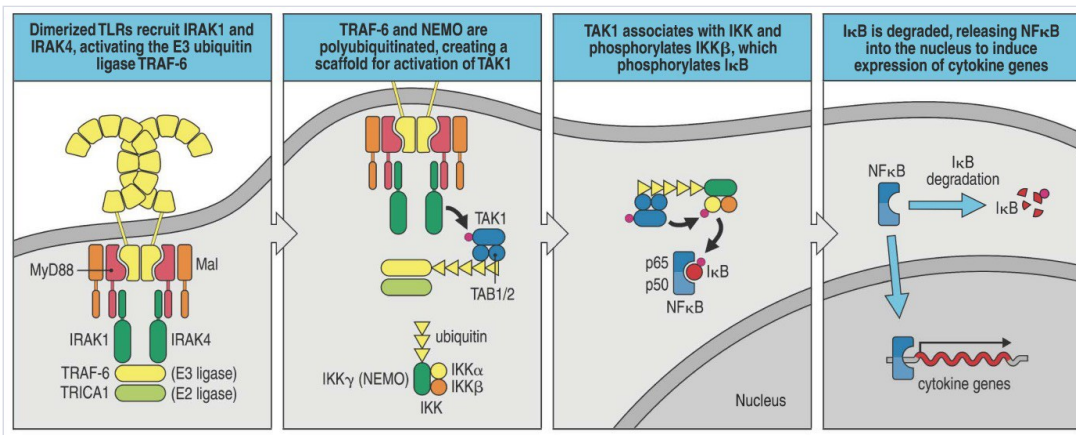


Fig 6.8 The same four steps as the textbook draws them, panel by panel, ending with IκB degraded and the p50/p65 NF-κB dimer walking into the nucleus to switch on the cytokine genes.

— ABSENCE PROVES FUNCTION — THE MYD88 KNOCKOUT MOUSE

MyD88 is the adaptor that binds the **TIR domain** of the TLR and recruits IRAK-4, IRAK-1 and TRAF-6. It is used by **every TLR except TLR3**, and the knockout mouse shows what that means:

- **MyD88^{-/-} mice have no response to LPS at all** — the single cleanest demonstration that this adaptor is the bottleneck.
- The lecturer's stronger claim: **MyD88 is essential to all inflammatory signalling pathways** — it is also the adaptor for the IL-1 receptor family, whose cytoplasmic domain is a TIR domain too.
- A splice variant, **MyD88s**, lacks part of the protein and acts as a natural **down-regulator** of the response — the same self-limiting design as SOCS and CTLA-4.

The surprising result. Rakoff-Nahoum and colleagues (*Cell* 2004) showed that MyD88^{-/-} mice are not merely infection-prone — their **intestines bleed after injury and fail to repair**. The conclusion in the paper's title is the point: *recognition of commensal microflora by Toll-like receptors is required for intestinal homeostasis*. **TLR signalling is not only a danger alarm; it is a continuous, low-level conversation with your own microbiota that maintains the gut epithelium.** Switching off inflammation entirely is not health.

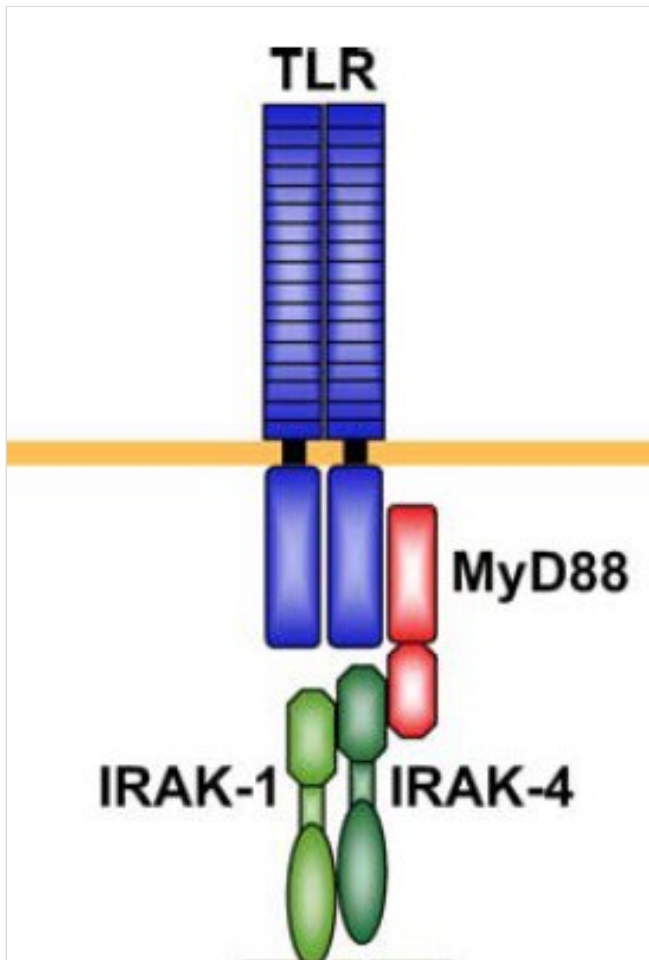


Fig 6.9 The receptor-proximal complex: the **TLR** in the membrane, **MyD88** docked on its TIR domain, and **IRAK-4** and **IRAK-1** assembled beneath — the structure sometimes called the Myddosome.

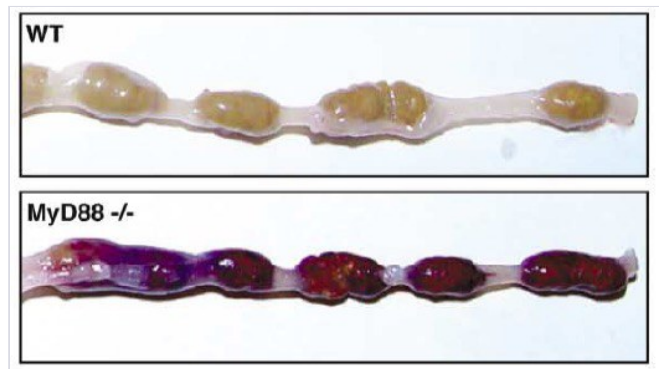


Fig 6.10 The experiment. Above, wild-type intestine after injury; below, **MyD88^{-/-}** intestine — dark with blood. Without TLR signalling the epithelium cannot repair itself.

6.6 Sensing the Cytosol: RIG-I-like Receptors

The lecturer sets up this section with a precise problem. **TLR3, TLR7 and TLR9 can detect viral RNA and DNA — but they sit in the endosome**, so they interact primarily with material that has entered by endocytosis. They are blind to what is happening in the **cytoplasm of an already-infected cell**, which is precisely where a replicating virus lives.

The solution is a second family of sensors, free in the cytosol: the **RIG-I-like receptors (RLRs, also called RLHs — RIG-I-like helicases)**. Named after **retinoic acid-inducible gene I**, they carry an **RNA helicase-like domain** that binds viral RNA directly in the cytoplasm.

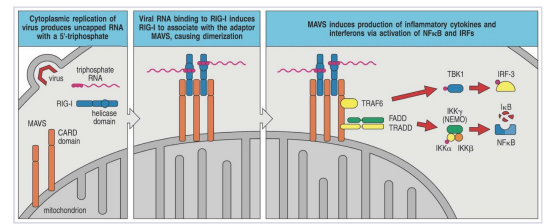


Fig 6.11 RIG-I binds viral RNA bearing an uncapped 5'-triphosphate — a chemical signature our own cytoplasmic mRNA does not carry. It then associates with the adaptor MAVS on the mitochondrial outer membrane, which nucleates two arms: **TBK1 → IRF3 → type I interferon**, and **IKK → NF-κB → inflammatory cytokines**.

REMEMBER IT AS

Three addresses, three sensor families. Outside the cell → surface TLRs. Inside a vesicle → endosomal TLRs. Free in the cytosol → RLRs for RNA, cGAS–STING for DNA, and NLRs for everything else (Part 7). If a question asks which sensor detects a given pathogen, first ask where in the cell that pathogen is.

6.7 The Effectors: Cytokines of Innate Immunity

The sensors have fired. What comes out? **Cytokines** — and the same molecules you studied in Parts 4 and 5, now seen from the innate end.

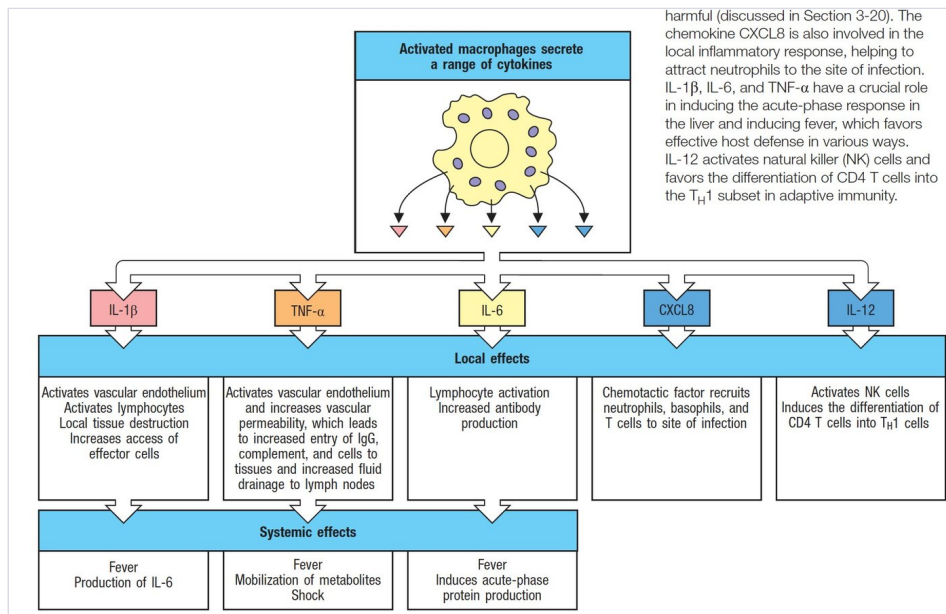


Fig 6.12 What an activated macrophage secretes, with **local effects** above and **systemic effects** below. **IL-1β** and **TNF-α** activate vascular endothelium and raise permeability; **IL-6** drives the acute-phase response and antibody production; **CXCL8** is the chemotactic factor that brings in neutrophils, basophils and T cells; **IL-12** activates NK cells and pushes CD4 cells towards Th1. That last box is **signal 3** — the innate cell choosing the adaptive response.

Homodimeric receptors		Receptors for erythropoietin and growth hormone
Heterodimeric receptors with a common chain		Receptors for IL-3, IL-5, GM-CSF share a common chain, CD131 or β_c (common β chain)
		Receptors for IL-2, IL-4, IL-7, IL-9, and IL-15 share a common chain, CD132 or γ_c (common γ chain). IL-2 receptor also has a third chain, a high-affinity subunit IL-2R α (CD25)
Heterodimeric receptors (no common chain)		IL-1 family receptors
		Receptors for IL-13, IFN- α , IFN- β , IFN- γ , IL-10
TNF receptor family		Tumor necrosis factor (TNF) receptors I and II, CD40, Fas (Apo1, CD95), CD30, CD27, nerve growth factor receptor
Chemokine receptor family		CCR1-10, CXCR1-5, XCR1, CX3CR1

Fig 6.13 Cytokines are classified by **receptor structure**. **Homodimeric** receptors (erythropoietin, growth hormone); **heterodimeric with a common chain** — the β_c family (IL-3, IL-5, GM-CSF) and the γ_c family of §4.4 (IL-2, IL-4, IL-7, IL-9, IL-15, IL-21, with IL-2R α = CD25 as the third chain); **heterodimeric with no common chain** (IL-1 family; IL-13, IFN- α , IFN- β , IFN- γ , IL-10); the **TNF receptor family** (TNFR1/II, CD40, Fas/CD95, CD30, CD27, NGF receptor); and the **chemokine receptor family** (CCR1-10, CXCR1-5, XCR1, CX3CR1).

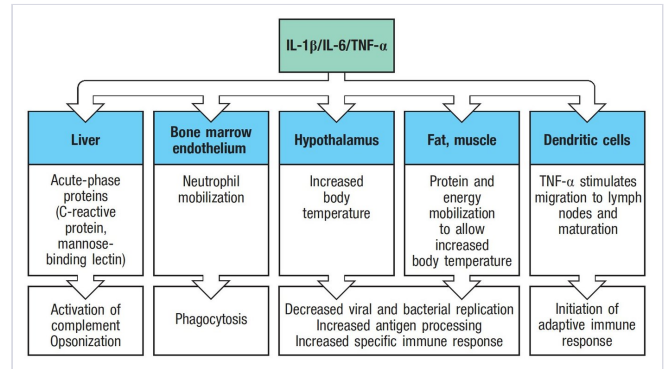


Fig 6.14 What **IL-1 β** , **IL-6** and **TNF- α** do to the whole body. **Liver**: acute-phase proteins — CRP and mannose-binding lectin (the lectin pathway of Session 2). **Bone marrow endothelium**: neutrophil mobilisation. **Hypothalamus**: fever. **Fat and muscle**: protein and energy mobilisation — the cachexia of chronic disease. **Dendritic cells**: TNF drives their migration to lymph nodes and their maturation, which is how they come to express the B7 of §2.2.

WHERE PART 7 BEGINS

Look again at Fig 6.12 and notice something. **TNF- α , IL-6 and CXCL8 are secreted by the ordinary route** — made with a signal peptide, pushed through the endoplasmic reticulum and Golgi, released. But **IL-1 β is not**. It has **no signal peptide**, it is made as an inactive precursor, and it cannot leave the cell by the normal door. NF- κ B — the endpoint of every pathway in this Part — **writes the pro-form and stops**.

So the most powerful inflammatory cytokine in the body sits inside the cell, fully transcribed and completely inert, waiting for a **second signal**. That machine is the inflammasome, and you have seen its logic before.

PART SEVEN

07

The Inflammasome — Two Signals Again

The most powerful inflammatory cytokine you make cannot get out of the cell that makes it, and is inactive even if it does. That is not a design flaw — it is a **second safety catch** on a molecule dangerous enough to cause shock. The machine that releases it is the inflammasome, and it works on exactly the logic you already know from Part 2: **one signal to prepare, a second to commit**.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Explain why IL-1 β needs a mechanism no other cytokine needs.
- Name what signal 1 and signal 2 each produce in the inflammasome pathway.
- Name the three components of the NLRP3 inflammasome and the domain each contributes.
- Give six chemically unrelated NLRP3 activators and the one event most of them share.
- Distinguish the canonical from the non-canonical pathway, and say which caspases are human.
- Explain how IL-1 β physically leaves the cell.

7.1 The Problem: IL-1 β Has No Signal Peptide

Every protein destined for secretion carries an N-terminal **signal peptide** that directs it into the endoplasmic reticulum. **The IL-1 superfamily does not have one.** It is made on free ribosomes in the cytosol and stays there.

There is a second problem on top of the first. It is made as **pro-IL-1 β** , a 31 kDa precursor that is **biologically inactive**. Even if it escaped, it would do nothing. So a cell that has detected a pathogen and switched on NF- κ B ends up holding a stockpile of an inert cytokine it cannot release — and that is exactly the state the system intends.

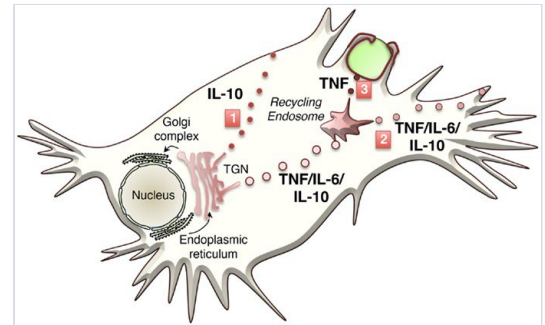


Fig 7.1 How ordinary cytokines leave. Anything with a **signal peptide** enters the endoplasmic reticulum, passes through the Golgi and is released by one of three routes: **direct from the trans-Golgi network** (IL-10), **via the recycling endosome** (TNF, IL-6, IL-10), or **routed to the phagocytic cup during phagocytosis** (TNF) so that it is delivered exactly where the microbe is.

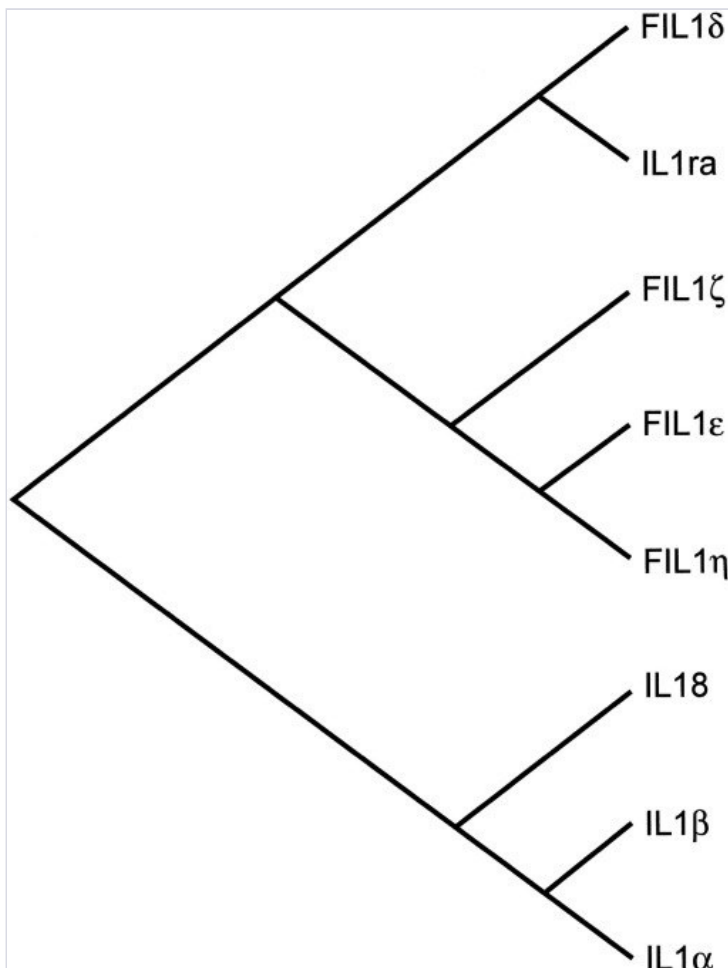


Fig 7.2 The IL-1 superfamily by sequence relatedness: **three subfamilies** — the **IL-1 family** (IL-1 α , IL-1 β , IL-1 α), the **IL-18 family** and the **IL-33 family** — with **more than ten members** in all. Note **IL-1 α** on the tree: a naturally occurring receptor antagonist, a member of the family whose job is to block it. That molecule became the drug **anakinra** (§8.4).

7.2 Signal 1 and Signal 2, Again

SIGNAL 1 — PRIMING

Write the parts

A **PRR** — a TLR seeing LPS, or TNFR, or IL-1R — activates **NF- κ B**, which transcribes **pro-IL-1 β** , **pro-IL-18** and **NLRP3** itself. Nothing is released. The cell is now **primed** and nothing more.

SIGNAL 2 — ACTIVATION

Assemble and cut

A second, independent stimulus — **ATP**, a **pore-forming toxin**, a **crystal** — triggers **assembly of the inflammasome**. Caspase-1 is activated, pro-IL-1 β is cleaved to its mature 17 kDa form, and GSDMD opens the door.

WHY TWO

The same reason as anergy

Mature IL-1 β causes fever, hypotension and, in quantity, shock. A cell must not be able to release it on the strength of a single, possibly mistaken, signal. **Requiring two independent inputs makes an accident improbable.**

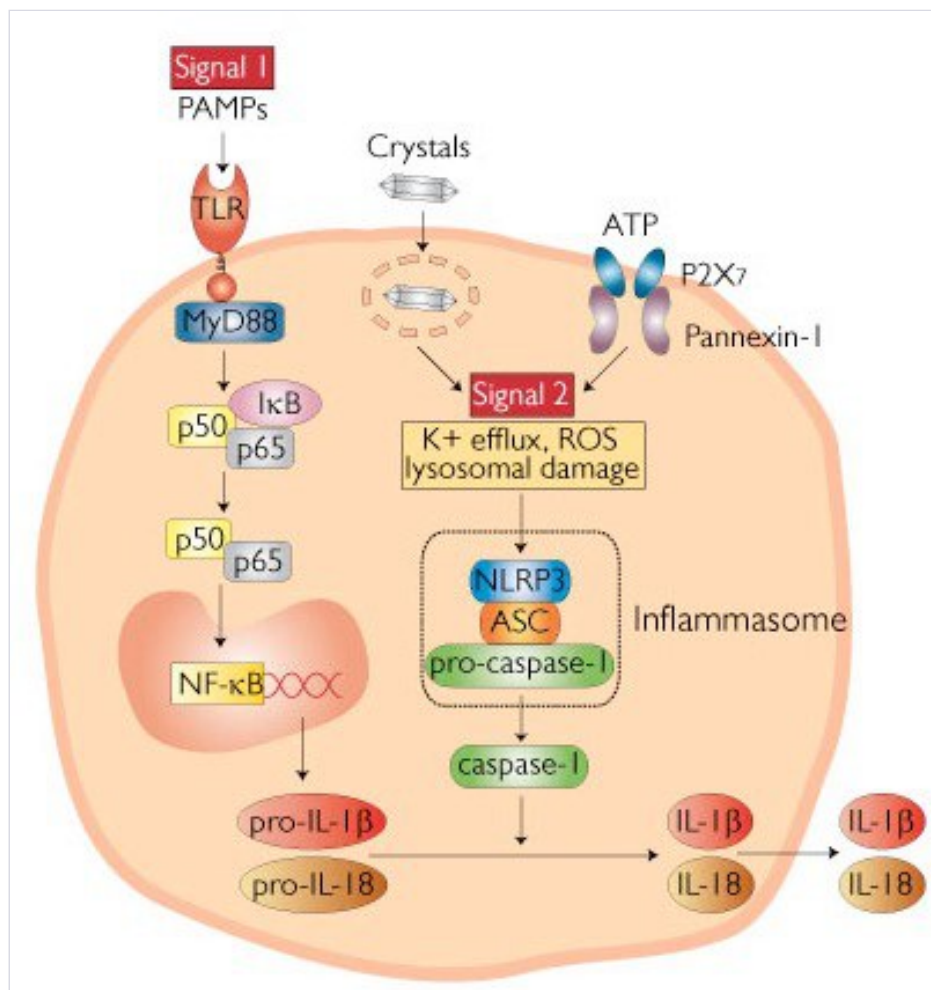


Fig 7.3 The two signals in one cell. **Signal 1** (left, red label): PAMPs engage a TLR → MyD88 → I κ B degraded → p50/p65 NF- κ B into the nucleus → transcription of pro-IL-1 β and pro-IL-18. **Signal 2** (right, red label): crystals, or ATP acting on P2X₇ and pannexin-1, cause K⁺ efflux, ROS and lysosomal damage → the NLRP3 / ASC / pro-caspase-1 complex assembles → caspase-1 cleaves the pro-forms → mature IL-1 β and IL-18 leave the cell.

THE PARALLEL YOU SHOULD SAY OUT LOUD

Put the two systems side by side and they are the same design:

- **T cell:** signal 1 = TCR sees peptide–MHC (specific). Signal 2 = CD28 sees B7 (permission). **One alone → anergy.**
- **Macrophage:** signal 1 = PRR sees PAMP (specific). Signal 2 = a danger event — ATP, crystal, toxin (permission). **One alone → a primed cell that releases nothing.**

In both cases the first signal says “**something is here**” and the second says “**and it is actually dangerous**”. In both cases the irreversible, expensive, potentially self-destructive response is withheld until both have arrived. **This is not a coincidence of teaching; it is how the immune system avoids being its own worst enemy.**

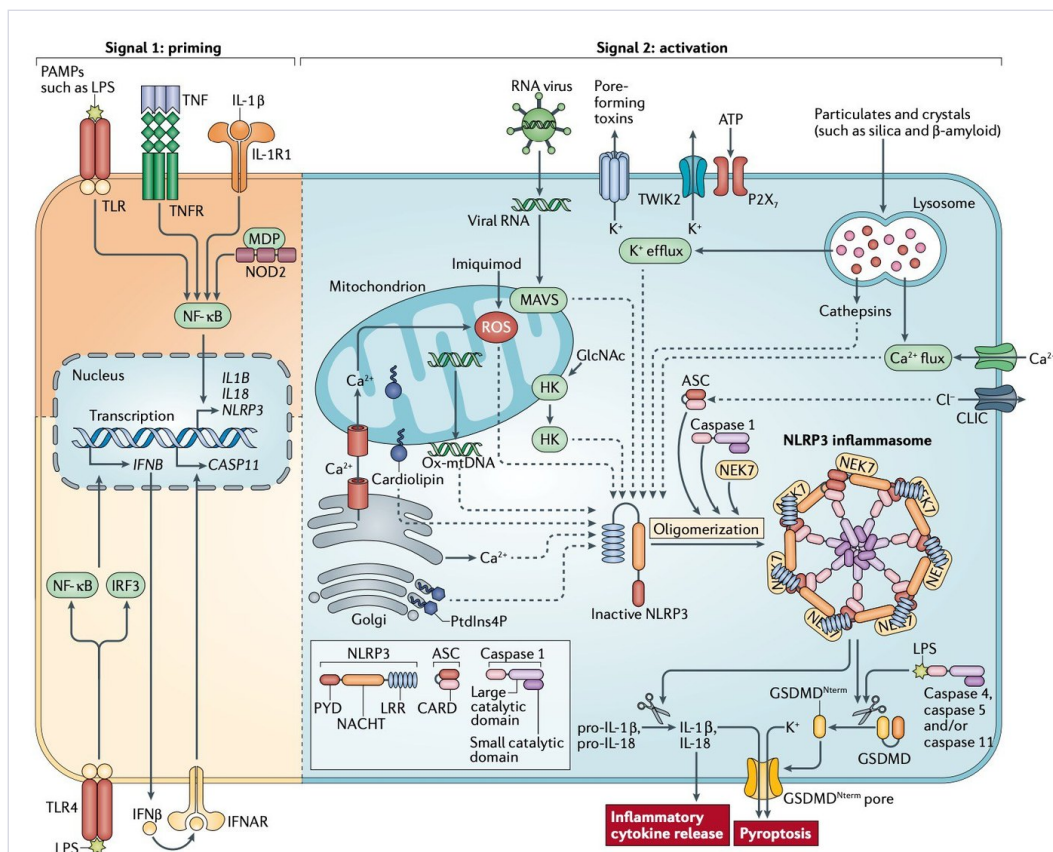


Fig 7.4 The same two signals at research resolution. **Left half — signal 1, priming:** LPS, TNF or IL-1β → NF-κB → transcription of IL1B, IL18 and NLRP3. **Right half — signal 2, activation:** RNA virus, pore-forming toxins, ATP via P2X₇, and particulates such as silica and β-amyloid, converging on K⁺ efflux, mitochondrial ROS, oxidised mitochondrial DNA, cardiolipin and lysosomal cathepsins → **NLRP3 oligomerisation** → caspase-1 → **inflammatory cytokine release and pyroptosis** through the GSDMD pore.

7.3 The Complex: Sensor, ASC, Caspase-1

DEFINITION

The **inflammasome** is a **multiprotein oligomer responsible for the activation of inflammatory responses**. The consequence of its assembly is **caspase-1 activation, by auto-processing**. Active caspase-1 then drives the **maturation and secretion of IL-1β and IL-18**.

COMPLEX CARD · S2 MΦ

The NLRP3 inflammasome

THREE PARTS, THREE JOBS

The sensor — NLRP3. Domains: **LRR** (leucine-rich repeat, the sensing end), **NACHT/NBD** (the nucleotide-binding domain that drives oligomerisation) and **PYD** (pyrin domain, which recruits ASC).

The adaptor — ASC. Two domains and nothing else: a **PYD** that binds NLRP3's PYD, and a **CARD** that binds caspase-1's CARD. It is a pure connector.

The executioner — pro-caspase-1. A **CARD** plus the catalytic subunits **p20** and **p10**.

HOW IT SWITCHES ON

NLRP3 oligomerises; its PYDs nucleate an **ASC filament**; the ASC CARDS in turn nucleate **caspase-1 filaments**.

Because: *The consequence is proximity-induced activation. Pro-caspase-1 has weak intrinsic activity; forced into a dense filament, the molecules cleave each other. This is why the lecture says caspase-1 is activated “by auto-processing”, and why the whole thing is called a signalling by polymerisation mechanism — the complex is not a switch, it is a nucleation seed.*

WHAT YOU SEE DOWN A MICROSCOPE

A single, large, perinuclear **ASC speck** — one per cell. Its appearance is the visual readout that the inflammasome has fired, and it is used as an assay.

Domains:

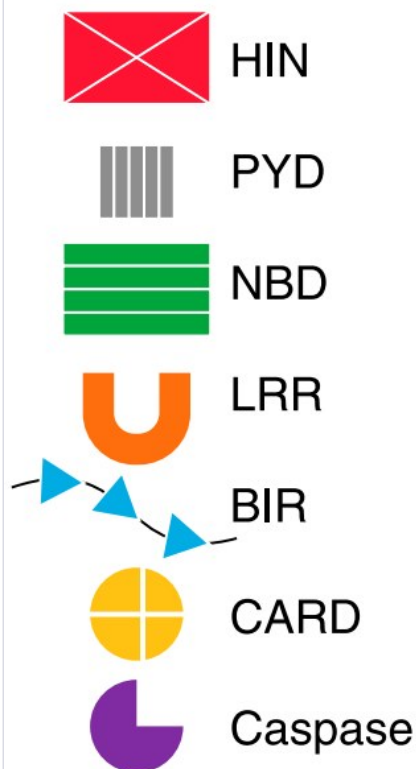


Fig 7.5 The domain alphabet. **HIN** (DNA-binding), **PYD** (pyrin), **NBD** (nucleotide-binding), **LRR** (leucine-rich repeat), **BIR**, **CARD** (caspase recruitment) and the **caspase** domain itself. Every inflammasome in the next figure is a different arrangement of these seven pieces.

NLRP3 Inflammasome

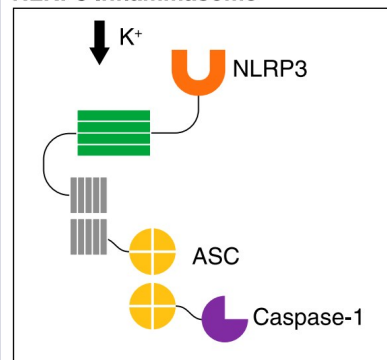


Fig 7.6 NLRP3 — activated by **K⁺ efflux** among many things. Sensor (PYD-NBD-LRR) → ASC (PYD-CARD) → caspase-1. The three-layer arrangement to learn.

AIM2 Inflammasome

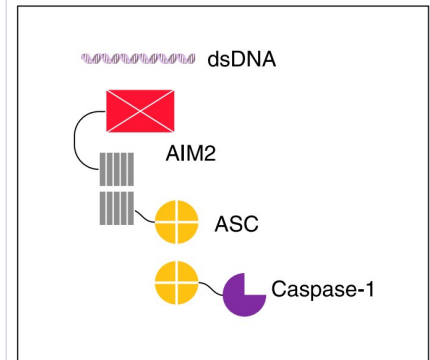


Fig 7.7 AIM2 — the same architecture with a different sensor. Its **HIN** domain binds **cytosolic double-stranded DNA** directly, so AIM2 is the inflammasome for DNA viruses and for intracellular bacteria that leak DNA into the cytosol.

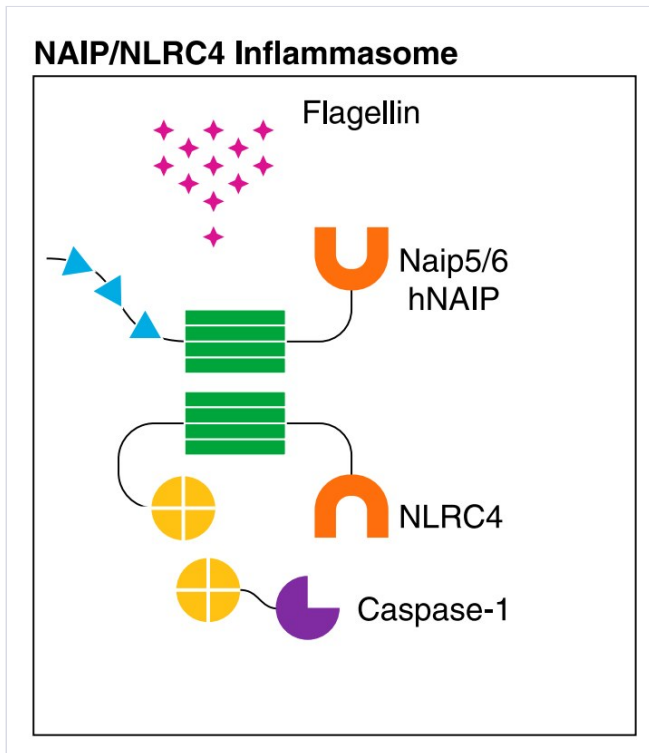


Fig 7.8 NAIP/NLRC4 — the inflammasome for **cytosolic flagellin** and bacterial type-III secretion-system components. Note the logic: TLR5 detects flagellin outside the cell, NAIP detects it inside. Outside means “a bacterium is nearby”; inside means “a bacterium is injecting things into me”, which is a far more urgent message.

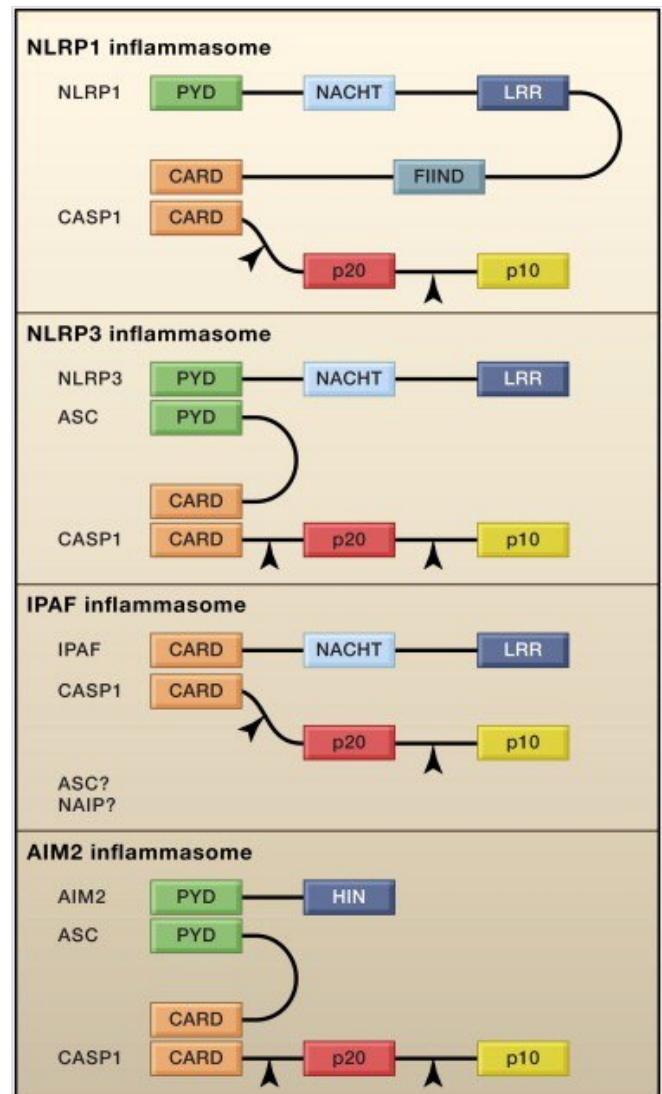


Fig 7.9 Four inflammasomes as domain diagrams. **NLRP1** and **IPAF/NLRC4** have their own CARD and can engage caspase-1 **without** **ASC**; **NLRP3** and **AIM2** have only a PYD and therefore **must** use **ASC** as the adaptor. This is the single structural fact that predicts which sensors need ASC.

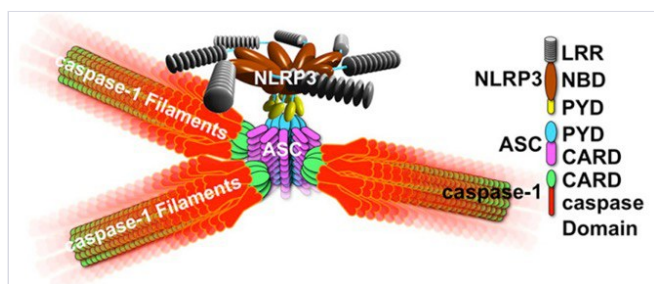


Fig 7.10 The structural model: **NLRP3 nucleates ASC filaments, which nucleate caspase-1 filaments.** A star-shaped polymer, not a tidy stoichiometric complex — which is how a handful of sensor molecules activate thousands of caspase molecules.

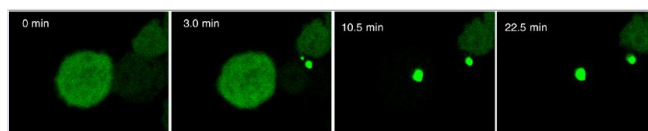


Fig 7.11 The same event in a living cell, four frames over 22 minutes. Diffuse green ASC in the cytoplasm suddenly condenses into **one bright speck**. Inflammasome activation is not gradual — it is a **phase transition**, all-or-none, per cell.

Caspase-1: the executioner

TABLE 7.1 THE CASPASES YOU NEED TO SEPARATE

Group	Members	What they do
Inflammatory caspases	Caspase-1, 4, 5 (human) Caspase-1, 11 (mouse)	Cleave pro-IL-1β and pro-IL-18 to the mature cytokines, and cleave gasdermin D
Apoptotic caspases	Initiators: caspase-8 (extrinsic), caspase-9 (intrinsic) Executioners: caspase-3, 6, 7	Mediate cell death — quiet, non-inflammatory, membrane-intact apoptosis

EXAM TRAP — CASPASE-11 IS A MOUSE GENE

The literature on the non-canonical inflammasome is almost all mouse work, so **caspase-11** appears everywhere. **Humans do not have caspase-11.** Its functional equivalents are **caspase-4 and caspase-5.** If a question names caspase-11, it is describing a mouse experiment; if it asks what senses intracellular LPS in a human cell, the answer is **caspase-4 and caspase-5.**

ARTICLES

A novel heterodimeric cysteine protease is required for interleukin-1 β processing in monocytes

Nancy A. Thornberry¹, Herbert G. Bial², Jimmy R. Calaycay¹, Kevin T. Chappman¹, Andrew D. Howard¹, Matthew J. Kostura¹, Douglas K. Miller¹, Susan M. Molinax¹, Jeffrey R. Weidner¹, John Amara¹, Keith O. Elskott¹, Julia M. Ayala¹, Francesca J. Casanovi¹, Jayne Chin¹, Gloria J.-F. Ding¹, Linda A. Egger¹, Erin P. Gaffney¹, Guohua Jiang¹, Qianqian C. Pinyan¹, S. M. Raju¹, Anna M. Robinson¹, J. Paul Salter¹, Ting-Ting Yarnitzky¹, Terry D. Lee¹, John E. Shively¹, Malcolm MacCoss¹, Richard A. Mamford¹, John A. Schmidt¹ & Michael J. Tocci¹

Fig 7.12 The 1992 Nature paper. The title is the whole discovery: “A novel heterodimeric cysteine protease is required for interleukin-1 β processing in monocytes.” The enzyme was named **interleukin-1 β converting enzyme (ICE)** and later renamed **caspase-1**.

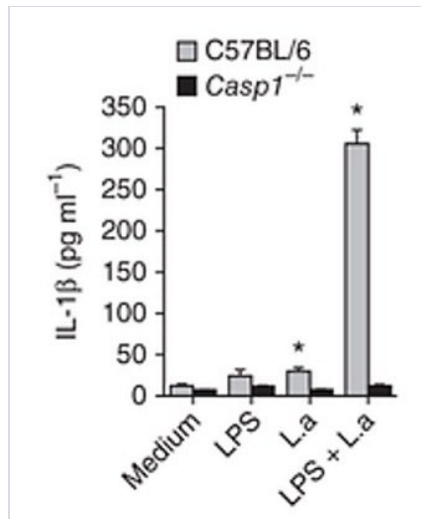


Fig 7.13 The proof, in one bar chart. Wild-type cells (grey) release a great deal of IL-1 β when given **LPS plus a second stimulus**; **Casp1^{-/-}** cells (black) release essentially none under any condition. Note also that LPS alone gives almost nothing even in wild-type — **that is signal 1 without signal 2**.



Fig 7.14 Jürg Tschopp (1951–2011), whose group at Lausanne — with Fabio Martinon and Kimberly Burns — described and named the inflammasome in 2002.

7.4 What Activates NLRP3

TABLE 7.2 NLRP3 ACTIVATORS — AND WHAT EACH ONE MEANS CLINICALLY

Activator	Where you meet it
ATP	Released by dying cells; acts on the P2X₇ receptor. The canonical laboratory signal 2, and a genuine DAMP in any tissue injury
Pore-forming toxins	Nigericin (experimental), and bacterial toxins such as streptolysin and α -haemolysin
Alum	The adjuvant in many vaccines. This is how alum works — an answer that was unknown for seventy years
MSU — monosodium urate	Gout . The crystal is phagocytosed, ruptures the lysosome, and the resulting IL-1 β is the pain and swelling of an acute attack
Cholesterol crystals, oxLDL	Atherosclerosis . The plaque macrophage is running an inflammasome
Glucose, palmitate (LCFA), ceramide	Type 2 diabetes and metabolic syndrome — the molecular link between obesity and chronic low-grade inflammation
Islet amyloid polypeptide	Deposited in the pancreatic islet in type 2 diabetes; also β -amyloid in Alzheimer's disease
Silica, asbestos	Silicosis and asbestosis — inhaled particles that a macrophage cannot digest

ONE RECEPTOR, CHEMICALLY UNRELATED LIGANDS — HOW?

Look at that list. ATP, a crystal of uric acid, a fatty acid, a bacterial toxin and a mineral fibre have nothing chemically in common, and no single binding site could accommodate them all. **NLRP3 therefore does not sense them directly.** It senses a **cellular consequence** they all produce.

The dominant one is **potassium efflux**. A drop in intracellular K^+ is the common denominator of pore formation, P2X₇ opening and membrane damage — and it is sufficient on its own to trigger NLRP3. There are also **K^+ -efflux-independent** routes: **lysosomal rupture** releasing cathepsin B (the crystal route), **mitochondrial ROS** and **oxidised mitochondrial DNA**, **Ca^{2+} influx**, and **NEK7**, a kinase that binds NLRP3 and licenses its oligomerisation.

The design principle: NLRP3 is not a receptor for a molecule; it is a **detector of cellular distress**. That is why it appears in so many diseases that have nothing to do with infection.

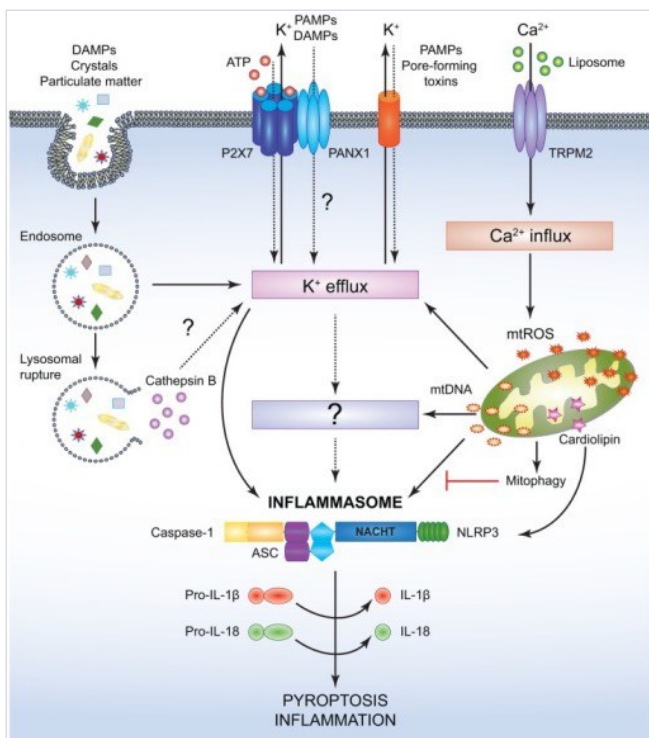


Fig 7.15 Everything converging on one box. DAMPs, crystals, particulates, ATP through **P2X₇** and **pannexin-1**, pore-forming toxins, liposomes and **TRPM2**-mediated Ca^{2+} influx all funnel into **K^+ efflux**; lysosomal rupture supplies **cathepsin B**; damaged mitochondria supply **mtROS**, **mtDNA** and **cardiolipin**. Output: caspase-1, IL-1 β , IL-18 and **pyroptosis**. Note **mitophagy** at the right, inhibiting the pathway by clearing the damaged mitochondria.

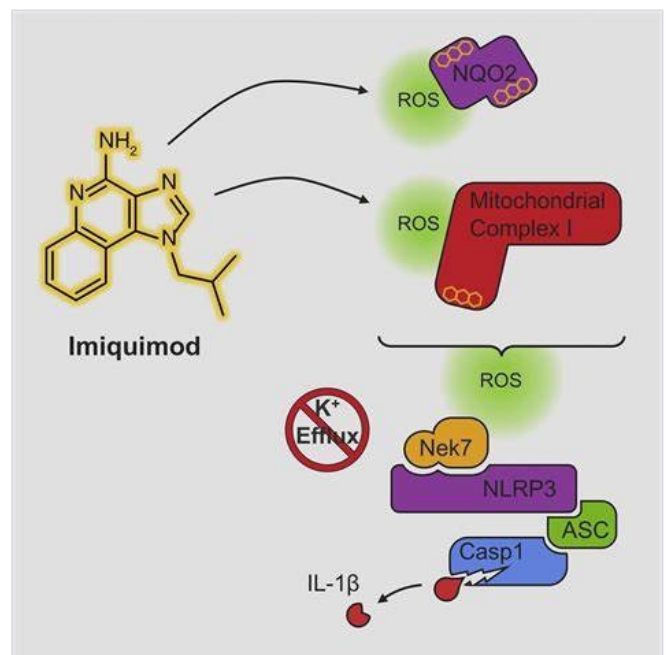


Fig 7.16 A worked example of a **K^+ -efflux-independent** activator. **Imiquimod** inhibits NQO2 and mitochondrial complex I, generating **ROS**, which licenses the **NEK7-NLRP3** interaction and assembles the complex with a crossed-out K^+ efflux. Imiquimod is a real drug — a topical treatment for genital warts and basal cell carcinoma — and this is part of how it works.

7.5 The Non-canonical Route: Caspase-11, 4 and 5

The lecturer asks: *what happens when cells are infected with intracellular bacteria?* TLR4 is on the cell surface and in endosomes. A bacterium that has escaped into the cytosol is carrying LPS **on the wrong side of every membrane TLR4 can see**.

TABLE 7.3 CANONICAL VERSUS NON-CANONICAL

	Canonical NLRP3	Non-canonical
Trigger	PAMPs and DAMPs — the Table 7.2 list	LPS free in the cytosol , from Gram-negative bacteria that have escaped the vacuole
Sensor	NLRP3 + NEK7 , then ASC	The caspase itself. Pro-caspase-11 (mouse) or pro-caspase-4/5 (human) binds LPS directly through its CARD — no separate receptor at all
Caspase	Caspase-1	Caspase-11 (mouse) · Caspase-4, -5 (human)
Direct products	IL-1β, IL-18 and cleaved GSDMD	Cleaved GSDMD only. These caspases cannot process pro-IL-1 β
How IL-1 β still gets made	Directly	Indirectly. GSDMD pores cause K⁺ efflux , which activates NLRP3 and caspase-1 in the same cell. The non-canonical pathway switches the canonical one on

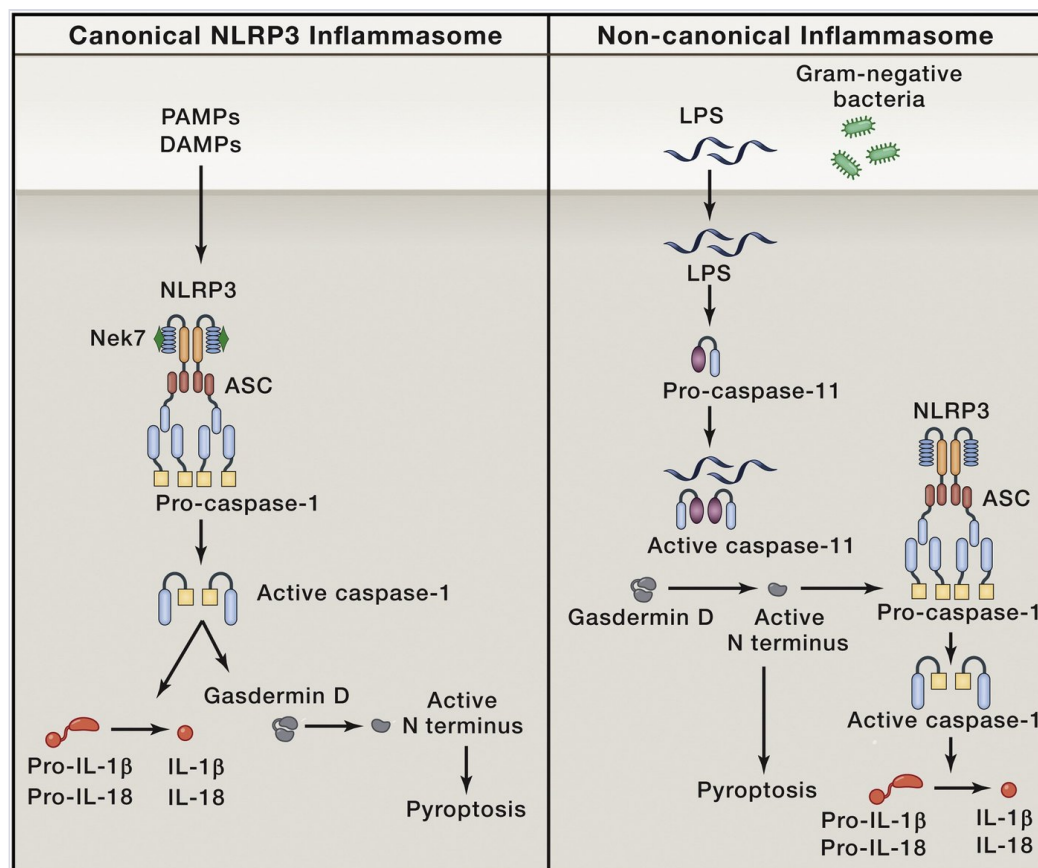


Fig 7.17 The two pathways side by side. **Left, canonical:** PAMPs/DAMPs → NLRP3 with **Nek7** and ASC → caspase-1 → both gasdermin D and the cytokines. **Right, non-canonical:** cytosolic LPS → pro-caspase-11 directly → gasdermin D → **pyroptosis**, and — via the resulting membrane damage — NLRP3, caspase-1 and the cytokines after all. Follow the arrow that loops back: **the two pathways are not alternatives, they are in series.**

7.6 Gasdermin D, the Pore and Pyroptosis

○ MECHANISM — HOW IL-1 β ACTUALLY LEAVES THE CELL

- 1 **Gasdermin D sits inert in the cytosol**, its pore-forming N-terminal domain held shut by an autoinhibitory C-terminal domain.
- 2 An **inflammatory caspase cuts between the two domains**, releasing the N-terminal fragment.
WHY: Note which enzymes can do this: caspase-1, 4, 5 and 11. The apoptotic caspases cannot — which is precisely why apoptosis is silent and pyroptosis is not.
- 3 The freed **N-terminal fragments oligomerise in the plasma membrane** and form a **ring-shaped pore**.
WHY: The structure is remarkable: a 27-fold ring built from 108 β -strands, with an inner diameter of about 180 Å — enormous by the standards of a membrane channel, and comfortably wide enough for a 17 kDa cytokine to pass.
- 4 **Mature IL-1 β and IL-18 pass out through the pore.** This is the door that the missing signal peptide made necessary.
- 5 If enough pores form, the cell can no longer hold its osmotic balance. It swells and bursts: **pyroptosis**.
WHY: Contrast with apoptosis. Apoptosis keeps the membrane intact and packages the contents tidily, so it is immunologically silent. Pyroptosis ruptures the cell and spills every DAMP it contains into the tissue, so it is maximally inflammatory — it recruits the neighbours. It is the right answer to an intracellular pathogen: destroy its niche and raise the alarm at the same time.

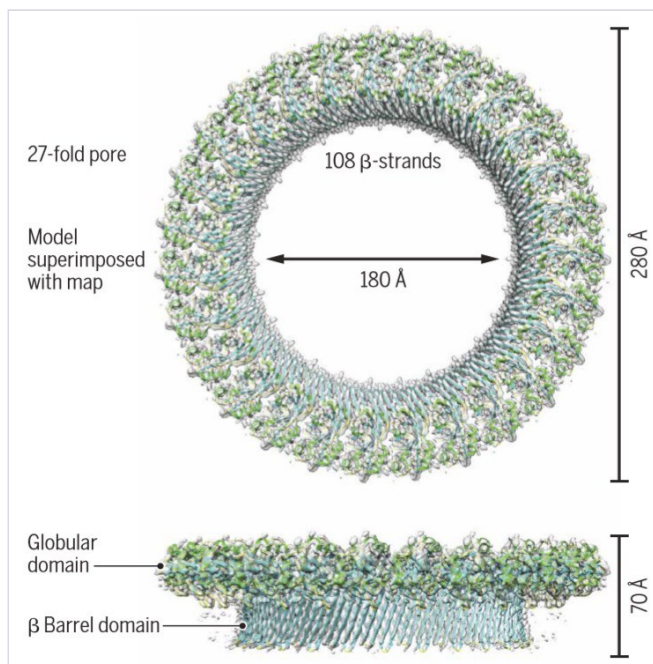


Fig 7.18 The pore, solved structurally: a 27-fold ring of 108 β -strands, inner diameter 180 Å, outer 280 Å, with a globular domain above and a β -barrel of about 70 Å crossing the membrane. Seen from the side below.

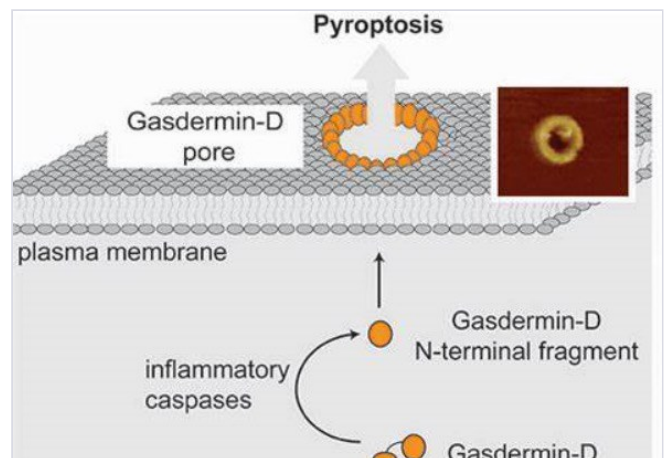


Fig 7.19 The mechanism in one picture: inflammatory caspases cleave gasdermin D, the N-terminal fragment inserts into the plasma membrane, and the gasdermin-D pore opens — with an atomic-force micrograph of a real pore inset. Arrow upwards: pyroptosis.



Fig 7.20 Feng Shao (邵峰), Beijing — identified gasdermin D as the substrate that executes pyroptosis.



Fig 7.21 Vishva M. Dixit — whose group independently identified gasdermin D and much of the non-canonical pathway.



Fig 7.22 Hao Wu — structural biologist who solved the filament and pore architectures that make the polymerisation mechanism visible.

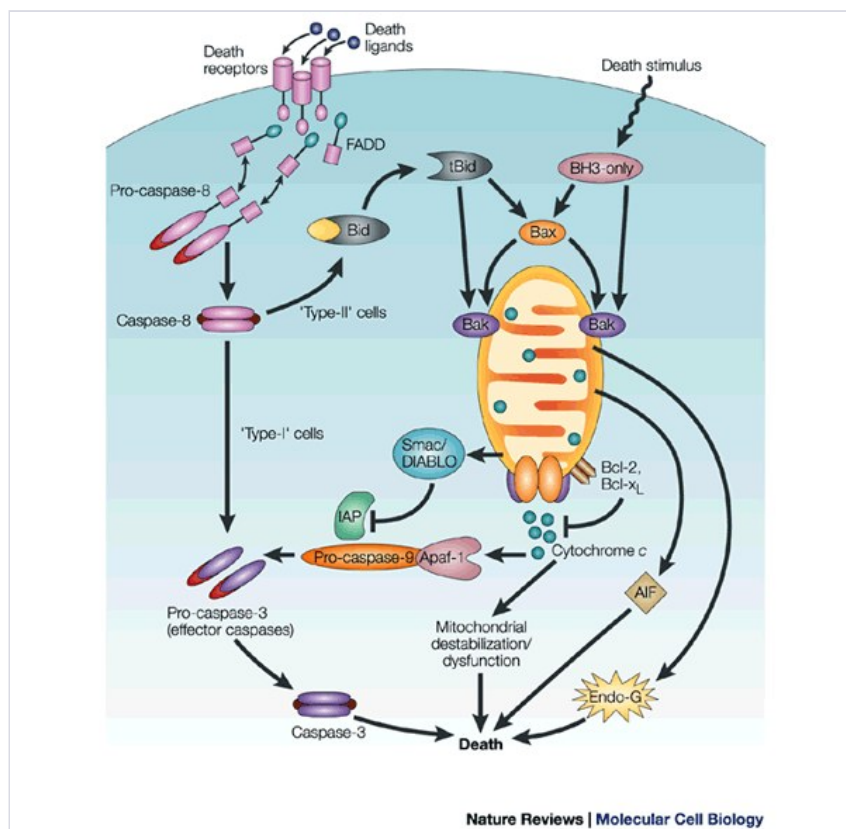


Fig 7.23 For contrast, the **apoptotic** caspases. **Extrinsic:** death ligands → death receptors → FADD → **caspase-8**. **Intrinsic:** a death stimulus → BH3-only proteins → Bax/Bak → mitochondrial cytochrome c → Apaf-1 → **caspase-9**. Both converge on **caspase-3**. None of these enzymes cleaves gasdermin D, and none releases IL-1 β . **Same protein family, opposite immunological meaning.**

7.7 Pathogens Fight Back

The pattern is the one you met at the end of Part 5. A host defence mechanism is confirmed as important by the fact that **pathogens spend genes on disabling it**. Poxviruses encode decoy cytokine receptors; *Shigella* encodes a ligase that destroys gasdermin D. **If you want to know which parts of immunology matter, read the list of what microbes have evolved to switch off.**

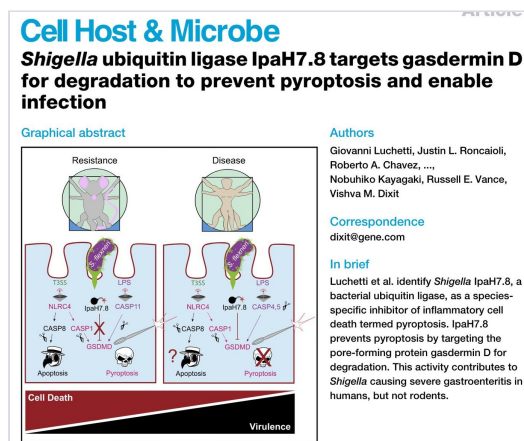


Fig 7.24 *Shigella* injects IpaH7.8, a bacterial ubiquitin ligase, which tags gasdermin D for proteasomal destruction. No gasdermin, no pore, no pyroptosis — and the bacterium keeps the cell it is living in. The effect is **species-specific**: IpaH7.8 targets human gasdermin D but not the rodent protein, which contributes to *Shigella* causing severe gastroenteritis in humans but not in rodents.

PART EIGHT

08

When It Goes Wrong — Storm, Disease, Therapy

Everything in Parts 6 and 7 is a mechanism for making inflammation. Nothing in them is a mechanism for making *too much* inflammation — that requires only that the same machinery run without limit. This Part is the clinical consequence, and it is where the whole of these two lectures becomes medicine.

BY THE END OF THIS PART YOU SHOULD BE ABLE TO

- Give five beneficial and five deleterious effects of inflammation.
- Define cytokine storm by its three obligatory components.
- Explain two routes by which SARS-CoV-2 activates the inflammasome.
- Match at least six anti-cytokine drugs to their targets and their diseases.

8.1 Inflammation as a Double-Edged Sword

TABLE 8.1 THE LECTURER'S TWO COLUMNS

Beneficial effects	Deleterious effects
1. Host defence against pathogens — the job it exists for	1. Immunopathology — the tissue damage is done by the response, not the microbe
2. Anti-tumour immunity	2. Autoimmunity
3. Immune homeostasis — including the MyD88 lesson of §6.5	3. Sepsis
4. Tissue repair and wound healing	4. Fibrosis — repair that does not stop
5. Adaptation to stress and restoration of homeostasis	5. Tumorigenesis and 6. allergy

READ THE TWO COLUMNS AGAINST EACH OTHER

They are not two lists; they are **the same list seen twice**. *Tissue repair* and *fibrosis* are one process with and without an off-switch. *Anti-tumour immunity* and *tumorigenesis* are the same inflammation at different durations — acute inflammation kills tumour cells, chronic inflammation causes them. *Host defence* and *immunopathology* are the same response judged by different outcomes: in tuberculosis and in severe COVID-19, most of the lung damage is yours, not the pathogen's.

The clinical corollary is uncomfortable and important: there is no drug that gives you column one without touching column two. Every anti-inflammatory you will prescribe trades some host defence for some relief, and the whole skill is judging the exchange rate in the patient in front of you.

8.2 Cytokine Storm

DEFINITION — THREE COMPONENTS, ALL OBLIGATORY

A **cytokine storm** requires all three of: (1) **elevated circulating cytokine levels**, (2) **acute systemic inflammatory symptoms**, and (3) **secondary organ dysfunction** — most often renal, hepatic or pulmonary — **beyond that attributable to the inciting pathogen itself**.

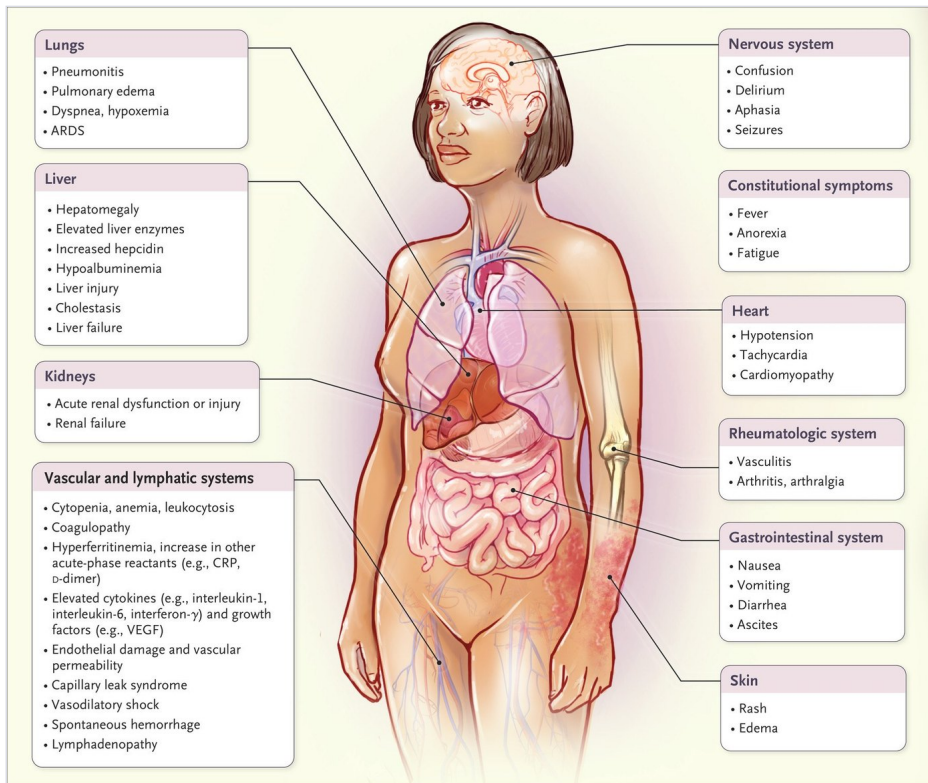


Fig 8.1 The clinical presentation, organ by organ. **Constitutional:** fever, anorexia, fatigue. **Lungs:** pneumonitis, pulmonary oedema, ARDS. **Nervous system:** confusion, delirium, seizures. **Heart:** hypotension, tachycardia, cardiomyopathy. **Liver:** hepatomegaly, transaminitis, hypoalbuminaemia, hyperferritinaemia. **Kidneys:** acute renal dysfunction. **Vascular and lymphatic:** cytopenias, coagulopathy, capillary leak, vasodilatory shock, spontaneous haemorrhage, lymphadenopathy, raised CRP, IL-6, IFN- γ , D-dimer and VEGF. **Skin:** rash, oedema. Note how many of these you can now derive: hypotension is TNF and IL-1 on the endothelium; fever is IL-1 and IL-6 on the hypothalamus; hypoalbuminaemia and raised CRP are the hepatic acute-phase response of Fig 6.14.

CLINICAL CORRELATION — THE FIVE SETTINGS YOU WILL ACTUALLY MEET

- **Sepsis** — the original and commonest. LPS on TLR4 in every macrophage at once.
- **CAR-T cell therapy** — **cytokine release syndrome** is the signature toxicity of the CD19 CAR-T cells you met in §1.3, and the standard treatment is **tocilizumab**, an anti-IL-6 receptor antibody.
- **HLH and macrophage activation syndrome** — often driven by defects in perforin-mediated killing, so the activated macrophage is never switched off. One of its diagnostic criteria is a raised **soluble CD25** (§4.3).
- **Severe influenza and severe COVID-19** — §8.3.
- **Anti-CD28 superagonist (TGN1412), 2006** — six healthy volunteers given a drug designed to deliver **signal 2 without signal 1** developed multi-organ failure within hours. The three-signal model of Part 2 is not an abstraction.

8.3 The Inflammasome in Severe COVID-19

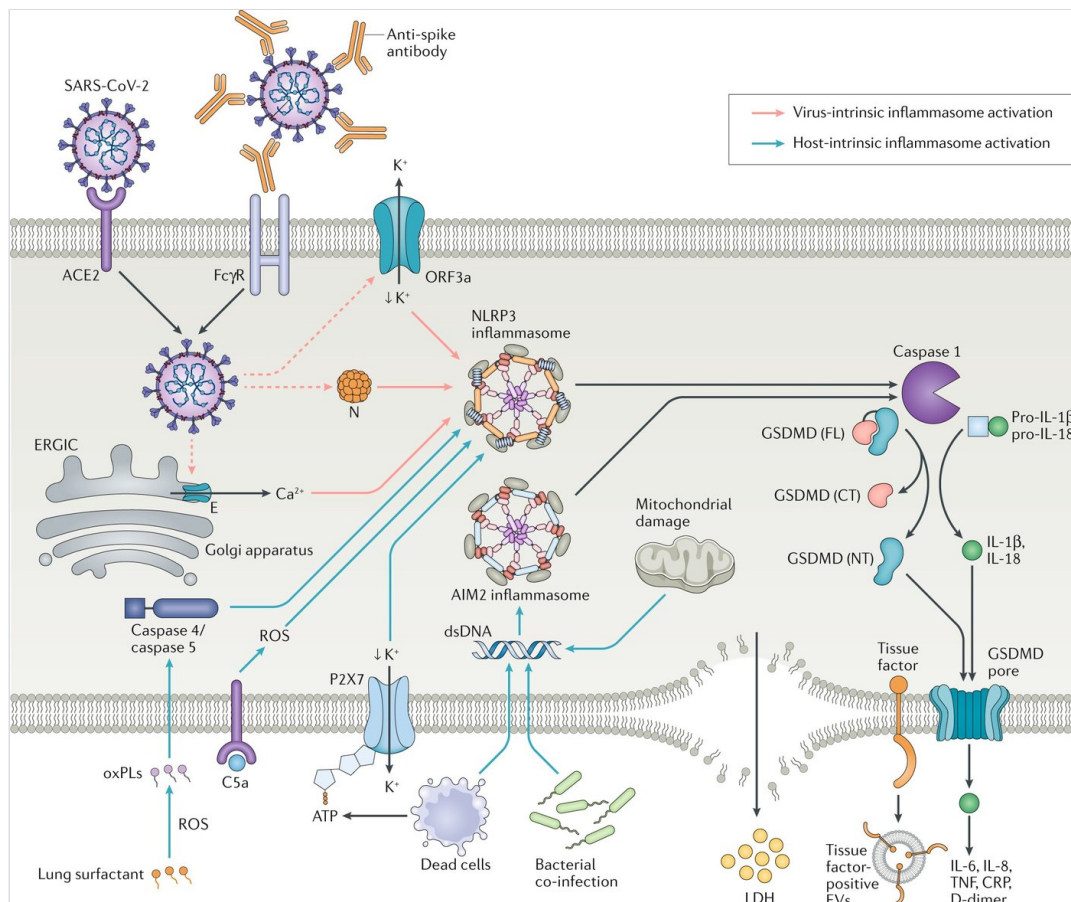


Fig 8.2 Two routes, drawn in two colours. **Virus-intrinsic (red):** SARS-CoV-2 enters via **ACE2**; its **ORF3a**, **E** and **N** proteins disturb ion flux and Golgi and mitochondrial integrity, giving **K⁺ efflux**, **Ca²⁺ flux**, **ROS** and **released mitochondrial DNA** — which activate **NLRP3** and **AIM2**. **Host-intrinsic (blue):** antibody-coated virus taken up through **FcγR**, complement **C5a**, dying cells, bacterial co-infection and lung surfactant. Both converge on **caspase-1**, **GSDMD pores**, **IL-1β** and **IL-18**, and — importantly — on **tissue factor** and **tissue-factor-positive extracellular vesicles**, which is the mechanistic link to the raised **D-dimer** and the thrombosis of severe COVID-19.

THREE THINGS THIS FIGURE LETS YOU EXPLAIN

- **Why severe COVID-19 is a thrombotic disease.** Inflammasome activation induces tissue factor. Inflammation and coagulation are not neighbours; they are the same pathway.
- **Why antibody can make it worse.** Uptake of antibody-coated virus through FcγR is a **host-intrinsic** activator — the antibody-dependent enhancement you met in Session 2 §6.4, arriving here by a different route.
- **Why dexamethasone works and antivirals given late do not.** By the time a patient is hypoxic, the problem is no longer viral replication — it is this diagram. Treat the response, not the virus.

8.4 Targeting Cytokines and the Inflammasome

TABLE 8.2 THE ANTI-CYTOKINE PHARMACOPOEIA — EVERY ONE OF THESE TARGETS A MOLECULE IN THIS GUIDE

Drug	Target	Used in
Infliximab, adalimumab (mAbs) · etanercept (receptor decoy)	TNF-α	Rheumatoid arthritis, psoriasis, psoriatic arthritis, ankylosing spondylitis, Crohn's disease. Screen for latent TB first
Tocilizumab	IL-6 receptor	Rheumatoid arthritis, giant-cell arteritis, CAR-T cytokine release syndrome , severe COVID-19
Anakinra (recombinant IL-1ra) · canakinumab (anti-IL-1β) · rilonacept (trap)	IL-1	CAPS , familial Mediterranean fever, gout, adult-onset Still's disease, pericarditis, HLH
Ustekinumab	IL-12/IL-23 p40 — the shared subunit of $\S 4.4$	Psoriasis, psoriatic arthritis, Crohn's disease
Secukinumab, ixekizumab	IL-17A — the Th17 output of $\S 5.1$	Psoriasis, ankylosing spondylitis
Dupilumab	IL-4Rα — blocks IL-4 and IL-13	Atopic dermatitis, asthma, chronic rhinosinusitis with nasal polyps
Tofacitinib, baricitinib, ruxolitinib	JAK1/2/3 — $\S 4.5$	Rheumatoid arthritis, ulcerative colitis, atopic dermatitis, myelofibrosis, GvHD, severe COVID-19
Colchicine	Microtubules → impairs NLRP3 assembly	Gout, familial Mediterranean fever, pericarditis, secondary prevention after myocardial infarction
MCC950, dapansutride (investigational) · disulfiram	NLRP3 · gasdermin D	Trials across gout, heart failure, neuroinflammation and sepsis

— EXCESS PROVES FUNCTION — CAPS AND THE NLRP3 GAIN-OF-FUNCTION SYNDROMES

Everywhere else in these two guides, function has been proved by **absence**. Here it is proved by **excess**. **Cryopyrin-associated periodic syndromes (CAPS)** are caused by **gain-of-function mutations in NLRP3**, which make the inflammasome assemble without waiting for signal 2. The three phenotypes form a severity spectrum:

- **FCAS** — familial cold autoinflammatory syndrome: urticarial rash and fever after cold exposure.
- **Muckle-Wells syndrome** — adds **sensorineural deafness** and **AA amyloidosis**.
- **NOMID / CINCA** — neonatal-onset multisystem inflammatory disease: chronic aseptic meningitis, arthropathy, and severe developmental impairment.

The decisive observation is therapeutic. These children respond **dramatically** to IL-1 blockade — anakinra or canakinumab — often within days. A single mutation in one sensor, producing one cytokine in excess, blocked by one drug. **There is no cleaner demonstration in clinical immunology that a pathway described on a slide is the pathway operating in a patient.**

CLINICAL CORRELATION — CANTOS, AND INFLAMMATION AS A CARDIAC RISK FACTOR

If cholesterol crystals activate NLRP3 (§7.4), then IL-1 β should be driving atherosclerosis, and blocking it should reduce heart attacks **independently of cholesterol**. The **CANTOS** trial tested exactly that: **canakinumab** (anti-IL-1 β) in patients with a previous myocardial infarction and a raised CRP. It reduced recurrent cardiovascular events with **no change in LDL cholesterol at all** — and increased fatal infections, the cost of column two in Table 8.1. The trial converted "inflammation causes atherosclerosis" from a plausible story into a demonstrated, drug-modifiable mechanism.

Trial	Type, size and inclusion	Intervention and end points
Targeting NLRP3		
Novartis DfV890 (NCT04382053)	Multicentre, randomized, controlled; 143 participants	Administered for 14 days in addition to standard of care; day 15 APACHE II severity or day of discharge
Olatec Therapeutics Dapansutrole (NCT04540120)	Randomized, double blind, placebo controlled; 80 participants	Initial dose of 8 × 250 mg on first day; 4 × 250-mg capsules for 14 days; proportion of participants with clinical deterioration at day 15, defined as COVID-19-related hospitalization or both worsening shortness of breath and oxygen saturation less than 92% on room air
Targeting gasdermin D		
University of California, San Francisco Disulfiram for COVID-19 (DISCO) trial (NCT04485130)	Randomized, dose escalation, placebo controlled, double blind; 60 participants	1,000 mg per day for 5 days or 2,000 mg per day for 5 days; plasma levels of IL-1 β , IL-6 and other cytokines at 31 days
ETICA Disulfiram (NCT04594343)	Randomized, double blind, placebo controlled; 200 participants	500 mg per day for 14 days; plasma IL-18 level at day 7
University of Oxford RECOVERY trial Dimethyl fumarate (NCT04381936)	Randomized, open label	120 mg every 12 hours for 2 days and 240 mg every 12 hours for 8 days; mortality 28 days after randomization

APACHE II, acute physiology and chronic health evaluation II.

Fig 8.3 Trials of inflammasome-pathway inhibition in COVID-19, split by target: **targeting NLRP3** (dapansutrole) and **targeting gasdermin D** (disulfiram, dimethyl fumarate). Note the endpoints — clinical deterioration, plasma IL-1 β and IL-6, mortality. The pathway you learned in Part 7 was being tested in patients while you were in secondary school.

PART NINE

09

Master Tables, Self-Test and Glossary

No new material. This is the compression: two lectures reduced to four master tables, a set of deliberate look-alikes, thirty questions and a glossary. On the night before the exam, read nothing else.

HOW TO USE THIS PART

- Cover the right-hand columns of every master table and rebuild them from the left.
- Do the thirty questions **before** re-reading anything; the ones you miss name the section to return to.
- Every answer carries a section reference. Follow it — that is what it is for.

9.1 Master Table — Every CD Molecule in These Lectures

TABLE 9.1 CD NUMBER, ALIAS, CELL, PARTNER, JOB

CD	Also called	On which cell	Binds	What it does
CD3	—		Nothing external — it is bolted to the TCR	Signal 1 transducer. Carries the ITAMs. Pan-T marker; anti-CD3 alone activates T cells
CD4	—	 helper	MHC class II, $\beta 2$ domain (non-polymorphic)	Co-receptor; brings Lck . Raises sensitivity ~100×. Also the HIV receptor
CD8	—	 cyto-toxic	MHC class I, $\alpha 3$ domain (non-polymorphic)	Co-receptor; brings Lck . $\alpha\alpha$ homodimer or $\alpha\beta$ heterodimer
CD14	—		LPS , with LBP	Co-receptor that delivers LPS to TLR4/MD-2 . Monocyte/macrophage marker
CD19	—		Part of the CD19/CD21/CD81 complex	The signalling subunit of the B-cell co-receptor. Pan-B marker; CAR-T target
CD20	—		—	Mature B cells, not plasma cells . Target of rituximab
CD21	CR2, C3dR, EBV-R	 , FDC	C3d deposited on antigen	Lowers the B-cell activation threshold up to 1000× . Entry receptor for Epstein-Barr virus
CD25	IL-2Rα	 activated	IL-2	Converts the IL-2 receptor to high affinity . Activation marker ; target of basiliximab; also on Tregs
CD28	—		CD80 / CD86	Signal 2 — the accelerator. Constitutive. Without it: anergy
CD35	CR1	 , RBC	C3b, C4b	Complement regulation and immune-complex transport on erythrocytes
CD40	—	 	CD40L (CD154)	Receives T-cell help → proliferation, class switching, germinal centres
CD64	FcγRI		IgG Fc , high affinity	The receptor end of opsonisation

CD	Also called	On which cell	Binds	What it does
CD79a/b	Igα / Igβ		Bolted to membrane Ig	The B-cell ITAM unit — exactly what CD3 is to the T cell
CD80 / CD86	B7-1 / B7-2	 	CD28 and CTLA-4	Provides signal 2. Up-regulated only when the APC has been activated by innate signals
CD152	CTLA-4	 activated	CD80 / CD86 , higher affinity than CD28	The brake. Induced after TCR firing → cell-cycle arrest. Target of ipilimumab
CD154	CD40L	 activated	CD40	Delivers help. Loss → X-linked hyper-IgM syndrome
CD256 / CD257	APRIL / BAFF		BCMA, TACI, BAFF-R (CD268)	Surface cytokines that keep B cells alive

9.2 Master Table — MHC Class I versus Class II

TABLE 9.2 THE TABLE TO BE ABLE TO WRITE FROM MEMORY

	Class I	Class II
Genes	HLA-A, -B, -C	HLA-DP, -DQ, -DR
Chains	α (45 kDa) + β₂-microglobulin (12 kDa)	α (35 kDa) + β (28 kDa)
Both chains in the MHC?	No — β ₂ m is on chromosome 15	Yes — both are MHC genes
Peptide-binding cleft	α1 + α2	α1 + β1
Cleft shape	Closed at both ends — <i>hot dog</i>	Open at both ends — <i>pita</i>
Peptide length	8–10 amino acids	Over 20 possible
Co-receptor site	α3 → binds CD8	β2 → binds CD4
Distribution	All nucleated cells	APCs (B, DC, MΦ), thymic epithelium, activated human T cells
Peptide source	Endogenous — cytosolic, via proteasome and TAP	Exogenous — endocytosed, via endosomal cathepsins
Presents to	CD8⁺ cytotoxic T cells	CD4⁺ helper T cells
Message	“Here is what I am making ”	“Here is what I found ”
Disease of down- or up-regulation	Tumours down-regulate it to escape CD8 cells — and become NK targets	Autoimmune disease up-regulates it on cells that should not present

9.3 Master Table — The Cytokines

TABLE 9.3 EVERY CYTOKINE NAMED IN THESE TWO LECTURES

Cytokine	Made by	Main action	Clinical hook
IL-1 β	Macrophages	Fever, endothelial activation, acute-phase response. No signal peptide; needs the inflammasome and a GSDMD pore	CAPS, gout → anakinra, canakinumab
IL-2	Activated T cells	Autocrine T-cell growth and survival; Treg development. The pleiotropy example	Proleukin; anti-CD25 basiliximab
IL-4	Th2, mast cells	Drives Th2 via STAT6 → GATA-3. Induces IgE ; inhibits Th1	Atopy → dupilumab (anti-IL-4R α)
IL-5	Th2	Eosinophils ; augments IgA production	Eosinophilic asthma → mepolizumab
IL-6	Macrophages, T cells	Acute-phase response (CRP), fever, antibody production; with TGF- β drives Th17	Cytokine storm, CRS → tocilizumab
IL-7	Stromal cells	Survival during differentiation ; T-cell development; memory maintenance	IL-7R α deficiency → T⁺B⁺NK⁺ SCID
IL-12	Macrophages, DCs	Drives Th1 via STAT4 → T-bet; activates NK cells. Shares p40 with IL-23	Psoriasis → ustekinumab (anti-p40)
IL-15	Macrophages, stroma	NK-cell development and survival; memory CD8 maintenance. The redundancy example with IL-2	γ c family; lost in X-SCID
IL-17	Th17	Recruits neutrophils ; defence against extracellular bacteria and fungi	Psoriasis, AS → secukinumab
IL-18	Macrophages	With IL-12 induces IFN-γ . Like IL-1 β : no signal peptide, inflammasome-dependent	Raised in HLH and MAS
IL-21	Tfh, Th17	B-cell differentiation and antibody production; germinal-centre function	γ c family; loss contributes to the B⁺ but use-less phenotype of X-SCID
IFN- γ	Th1, NK cells	Activates macrophages ; up-regulates MHC I and II; inhibits Th2; induces opsonising IgG subclasses	IFN- γ receptor defects → mycobacterial disease
IFN- α/β	Most cells; pDCs	Antiviral state ; downstream of RIG-I and endosomal TLRs	Hepatitis therapy; IFN signature in SLE
TNF- α	Macrophages, T cells	Vasodilation, permeability, fever, cachexia ; maintains granulomas; DC maturation	RA, psoriasis, IBD → infliximab, adalimumab, etanercept. Screen for TB

Cytokine	Made by	Main action	Clinical hook
LT-α / LT-β	Lymphocytes	Lymphoid organogenesis — lymph nodes, Peyer's patches, FDC networks	Knockout → no lymph nodes at all
TGF-β	Many cells, Tregs	Immunosuppressive ; induces IgA ; with IL-6 drives Th17; fibrosis	The switch between tolerance and Th17 depends on IL-6
FasL, TRAIL	Activated T cells	Instructed apoptosis via caspase-8	Fas mutation → ALPS ; lpr and gld mice
CXCL8 (IL-8)	Macrophages	Chemotaxis of neutrophils, basophils and T cells	The prototype chemokine

9.4 Master Table — The Two Two-Signal Systems Compared

THIS TABLE IS THE REASON THE TWO LECTURES ARE IN ONE GUIDE

Lecture 3 and Lecture 4 describe two different cells doing two different things, and they are built on the same control architecture. If you can fill in this table you have understood both lectures at once, and you will never again mix up which signal is which.

TABLE 9.4 T-CELL ACTIVATION AND THE INFLAMMASOME, SIDE BY SIDE

	T-cell activation (Lecture 3)	The inflammasome (Lecture 4)
Cell	Naïve T lymphocyte	Macrophage or dendritic cell
Signal 1 — the specific one	TCR/CD3 + CD4 or CD8 reads peptide-MHC	PRR reads a PAMP — TLR4 and LPS
What signal 1 produces	NFAT, NF- κ B, AP-1 → IL-2 and CD25	NF- κ B → pro-IL-1β, pro-IL-18 and NLRP3
Signal 2 — the permission	CD28 binds B7 (CD80/86) on an activated APC	A danger event : ATP, a crystal, a pore-forming toxin → K⁺ efflux
Result of both together	Proliferation, differentiation, effector function	Caspase-1 activation, mature IL-1β and IL-18, pyroptosis
Result of signal 1 alone	Anergy — the cell survives but is permanently unresponsive	A primed cell that releases nothing — full of inert pro-cytokine
The built-in brake	CTLA-4 , induced by the activation it then limits	Mitophagy, MyD88s, IL-1ra — same self-limiting design
Disease of losing the brake	CTLA-4 haploinsufficiency ; checkpoint-blockade autoimmunity	CAPS — NLRP3 gain of function
Why two signals at all	Because the committed response is irreversible and dangerous. One signal says “something is here”; two say “and it is actually a threat”. Requiring both makes an accidental attack on self improbable	

9.5 Tell Them Apart — the Look-Alikes

TABLE 9.5 FOURTEEN CONFUSABLE PAIRS AND THE FACT THAT SEPARATES THEM

The pair	The separating fact	Where
CD28 vs CTLA-4	Same ligand (B7). CD28 is constitutive and stimulatory; CTLA-4 is induced, higher-affinity and inhibitory	\$2.2
CD40 vs CD40L	CD40 is on the B cell/APC; CD40L (CD154) is on the T cell. Losing the ligand gives hyper-IgM	\$1.4
CD3 vs CD79a/b	The same job in different cells: CD3 is the T-cell ITAM unit, CD79 the B-cell one	\$1.4, \$1.5
CD21 vs CD35	CD21 = CR2, binds C3d (B-cell co-receptor). CD35 = CR1, binds C3b (regulation and transport)	\$1.2
MHC I vs MHC II	Inside-eight, outside-four. And the rule of eight: $CD4 \times II = CD8 \times I = 8$	\$3.4, Table 9.2
β_2 -microglobulin vs the class II β chain	β_2m is chromosome 15, invariant, not an MHC gene . The class II β chain is an MHC gene and is polymorphic	\$3.2
Pleiotropy vs redundancy	One cytokine, many effects versus many cytokines, one effect	\$4.2
γc vs JAK3	The receptor chain versus the kinase it carries . Losing either gives SCID — but γc is X-linked and JAK3 is autosomal recessive	\$4.4, \$4.5
T-bet vs GATA-3 vs ROR γ t	Master transcription factors of Th1, Th2, Th17 . Driven by IL-12, IL-4, TGF-β+IL-6 respectively	\$5.1
PAMP vs DAMP	Microbial molecule versus host molecule in the wrong place. Same receptors, same cytokines	\$6.3
Surface TLRs vs endosomal TLRs	Surface read surface molecules (LPS, flagellin, peptidoglycan); endosomal read nucleic acids . The location is the self-non-self discrimination	\$6.4
MyD88 vs TRIF	MyD88 is used by every TLR except TLR3. TLR3 uses TRIF	\$6.5
Caspase-1 vs caspase-3	Inflammatory (cuts IL-1 β and GSDMD, causes pyroptosis) versus apoptotic (silent cell death). Same family, opposite meaning	\$7.3
Apoptosis vs pyroptosis	Membrane intact, immunologically silent versus membrane ruptured by GSDMD pores, maximally inflammatory	\$7.6

9.6 Self-Test — 30 Questions with Answers

Answer each one out loud or on paper before reading on. An answer you can recognise but not produce is not learned.

Q1 What exactly is clustered in a "cluster of differentiation"?

The monoclonal antibodies, not the molecules. When antibodies raised in different laboratories all turn out to react with the same membrane molecule, they are grouped as one cluster and that cluster gets a number. The definition is therefore **operational** — it says how the molecule was detected, not what it is or does. (§1.2)

Q2 Why can neither the BCR nor the TCR signal on its own, and what solves the problem in each case?

Both have **cytoplasmic tails too short for any kinase to act on** — membrane IgM's is three amino acids. Each is therefore non-covalently bolted to a separate signalling unit whose long tails carry **ITAMs: CD79a/CD79b (Igα/Igβ)** for the BCR and **CD3** for the TCR. Antenna and wire. (§1.4, §1.5)

Q3 Give four names for CD21 and say what it binds.

CD21 = CR2 = C3d receptor = EBV receptor. It binds **C3d** left on antigen by complement. With CD19 and CD81 it forms the B-cell co-receptor, and engaging it together with the BCR lowers the activation threshold by up to **1000-fold**. Epstein-Barr virus uses the same molecule to enter B cells. (§1.2, §1.4)

Q4 A patient makes IgM but no IgG, IgA or IgE. Which molecule is defective, and on which cell does it sit?

CD40 ligand (CD154), on the T cell — X-linked hyper-IgM syndrome. Class switching requires the B cell's CD40 to be engaged by CD40L on an activated helper T cell. Without that signal the B cell is never instructed to switch, so it stays on its default isotype. Note that the defect is in the **T** cell even though the visible abnormality is in the antibody. (§1.4)

Q5 Why do CD4 and CD8 bind non-polymorphic regions of the MHC molecule?

Because if they bound a variable region, each co-receptor would have to be re-matched to every MHC allele in every person, which is impossible. By binding **α3** (class I) and **β2** (class II) — constant across all alleles — one CD8 works with every HLA-A, -B and -C a person carries. **Polymorphism is confined to the peptide-binding cleft, where variation is needed; the rest is kept constant, where variation would break the machinery.** (§1.5)

Q6 Name the three signals of T-cell activation, the molecule that delivers each, and the cell it sits on.

Signal 1: TCR/CD3 with CD4 or CD8, on the **T cell**, reading peptide-MHC on the APC. **Signal 2:** CD28 on the **T cell**, binding B7-1/B7-2 (CD80/CD86) on the **APC**. **Signal 3:** cytokines — IL-12, IL-4, IL-6/TGF-β — secreted by the **APC**. Key, accelerator, steering wheel. (§2.1–§2.4)

Q7 A T cell meets its antigen on a resting tissue cell. What happens, and why is that useful?

It receives **signal 1 without signal 2** and becomes **anergic** — it survives but is permanently unresponsive to that antigen, even if it is later presented properly. This is useful because **resting tissue cells carry no B7**, so a self-reactive T cell that escaped thymic selection is **silenced by meeting its self-antigen rather than activated by it**. The two-signal requirement is peripheral tolerance. (§2.2)

Q8 CD28 and CTLA-4 bind the same ligand. Give three differences between them.

Timing: CD28 is constitutive; CTLA-4 is **induced after TCR stimulation**. **Affinity:** CTLA-4 binds B7 much more tightly, so once present it outcompetes CD28. **Effect:** CD28 promotes proliferation and IL-2; CTLA-4 causes cell-cycle arrest. The design is **self-limiting** — switching the cell on also switches on the molecule that will switch it off. (§2.2)

Q9 Trace the TCR signal from ITAM to gene, and name the drug class that blocks it.

Lck (brought by CD4/CD8) phosphorylates the CD3 ITAMs → **ZAP-70** docks on the phospho-ITAMs → **PLCγ1** makes IP₃ and DAG, and Ras/Rac are loaded with GTP → **Ca²⁺ with calcineurin, PKC, ERK and JNK** → the transcription factors **NFAT, NF-κB and AP-1** → the **IL-2** gene. **Calcineurin inhibitors** — **cyclosporin and tacrolimus** — block the NFAT arm, which is how transplant rejection is prevented. (§2.3)

Q10 Where is the MHC locus, what is it called in humans and in mice, and what unexpected genes sit in its class III region?

Chromosome 6 in humans, where it is called **HLA**; in the mouse it is the **H-2** locus. The **class III** region contains neither class I nor class II genes but the complement components **C4, factor B and C2**, and the cytokines **TNF, LT-α and LT-β**. (§3.1)

Q11 Give the genes, the chains, the cleft, the co-receptor site, the distribution and the function of MHC class I.

Genes: HLA-A, -B, -C. **Chains:** α (45 kDa) plus **β₂-microglobulin** (12 kDa, encoded on **chromosome 15**, not in the MHC). **Cleft:** α1 + α2, **closed at both ends, 8–10 residues**. **CD8 binds α3**. **Distribution:** every nucleated cell. **Function:** present **endogenous** antigen to **CD8⁺** cytotoxic T cells. (§3.2)

Q12 Why can class II hold a peptide of 20 residues when class I cannot?

Because the **class I cleft is closed at both ends** (a hot-dog bun) and grips the peptide's N- and C-termini, so only 8–10 residues fit; the **class II cleft is open at both ends** (an open pita) and grips the middle, letting the ends hang out. The shapes follow the machinery: class I peptides are cut to a standard size by the **proteasome**, whereas class II peptides are produced in ragged lengths by **endosomal cathepsins**. (§3.4)

Q13 A tumour has lost surface MHC class I. Why is that both a good and a bad strategy for it?

Good: without class I it cannot present its tumour peptides, so **CD8⁺ cytotoxic T cells cannot see it**. **Bad:** a cell with no class I is exactly what an **NK cell** kills, by the *missing-self* rule. Tumour escape from one arm of immunity exposes it to the other. (§3.5)

Q14 Name the HLA allele associated with ankylosing spondylitis, with its reported frequency, and say what “association” does and does not mean.

HLA-B27, present in **58–97%** of patients depending on the population, against roughly 8% of the general European population. Association means **increased risk**, not causation: B27 is **neither necessary nor sufficient** — most B27-positive people never develop the disease, and some patients are B27-negative. It is a risk allele, not a diagnostic test. (§3.5)

Q15 Define pleiotropy, redundancy, synergy and antagonism, with one cytokine example each.

Pleiotropy — one cytokine, many effects: **IL-2** controls proliferation, apoptosis and Treg development. **Redundancy** — different cytokines, the same effect: **IL-2 and IL-15** both sustain NK-cell proliferation. **Synergy** — together more than the sum: **IL-12 and IFN- γ** driving Th1. **Antagonism** — one blocks the other: **IL-12/IFN- γ versus IL-4** in Th1/Th2 development. (§4.2)

Q16 Describe the three affinity states of the IL-2 receptor and explain why the design matters.

α alone — low affinity ($\sim 10^{-8}$ M), binds but **cannot signal**, because CD25 has no signalling tail. **$\beta + \gamma$** — intermediate ($\sim 10^{-9}$ M), can signal; the resting form. **$\alpha + \beta + \gamma$** — high affinity ($\sim 10^{-11}$ M), a hundred-fold tighter. Since **CD25 is induced by activation**, only a T cell that has just seen its antigen can capture IL-2 at physiological concentrations. **That is how a diffusible growth factor produces a clonal response.** (§4.3)

Q17 A boy has almost no T cells, no NK cells, and normal numbers of B cells that make no useful antibody. What is the defect, and derive the phenotype from it.

X-linked SCID — mutation in the **common γ chain (γ c)**. γ c is shared by the receptors for **IL-2, IL-4, IL-7, IL-9, IL-15 and IL-21**. Losing it removes six signals: **no IL-7 $\rightarrow$ no T cells; no IL-15 $\rightarrow$ no NK cells; no IL-21 $\rightarrow$ B cells present but unable to make useful antibody**. Hence **T⁻ B⁺ NK⁻**. X-linked because the gene is on the X chromosome, so it presents in boys. Cure: haematopoietic stem-cell transplant. A clinically identical but **autosomal recessive** SCID is caused by **JAK3** deficiency, because JAK3 is the kinase γ c carries. (§4.4)

Q18 Why is JAK-STAT called the shortest signalling pathway, and name two ways the cell switches it off.

Because it goes from the outside of the membrane to DNA in five steps with **no second messengers at all**: ligand brings the receptor chains together $\rightarrow$ JAKs trans-phosphorylate $\rightarrow$ receptor tails are phosphorylated $\rightarrow$ a STAT docks via its SH2 domain and is phosphorylated $\rightarrow$ the STAT dimer enters the nucleus. The STAT is **both** transducer and transcription factor. Off-switches: **SOCS** (binds and inhibits JAKs, then recruits a Cullin E3 ligase so the JAK is destroyed in the proteasome), **CIS** (occupies the STAT docking site) and **phosphatases such as SHP-1**. (§4.5, §4.6)

Q19 For Th1, Th2 and Th17, give the driving cytokine, the master transcription factor, the main effector cytokine and the cell type recruited.

Th1: IL-12 $\rightarrow$ STAT4 $\rightarrow$ **T-bet** $\rightarrow$ **IFN- γ** $\rightarrow$ activates **macrophages**; intracellular pathogens. **Th2**: IL-4 $\rightarrow$ STAT6 $\rightarrow$ **GATA-3** $\rightarrow$ **IL-4, IL-5, IL-13** $\rightarrow$ **eosinophils** and IgE; helminths and allergy. **Th17**: TGF- β + IL-6 (+ IL-21, IL-23) $\rightarrow$ **ROR γ t** $\rightarrow$ **IL-17** $\rightarrow$ **neutrophils**; extracellular bacteria and fungi, and psoriasis when excessive. (§5.1)

Q20 Which cytokine drives IgE, which drives IgA, and what is the trap in the lecturer's switching table?

IL-4 (with IL-13) induces **IgE** — the Th2 cytokine making the allergy antibody. **TGF- β** induces **IgA**, and IL-5 augments IgA production. **IFN- γ** does the mirror image, inducing opsonising IgG subclasses and inhibiting IgE. **The trap**: the table lists **IgG2a and IgG2b, which are mouse subclasses**. Human IgG subclasses are IgG1–4; there is no human IgG2a. The biology transfers; the numbers do not. (§5.2)

Q21 Explain the “IL-2 pump” experiment and what it proves.

A superantigen expands all T cells carrying a particular **Vβ** segment; the population then **contracts**. Implanting an **IL-2 pump rescues the contraction** — the cells do not die. This proves the deaths were caused by the **absence** of a survival cytokine, not by anything the antigen did, which is why they are called **passive cell death** (death by neglect), regulated by the **Bcl-2 family** through the intrinsic mitochondrial pathway. (§5.4)

Q22 Distinguish IL-2 from IL-7 in one sentence each, and give the immunodeficiency caused by losing the IL-7 receptor.

IL-2: survival during proliferation. IL-7: survival during differentiation, and maintenance of memory — IL-7R^{hi} effector cells are the ones that become memory cells. Losing **IL-7Rα (CD127)** causes a **T⁻ B⁺ NK⁺ SCID**: only T cells are missing, because only they depend absolutely on IL-7. (§5.4)

Q23 A child has massive chronic lymphadenopathy, splenomegaly, autoimmune cytopenias and expanded double-negative T cells. What is the diagnosis and why does the defect cause too much immune system?

ALPS — autoimmune lymphoproliferative syndrome, from a **Fas** (or FasL) mutation. Fas-mediated apoptosis is the **active**, instructed death pathway that deletes repeatedly restimulated and self-reactive lymphocytes. Losing it does not cause immunodeficiency; it causes **accumulation** — lymphocytes that should have been deleted are not, and among them are the self-reactive ones. **Autoimmunity here is a failure of subtraction**. The mouse equivalents are the **lpr** (Fas) and **gld** (FasL) strains, both with lupus-like disease. (§5.5)

Q24 Give four differences between innate and adaptive immunity, and state how innate immunity controls adaptive immunity.

Timing (minutes-hours vs days), **specificity** (general molecular patterns vs one antigen), **receptor origin** (germline-encoded, a few hundred vs somatically rearranged, ~10¹¹) and **memory** (classically none vs the basis of vaccination). Innate immunity controls adaptive immunity by supplying **signal 2** — B7 appears on an APC only after its own PRRs have fired — and **signal 3**, the cytokine that decides which effector lineage results. **The innate system decides whether the adaptive response happens and what kind it is**. (§6.1)

Q25 Match each cardinal sign of inflammation to a mediator.

Rubor, calor and tumor — TNF and IL-1 acting on endothelium: vasodilation and increased permeability. **Calor** also systemically — IL-1 and IL-6 on the hypothalamus give **fever**. **Dolor** — prostaglandins and bradykinin sensitising nociceptors. **Functio laesa** — the sum of swelling, pain and cellular infiltrate. (§6.2)

Q26 Match six microbial ligands to their TLRs, and explain why nucleic-acid sensors are confined to the endosome.

LPS → TLR4 (with MD-2 and CD14); **peptidoglycan, porins, lipoarabinomannan → TLR2**; **triacyl lipopeptides → TLR1/2**; **diacyl lipopeptides and lipoteichoic acid → TLR2/6**; **flagellin → TLR5**; **dsRNA → TLR3**; **ssRNA → TLR7**; **CpG DNA → TLR9**. Nucleic acids are chemically almost identical whether they are yours or a microbe's, so chemistry alone cannot discriminate — **geography does**. The sensors sit in the endosome, a compartment your own DNA never enters, so anything found there arrived inside something that was swallowed. When that fails, you get the **type I interferon signature of SLE**, which is why hydroxychloroquine (which raises endosomal pH) works. (§6.4)

Q27 Draw the TLR to NF- κ B chain, and say what MyD88-knockout mice tell us beyond “no response to LPS”.

Ligand dimerises the TLR → **MyD88** binds its TIR domain → recruits **IRAK-4 and IRAK-1** → activates **TRAF-6**, an E3 ligase that builds a **polyubiquitin scaffold** → **TAB1/2 and TAK1** dock on it → TAK1 phosphorylates **IKK β** → I κ B is phosphorylated and degraded → **NF- κ B** enters the nucleus. MyD88 is used by every TLR **except TLR3**. Beyond LPS unresponsiveness, MyD88^{-/-} mice have **intestines that bleed and fail to repair after injury**: recognition of the **commensal** microflora by TLRs is required for intestinal homeostasis. **Inflammatory signalling is not only an alarm; switching it off entirely is not health.** (§6.5)

Q28 Why does IL-1 β need the inflammasome when TNF does not?

Two reasons, both structural. **IL-1 β has no signal peptide**, so it is never routed into the ER–Golgi secretory pathway and cannot leave by the normal door; and it is made as an **inactive 31 kDa pro-form**. So NF- κ B (signal 1) writes the pro-form and stops. A **second signal** must assemble the inflammasome, activate **caspase-1** to cut the pro-form to its mature 17 kDa form, and cleave **gasdermin D** to open a pore for it to pass through. TNF has a signal peptide and needs none of this. **The extra machinery is a second safety catch on a cytokine that can cause shock.** (§7.1, §7.6)

Q29 ATP, uric acid crystals, palmitate and a pore-forming toxin all activate NLRP3. Since they have nothing chemically in common, how is that possible?

Because **NLRP3 does not sense them directly**. It senses a **cellular consequence** they all produce — most importantly **potassium efflux**, the common denominator of pore formation, P2X₇ opening and membrane damage. There are also K⁺-independent routes: **lysosomal rupture releasing cathepsin B** (the crystal route), **mitochondrial ROS and oxidised mtDNA**, **Ca²⁺ influx**, and licensing by the kinase **NEK7**. **NLRP3 is a detector of cellular distress, not a receptor for a molecule** — which is why it appears in gout, atherosclerosis, type 2 diabetes and silicosis as well as in infection. (§7.4)

Q30 Compare apoptosis and pyroptosis, and say which caspases produce each.

Apoptosis: caspase-8 (extrinsic) or caspase-9 (intrinsic) → caspase-3/6/7. The membrane stays **intact**, the contents are packaged, and the death is **immunologically silent**. **Pyroptosis**: the inflammatory caspases — **caspase-1** canonically, and **caspase-4/5 in humans (caspase-11 in mice)** non-canonically — cleave **gasdermin D**; the N-terminal fragment forms a **27-fold ring pore of 108 β -strands, ~180 Å across**; IL-1 β and IL-18 escape through it, and the cell finally bursts, **spilling every DAMP into the tissue**. Maximally inflammatory, which is the correct response to an intracellular pathogen: destroy its niche and raise the alarm at once. (§7.3, §7.6)

9.7 Glossary of Key Terms

Anergy

Functional inactivation of a lymphocyte that received signal 1 without signal 2. Not absence of response — permanent unresponsiveness.

ASC

The inflammasome adaptor: a PYD plus a CARD and nothing else. Forms the visible *speck*.

B7 (CD80/CD86)

Costimulatory ligands on the APC; bind CD28 and CTLA-4. Expressed only after the APC is activated.

CAPS

Cryopyrin-associated periodic syndromes: NLRP3 gain-of-function. FCAS, Muckle–Wells, NOMID. Respond dramatically to IL-1 blockade.

CARD

Caspase recruitment domain; the protein–protein interaction module linking ASC to caspase-1.

Caspase-1

The inflammatory caspase (originally ICE). Cleaves pro-IL-1 β , pro-IL-18 and gasdermin D. Activated by auto-processing in the inflammasome.

CD (cluster of differentiation)

A membrane molecule recognised by a cluster of monoclonal antibodies. An operational, not structural, definition.

CTLA-4 (CD152)

The brake on T-cell activation. Induced after TCR firing, outcompetes CD28 for B7. Target of ipilimumab.

Cytokine storm

Raised circulating cytokines + acute systemic inflammation + secondary organ dysfunction beyond that caused by the pathogen.

DAMP

Damage-associated molecular pattern: a host molecule normally hidden inside cells — ATP, uric acid, HMGB1, mitochondrial DNA.

Gasdermin D

Cleaved by inflammatory caspases; its N-terminal fragment forms the membrane pore through which IL-1 β escapes, and causes pyroptosis.

 γ c (common γ chain, CD132)

Shared receptor subunit for IL-2, 4, 7, 9, 15 and 21; carries JAK3. Mutation causes X-linked SCID.

HLA

Human leukocyte antigens — the human MHC locus, on chromosome 6.

Inflammasome

Multiprotein oligomer — sensor + ASC + caspase-1 — whose assembly activates caspase-1 by auto-processing.

ITAM

Immunoreceptor tyrosine-based activation motif, YxxL/V. Phosphorylated by Src-family kinases; docking site for Syk or ZAP-70.

JAK

Janus kinase; JAK1, 2, 3 and Tyk2. Pre-bound to cytokine receptor tails. JAK3 is restricted to haematopoietic cells.

Lck

The Src-family kinase carried by CD4 and CD8 to the TCR/CD3 complex; starts the signal by phosphorylating the ITAMs.

MHC restriction

A T cell recognises peptide only when presented by the MHC allele it developed with. Doherty and Zinkernagel, Nobel 1996.

MyD88

The adaptor used by every TLR except TLR3, and by the IL-1 receptor family. Knockout: no LPS response, and failure of intestinal homeostasis.

NLRP3

The best-studied inflammasome sensor (PYD-NACHT-LRR). Detects cellular distress, chiefly K⁺ efflux, rather than a specific ligand.

PAMP

Pathogen-associated molecular pattern: microbial, essential to the microbe, shared across whole classes — LPS, flagellin, CpG DNA.

PRR

Pattern-recognition receptor: germline-encoded sensor on the surface, in endosomes or in the cytosol.

Pyroptosis

Lytic, highly inflammatory cell death through gasdermin-D pores. The opposite of apoptosis in immunological meaning.

RIG-I

Cytosolic sensor of viral RNA bearing a 5'-triphosphate; signals through MAVS on mitochondria to IRF3 and type I interferon.

SCID

Severe combined immunodeficiency. γ c mutation gives X-linked T⁻B⁺NK⁻; JAK3 gives the same picture autosomal recessively; IL-7R α gives T⁻B⁺NK⁺.

Signal 1 / Signal 2 / Signal 3

Recognition / permission / direction. The control architecture shared by T-cell activation and the inflammasome.

SOCS

Suppressor of cytokine signalling: binds and inhibits JAKs, then recruits an E3 ligase so they are destroyed. Induced by the cytokine it suppresses.

STAT

Signal transducer and activator of transcription — one protein that is both. STAT4 for IL-12/Th1, STAT6 for IL-4/Th2.

TLR

Toll-like receptor. Surface TLRs read microbial surface molecules; endosomal TLRs (3, 7, 8, 9) read nucleic acids. Nobel Prize 2011.

ZAP-70

The tandem-SH2 kinase that docks on phosphorylated CD3 ITAMs; the T-cell equivalent of Syk.

LAST WORD

Three sentences carry these two lectures. **A CD molecule is half of a pair** — name the partner and the cell it sits on, and you have named the fact. **An MHC molecule is a display case** — ask where the peptide came from and the class, the co-receptor and the T cell all follow. **A cytokine is an instruction** — and no cell obeys an instruction until a second, independent signal has confirmed that the danger is real. Everything else in these nine Parts is detail hung on those three frames.